Supplementary Material for "Private Minimum Hellinger Distance Estimation via Hellinger Distance Differential Privacy"

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Editor: Action Editor name

6 Proofs

In this section we provide proofs and detailed results in main-paper order. Unprefixed result numbers identify results in the main paper or statements relocated here; section-prefixed result numbers identify new supplementary results. Section numbers and equation references within the supplement refer to this document unless the main paper is named. Sections 7–11 contain regularity conditions, mechanism calculations, privacy comparisons, supporting proofs, and additional numerical results, respectively.

6.1 Proof of Proposition 6

Proof Put $\nu = P_1 + P_2$ and $q = dP_1/d\nu$, so $dP_2/d\nu = 1 - q$. For the proof, write $D_\lambda(P_1, P_2) = \int J_\lambda(q)\, d\nu$, where

$$J_\lambda(u) = \frac{u^{\lambda+1}(1-u)^{-\lambda} - (\lambda+1)u + \lambda(1-u)}{\lambda(\lambda+1)}, \qquad 0 < u < 1.$$

At $\lambda = 0, -1$, respectively, use $J_0(u) = u\log\{u/(1-u)\} - 2u + 1$ and $J_{-1}(u) = (1-u)\log\{(1-u)/u\} + 2u - 1$. Give these functions their lower-semicontinuous extensions to $[0, 1]$. They are nonnegative and convex: on $(0, 1)$, $J_\lambda(1/2) = J'_\lambda(1/2) = 0$ and $J''_\lambda(u) = u^{\lambda-1}(1-u)^{-\lambda-2} > 0$. The linear terms in the numerator integrate to -1, so the integral is the original power divergence, with the stated boundary conventions.

For a cell A of positive ν-measure, put $\bar{q}_A = P_1(A)/\nu(A)$. Jensen's inequality gives

$$D_\lambda(P_1^\pi, P_2^\pi) = \sum_{A \in \pi:\, \nu(A) > 0} \nu(A) J_\lambda(\bar{q}_A) \le \int J_\lambda(q)\, d\nu.$$

Applying this inequality within the cells of a coarser partition proves monotonicity under refinement.

Let $(I_{r,j})_j$ be the nested dyadic partitions of $[0,1]$, with interval length 2^{-r} and 1 included in the last interval, and set $A_{r,j} = \{q \in I_{r,j}\}$. Omit empty cells to obtain π_r, and define

$$s_r = \sum_j \min\left\{ r, \inf_{u \in \overline{I_{r,j}}} J_\lambda(u) \right\} \mathbf{1}_{A_{r,j}}.$$

These are nonnegative simple functions. Nestedness and lower semicontinuity give $s_r \uparrow J_\lambda(q)$, including points where the limit is infinite. Also, $\bar{q}_{A_{r,j}}$ belongs to $\overline{I_{r,j}}$, and hence

$$\int s_r \, d\nu \le D_\lambda(P_1^{\pi_r}, P_2^{\pi_r}) \le D_\lambda(P_1, P_2).$$

The monotone convergence theorem proves the finite-partition identity, also when the divergence is infinite. The privacy equivalence follows by applying the identity to each neighboring pair. ■

6.2 Proof of Theorem 8

We start with the proof of Theorem 8. We begin with the case $\lambda(\lambda+1) \neq 0$. Since $M_1(w, D)$ is (λ, ϵ)-PDP, the power divergence between $M_1(w, D)$ and $M_1(w, D')$ is atmost ϵ_1. For brevity, we denote the random variables $M_1(w, D)$ and $M_1(w, D')$ by X_1 and X_2 respectively. Let $p_1(\cdot)$ and $p_2(\cdot)$ denote their densities. Thus, by the $(\lambda, \epsilon)-$PDP property it follows that,

$$\frac{1}{\lambda(\lambda+1)} \mathbf{E}_{p_2}\left[\left(\frac{p_1(X_2)}{p_2(X_2)} \right)^{\lambda+1} \right] \le \epsilon_1 + \frac{1}{\lambda(\lambda+1)} \tag{1}$$

Next, let the random variables $Y_1|X_1$ and $Y_2|X_2$ represent the compositions of mechanisms $M_2(M_1(w, D), D)$ given $M_1(w, D)$ and $M_2(M_1(w, D'), D')$ given $M_1(w, D')$ with conditional densities $q_{Y_1|X_1}(\cdot)$ and $q_{Y_2|X_2}(\cdot)$ respectively. Again using the PDP property, it follows that for a generic random variable $V \sim q_{Y_2|X_2}(\cdot)$,

$$\frac{1}{\lambda(\lambda+1)} \mathbf{E}_{q_{Y_2|X_2}}\left[\left(\frac{q_{Y_1|X_1}(V)}{q_{Y_2|X_2}(V)} \right)^{\lambda+1} \right] \le \epsilon_2 + \frac{1}{\lambda(\lambda+1)} \tag{2}$$

Now, to calculate power divergence of $M^{(2)}(w, D)$ and $M^{(2)}(w, D')$, we need to calculate the joint power divergence of (X_1, Y_1) and (X_2, Y_2). Now, using that the joint density is the product of conditional density and the marginal density, it follows that

$$D_\lambda(M^{(2)}(w, D), M^{(2)}(w, D')) = \frac{1}{\lambda(\lambda+1)} \mathbf{E}_{p_2}\left\{ \left(\frac{p_1(X_2)}{p_2(X_2)} \right)^{\lambda+1} \mathbf{E}_{q_{Y_2|X_2}}\left[\left(\frac{q_{Y_1|X_1}(V)}{q_{Y_2|X_2}(V)} \right)^{\lambda+1} \right] - 1 \right\}.$$

Now, suppose $\lambda(\lambda+1) > 0$. Then, using (1) and (2) it follows that

$$\begin{aligned} D_\lambda(M^{(2)}(w, D), M^{(2)}(w, D')) &\le \frac{1}{\lambda(\lambda+1)} [\epsilon_1(\lambda(\lambda+1)) + 1][\epsilon_2(\lambda(\lambda+1)) + 1] - \frac{1}{\lambda(\lambda+1)} \\ &= \epsilon_1 + \epsilon_2 + \lambda(\lambda+1)\epsilon_1\epsilon_2. \end{aligned}$$

Next, if $\lambda(\lambda+1)<0$, then using the condition $0<\epsilon<-[\lambda(\lambda+1)]^{-1}$ the above inequality continues to hold. Finally, consider the case $\lambda=0$. In this case, the power divergence reduces to the Kullback Leibler (KL) divergence between the densities. Hence, using the PDP property, it follows that

$$D_0(p_1,p_2)=KL(p_1,p_2)\leq\epsilon_1,\quad\text{and}\quad D_0(q_{Y_1|X_1},q_{Y_2|X_2})=KL(q_{Y_1|X_1},q_{Y_2|X_2})\leq\epsilon_2.$$

Hence, with $V\sim q_{Y_1|X_1}(\cdot)$, we have

$$D_\lambda(M^{(2)}(w,D),M^{(2)}(w,D'))=\mathbf{E}_{p_1}\left[\log\frac{p_1(X_2)}{p_2(X_2)}\right]+\mathbf{E}_{q_{Y_1|X_1}}\left[\log\frac{q_{Y_1|X_1}(V)}{q_{Y_2|X_2}(V)}\right]\leq\epsilon_1+\epsilon_2.$$

The proof of the case $\lambda=-1$ is similar. This completes the proof of (1). Next, the proof of (2) follows exactly as in (1) except that $M_2(X_1,D)|X_1$ and $M_2(X_2,D')|X_2$ are now replaced by $M_2(w,D)$ and $M_2(w,D')$. That is, we replace $\frac{q_{Y_1|X_1}(V)}{q_{Y_2|X_2}(V)}$ in (2) by $\frac{q_{Y_1}(V)}{q_{Y_2}(V)}$, where q_{Y_1} and q_{Y_2} are the unconditional distributions of $M_2(w,D)$ and $M_2(w,D')$ respectively and V has the density q_{Y_2}.

To start the proof of part (3), we first notice that adjacent $\mathbf{D}^{(2)}$ and $\mathbf{D}^{(2)'}$ can be decomposed into two distinct cases. By definition,

$$||\mathbf{D}^{(2)}-\mathbf{D}^{(2)'}||_H=\sum_{i=1}^{2}||D_i-D_i'||_H=1.$$

Since the Hamming distance is a non-negative integer, the above equation holds if either: Case (1): $||D_1-D_1'||_H=1$ and $||D_2-D_2'||_H=0$; or Case (2): $||D_1-D_1'||_H=0$ and $||D_2-D_2'||_H=1$. Let, as before, $p_1(\cdot)$, $p_2(\cdot)$ denote the distributions of $M_1(w_1,D_1)$, $M_1(w_1,D_1')$. Also, let $q_1(\cdot)$, and $q_2(\cdot)$ are density functions of $M_2(w_2,D_2)$ and $M_2(w_2,D_2')$ respectively. The joint density of $M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)})$ and $M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)'})$ are therefore given by $h_1(x,y)$ and $h_2(x,y)$ respectively, where $h_1(x,y)=p_1(x)q_1(y)$ and $h_2(x,y)=p_2(x)q_2(y)$.

If case (1) happens, $q_1(\cdot)=q_2(\cdot)$ holds since $D_2=D_2'$. This leads to $\frac{h_1(x,y)}{h_2(x,y)}=\frac{p_1(x)}{p_2(x)}$. The PD between $M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)})$ and $M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)'})$ is reduced to the PD between $M_1(w_1,D_1)$ and $M_1(w_1,D_1')$, since

$$\begin{aligned}D_\lambda(M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)}),M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)'}))=&\frac{1}{\lambda(\lambda+1)}\mathbf{E}_{h_2}\left[\left(\frac{h_1(X,Y)}{h_2(X,Y)}\right)^{\lambda+1}-1\right]\\=&\frac{1}{\lambda(\lambda+1)}\mathbf{E}_{p_2}\left[\left(\frac{p_1(X)}{p_2(X)}\right)^{\lambda+1}-1\right]\\=&D_\lambda(M_1(w_1,D_1),M_1(w_1,D_1'))\leq\epsilon_1.\end{aligned}$$

The last inequality follows from $M_1(w_1,D_1)$ is $(\lambda,\epsilon_1)-$PDP. Similarly in Case (2),

$$D_\lambda(M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)}),M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)'}))=D_\lambda(M_2(w_2,D_2),M_2(w_2,D_2'))\leq\epsilon_2.$$

Combining Case (1) and Case (2), we get

$$D_\lambda(M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)}),M^{(2)}(\mathbf{W}^{(2)},\mathbf{D}^{(2)'}))\leq\max\{\epsilon_1,\epsilon_2\}$$

which implies that the parallel composition is $(\lambda,\max\{\epsilon_1,\epsilon_2\})-$PDP. This completes the proof of Theorem 8. ∎

6.3 Proof of Theorem 9

We begin with the case when Y_i's are $N(0, \sigma^2)$. In this case, using Lemma 8.1 we have that

$$D_\lambda(M(w,D), M(w,D')) \;=\; \frac{1}{\lambda(\lambda+1)}\left[e^{\frac{\lambda(\lambda+1)||\mathbf{v}||_2^2}{2\sigma^2}} - 1\right],$$

where $\mathbf{v}$ is the difference between the mean of $M(w,D)$ and the mean of $M(w,D')$, and for $r = 1, 2$, $||\mathbf{v}||_r \leq \Delta_{L_r} W$. Thus, a sufficient condition for $D_\lambda(M(w,D), M(w,D')) \leq \epsilon$ is

$$\sigma \geq ||\mathbf{v}||_2 \sqrt{\frac{\lambda(\lambda+1)}{2\log(1+\lambda(\lambda+1)\epsilon)}},$$

which is well-defined for all $\epsilon > 0$ when $\lambda(\lambda+1) > 0$, and for $0 < \epsilon < -[\lambda(\lambda+1)]^{-1}$ when $\lambda(\lambda+1) < 0$. Thus, we choose

$$\sigma^2_{\lambda,\epsilon} = (\Delta_{L_2} W)^2 \frac{\lambda(\lambda+1)}{2\log(1+\lambda(\lambda+1)\epsilon)}.$$

Next, when $\lambda = 0$,

$$D_0(M(w,D), M(w,D')) = \frac{||w_1 - w_2||_2^2}{2\sigma^2}$$

Again, $D_0 M(w,D), M(w,D') \leq \epsilon$ is ensured by choosing

$$\sigma^2_{0,\epsilon} = \frac{(\Delta_{L_2} W)^2}{2\epsilon}.$$

The case for $\lambda = -1$ is similar. Next, turning to the Laplace case, using Lemma 8.2, notice that

$$\begin{aligned} D_\lambda(M(w,D), M(w,D')) \;&=\; \frac{1}{\lambda(\lambda+1)}\left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i - v_i| - \lambda|y_i|}{b}} \mathrm{d}y_i\right) - 1\right] \\ &\leq\; \frac{1}{\lambda(\lambda+1)}\left[e^{\frac{sign(\lambda)(\lambda+1)||v||_1}{b}} - 1\right]. \end{aligned}$$

Similarly,

$$D_\lambda(M(w,D'), M(w,D)) \leq \frac{1}{\lambda(\lambda+1)}\left[e^{\frac{sign(\lambda+1)(\lambda)||v||_1}{b}} - 1\right].$$

Thus, $D_\lambda(M(w,D), M(w,D')) \leq \epsilon$ is ensured by choosing

$$b \geq \max\left\{\frac{sign(\lambda)(\lambda+1)||v||_1}{\log(\lambda(\lambda+1)\epsilon+1)}, \frac{sign(\lambda+1)(\lambda)||v||_1}{\log(\lambda(\lambda+1)\epsilon+1)}\right\}.$$

Thus, we choose $b_{\lambda,\epsilon}$ to be

$$b_{\lambda,\epsilon} = \max\left\{\frac{sign(\lambda)(\lambda+1)\Delta_{L_1} W}{\log(\lambda(\lambda+1)\epsilon+1)}, \frac{sign(\lambda+1)(\lambda)\Delta_{L_1} W}{\log(\lambda(\lambda+1)\epsilon+1)}\right\}.$$

Turning to the case $\lambda(\lambda+1)=0$, we note that

$$\begin{aligned} D_0(M(w,D),M(w,D')) &= \sum_{i=1}^{m}\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{|y_i|}{b}}\cdot\frac{|y_i-v_i|-|y_i|}{b}\mathrm{d}y_i \\ &\leq \frac{||w_1-w_2||_1}{b} \end{aligned}$$

Hence, $D_0(M(w,D),M(w,D'))\leq\epsilon$ holds if $b\geq\frac{\Delta_{L_1}W}{\epsilon}$. Thus,

$$b_{0,\epsilon}=\frac{\Delta_{L_1}W}{\epsilon}.$$

The proof for the case $\lambda=-1$ is similar. ∎

6.4 Proof of Theorem 10

The result follows from Theorem 8 by taking $\lambda=-\frac{1}{2}$ and using $D_{-1/2}=2D_{HD}$. ∎

6.5 Proof of Corollary 11

The proof is based on the following iterative argument for adaptive and sequential compositions. Setting $\epsilon_1=\epsilon$ and $\epsilon_2=h_1(\epsilon)$ it follows from Theorem 10 part 1. and part 2., that $h_2(\epsilon)=\epsilon+h_1(\epsilon)-\frac{1}{2}\epsilon h_1(\epsilon)$. Now iterating, we obtain $h_{j+1}(\epsilon)=\epsilon+h_j(\epsilon)-\frac{1}{2}\epsilon h_j(\epsilon)$. The proof for the parallel compositions follows from part 3. of Theorem 10. ∎

6.6 Proof of Proposition 12

The proof of the Proposition follows using a comparison argument. Recall that the total variation distance between two densities can be expressed as one-half the L_1-norm, which is bounded above by the Hellinger distance between the densities. That is,

$$TV(p_1,p_2)=\frac{1}{2}||p_1-p_2||_1\leq HD(p_1,p_2).$$

Now, if $HD^2(p_1,p_2)\leq\epsilon$, then $TV(p_1,p_2)\leq\sqrt{\epsilon}$ which implies that $M(\cdot,\cdot)$ satisfies $\sqrt{\epsilon}-$TV privacy. Hence, using Ghazi and Issa (2024) page 209, it follows that M also satisfies $(0,\sqrt{\epsilon})$ differential privacy. Turning to μ-GDP, suppose that M satisfies either of the additional mechanism assumptions in Proposition 12. Let $\eta=\sup_{D\sim D'}\mathrm{TV}(P_D,P_{D'})$, where P_D denotes the output law at D. For the Gaussian mechanism, the exact Gaussian testing formula in Dong et al. (2022) gives the smallest GDP parameter as $2\Phi^{-1}\{(1+\eta)/2\}$. The same identity holds for the scalar Laplace mechanism by Proposition 15.1. Since the first part gives $\eta\leq\sqrt{\epsilon}$, monotonicity of Φ^{-1} implies that M satisfies μ-GDP with $\mu=2\Phi^{-1}\left(\frac{1+\sqrt{\epsilon}}{2}\right)$. ∎

6.7 Proof of Theorem 13

For k-neighbor datasets D and D', we need to bound

$$D_{HD}(M(w,D),M(w,D')).$$

By the definition of k-neighbor datasets, there exist datasets $B_0, \ldots, B_k$ with $B_0 = D$ and $B_k = D'$, such that

$$||B_i - B_{i+1}||_H = 1, \qquad i = 0, 1, \ldots, k-1.$$

Also, since $M(w, B_i)$ is $\epsilon-$HDP for all $0 \le i \le k$, we get that

$$D_{HD}(M(w, B_i), M(w, B_{i+1})) \le \epsilon.$$

Now, using that $D_{HD}^{\frac{1}{2}}(\cdot, \cdot) = HD(\cdot, \cdot)$ is a metric, using the triangle inequality

$$D_{HD}^{\frac{1}{2}}(M(w, D), M(w, D')) \le \sum_{i=1}^{k} D_{HD}^{\frac{1}{2}}(M(w, B_{i-1}), M(w, B_i)) \le k\sqrt{\epsilon}. \tag{3}$$

The result follows by squaring both sides of (3). ∎

6.8 Proof of Proposition 14

The HDP mechanism bounds follow from Theorem 9 by taking $\lambda = -\frac{1}{2}$ and replacing the PDP budget by 2ϵ, using $D_{-1/2} = 2D_{HD}$. ∎

6.9 Proof of Proposition 16

Proof Writing $a = P_1(A)$ and $b = P_2(A)$, the partition identity implies

$$(\sqrt{a} - \sqrt{b})^2 + (\sqrt{1-a} - \sqrt{1-b})^2 \le h. \tag{4}$$

Set $a = \sin^2\theta$ and $b = \sin^2\phi$, with $\theta, \phi \in [0, \pi/2]$. The inequality becomes $\cos(\theta - \phi) \ge c_h$. With $\omega_h = \arccos c_h$, it follows that

$$\max\{0, \phi - \omega_h\} \le \theta \le \min\{\pi/2, \phi + \omega_h\}.$$

Taking squared sines gives the event-probability bounds of Proposition 16. Bernoulli laws with probabilities at the two endpoints attain the bounds. Setting $b = 0$ proves the zero-probability bound of Proposition 16. Dropping the second term in (4) gives $|\sqrt{a} - \sqrt{b}| \le \sqrt{h}$. Since $\sqrt{a} \le \sqrt{b} + \sqrt{h}$, we obtain $|a - b| \le 2\sqrt{hb} + h$, proving the relative-probability bound of Proposition 16. ∎

6.10 Proof of Proposition 17

Proof Let R, R_i be the actual and counterfactual joint laws of (D, Y). Since both have prior Π on D,

$$D_{HD}(R, R_i) = \int_{\mathcal{D}} D_{HD}(P_u, P_{u^{(-i)}})\, \Pi(du) \le h.$$

Thus their affinity ρ is at least c_h. Let m, m_i be the two output densities relative to a common dominating measure ζ, and let $a(y)$ be the affinity between the two posterior laws. Disintegration and Cauchy–Schwarz give

$$\rho = \int \sqrt{m(y) m_i(y)}\, a(y)\, d\zeta(y), \qquad \rho^2 \le \int m(y) a(y)^2\, d\zeta(y).$$

For any probability laws Q_1, Q_2, another application of Cauchy–Schwarz gives $\mathrm{TV}(Q_1, Q_2)^2 \le 1 - \rho(Q_1, Q_2)^2$. Hence the left side of the claimed expectation bound is at most $1 - \rho^2 \le 1 - c_h^2 = s_h^2$. Markov's inequality proves the asserted probability bound. Where the counterfactual marginal density is zero, any posterior version may be assigned; the inequalities remain valid. ■

6.11 HDP and approximate-DP conversions (Propositions 19–20)

From HDP to differential privacy. For $t \ge 0$, define the directional pairwise privacy profiles

$$\delta_{12}(t) = \sup_A \{P_1(A) - e^t P_2(A)\} = \int (p_1 - e^t p_2)_+ \, d\nu, \qquad \delta_{21}(t) = \int (p_2 - e^t p_1)_+ \, d\nu. \quad (5)$$

An (ϵ, δ)-DP guarantee bounds both quantities at $t = \epsilon$ for every neighboring pair. The profile formulation records all such guarantees; see Balle et al. (2020).

Proposition 19 (HDP to approximate DP). An h-HDP mechanism satisfies $(\epsilon, \Delta_h(\epsilon))$-DP for every $\epsilon \ge 0$, where

$$\Delta_h(\epsilon) = \frac{\sqrt{(e^\epsilon - 1)^2 + 4e^\epsilon s_h^2} - (e^\epsilon - 1)}{2}. \quad (6)$$

This conversion is sharp over the Hellinger class for each ϵ. In particular, $\Delta_h(0) = s_h$ and $\lim_{\epsilon \to \infty} \Delta_h(\epsilon) = s_h^2$.

Proof Put $a = e^\epsilon$ and $\rho = \rho(P_1, P_2) \ge c_h$. Cauchy–Schwarz gives

$$\int |p_1 - a p_2| \, d\nu \le \left\{(1 + a)^2 - 4a\rho^2\right\}^{1/2}.$$

Since $\delta_{12}(\epsilon) = \{\int |p_1 - a p_2| \, d\nu + 1 - a\}/2$, the asserted bound follows. Interchanging P_1, P_2 proves the reverse direction. For sharpness, suppose $0 < h < 2$, set $S = \{(a-1)^2 + 4a s_h^2\}^{1/2}$ and $d = \Delta_h(\epsilon)$, and let $b = (d - s_h^2)/S$ and $u = d + ab$. These numbers belong to $[0, 1]$, and direct substitution gives

$$D_{HD}\{\mathrm{Ber}(u), \mathrm{Ber}(b)\} = h, \qquad u - ab = d.$$

Thus the bound is attained on the success event. The cases $h = 0, 2$ follow from identical and mutually singular laws, respectively. The endpoint formulas follow from (6). ■

The function Δ_h is a universal envelope over the HDP class, not necessarily the attained profile of a specified mechanism. Exact Gaussian and Laplace calibrations retain that mechanism-specific information. For example, different divergence parameters can describe the same neighboring output laws; comparing their numerical values alone does not compare disclosure probabilities.

There is also a direct relation between power divergences and the pairwise profiles:

$$D_\lambda(P_1, P_2) = \int_0^\infty \left\{e^{\lambda t}\delta_{12}(t) + e^{-(\lambda+1)t}\delta_{21}(t)\right\} \, dt, \qquad \lambda \in \mathbb{R}. \quad (7)$$

Here the integral is allowed to be infinite. To verify the identity, when $a \geq b > 0$ and $\lambda \neq -1, 0$, integrate pointwise:

$$\int_0^{\log(a/b)} e^{\lambda t}(a - e^t b)\, dt = \frac{a^{\lambda+1}b^{-\lambda} - (\lambda+1)a + \lambda b}{\lambda(\lambda+1)}.$$

For $a < b$, use the second term in (7). At $\lambda = 0, -1$, integrate directly to obtain the corresponding logarithmic expression; zero-density cases use the same extended integrals. Tonelli's theorem and $\int p_1\, d\nu = \int p_2\, d\nu = 1$ prove the identity. In particular,

$$D_{HD}(P_1, P_2) = \frac{1}{2}\int_0^{\infty} e^{-t/2}\{\delta_{12}(t) + \delta_{21}(t)\}\, dt. \tag{8}$$

Thus HDP constrains a weighted integral of the pairwise disclosure profiles, whereas a specified approximate-DP guarantee constrains those profiles at a specified threshold. These identities are pairwise; no interchange of a supremum over neighboring datasets and an integral is being asserted.

From differential privacy to HDP. The converse conversion also follows directly from the neighboring output laws.

Proposition 20 (Approximate DP to HDP). Under symmetric adjacency, an (ϵ, δ)-DP mechanism, with $\epsilon \geq 0$ and $0 \leq \delta \leq 1$, satisfies h_*-HDP, where

$$h_* = 2\left\{1 - \frac{1-\delta}{\cosh(\epsilon/2)}\right\}. \tag{9}$$

The bound is sharp. In particular, pure ϵ-DP implies $2\{1 - 1/\cosh(\epsilon/2)\}$-HDP.

Proof Put $a = e^{\epsilon}$, $u = \min\{p_1, ap_2\}$, and $v = \min\{p_2, ap_1\}$. The two directional DP inequalities give $\int u\, d\nu \geq 1-\delta$ and $\int v\, d\nu \geq 1-\delta$. Where they are positive, $a^{-1} \leq u/v \leq a$, and hence

$$\sqrt{p_1 p_2} \geq \sqrt{uv} \geq \frac{\sqrt{a}}{1+a}(u+v).$$

Integration yields $\rho(P_1, P_2) \geq 2(1-\delta)\sqrt{a}/(1+a)$ and proves (9). Equality is attained by

$$P_1 = \left(\delta, \frac{(1-\delta)a}{1+a}, \frac{1-\delta}{1+a}, 0\right),$$
$$P_2 = \left(0, \frac{1-\delta}{1+a}, \frac{(1-\delta)a}{1+a}, \delta\right).$$

Indeed, $\sum_j (P_{1j} - aP_{2j})_+ = \sum_j (P_{2j} - aP_{1j})_+ = \delta$, and the affinity equals $2(1-\delta)\sqrt{a}/(1+a)$. This is the four-output extremal experiment used by Kairouz et al. (2015). Replacing the four points by uniform laws on disjoint unit intervals gives an attaining example with Lebesgue densities. ■

Distinguishing events and comparison of protections. Let U_0, U_1 be uniform laws on disjoint unit intervals. For $0 < h < 2$, set $r = h - h^2/4$ and take

$$P_2 = U_0, \qquad P_1 = (1-r)U_0 + rU_1.$$

Then $D_{HD}(P_1, P_2) = 2(1 - \sqrt{1-r}) = h$. The event $A = \mathrm{supp}(U_1)$ has $P_2(A) = 0$ and $P_1(A) = r > 0$, so it violates every finite pure-DP guarantee. Observing this event identifies the first of the two neighboring datasets with certainty. Its probability is nevertheless bounded sharply by the zero-probability bound of Proposition 16. The same pair satisfies $(0, r)$-DP, so this example is not excluded by approximate DP at all parameter values. It also rules out a general implication from positive-budget HDP to finite-parameter GDP: testing P_2 against P_1 has positive power at zero type-I error, whereas a finite Gaussian trade-off has value one at that endpoint (Dong et al., 2022).

The example can be realized as a mechanism on a binary database domain, or by selecting P_1 or P_2 according to a binary database attribute. It identifies a distinction between the constraints, not a universal ordering of all disclosure risks. In particular, mechanisms with the same attained Hellinger distance can have crossing privacy profiles; Appendix 13 gives an example. Relative protection must therefore be assessed at specified parameters and for specified disclosure quantities. The exact calibrations developed in Section 2.4 of the main paper provide the corresponding comparisons within particular mechanism families.

6.12 Proof and additional calculations for Proposition 21

For the Gaussian mechanism in Section 2.4 of the main paper, consider neighboring output laws $P_i = N(w_i, \sigma^2 I_m)$ with $d > 0$. Their log-likelihood ratio is

$$\ell_{12}(y) = \log \frac{p_1(y)}{p_2(y)} = \frac{(w_1 - w_2)^T \{y - (w_1 + w_2)/2\}}{\sigma^2}. \tag{10}$$

Its distribution under the two neighboring output laws is

$$\ell_{12}(Y) \sim \begin{cases} N(\mu_d^2/2, \mu_d^2), & Y \sim P_1, \\ N(-\mu_d^2/2, \mu_d^2), & Y \sim P_2. \end{cases} \tag{11}$$

Proof Substitution of $Y = w_i + \sigma Z$ into the preceding likelihood-ratio identity proves (11). Under P_2, for every $s \in \mathbb{R}$,

$$\mathbf{E}_{P_2} e^{s\ell_{12}(Y)} = \exp\{s(s-1)\mu_d^2/2\}. \tag{12}$$

Taking $s = 1/2$ gives the affinity $e^{-\mu_d^2/8}$ and proves the Hellinger formula of Proposition 21. For $\epsilon \geq 0$, the event maximizing $P_1(A) - e^\epsilon P_2(A)$ is $A_\epsilon = \{\ell_{12} > \epsilon\}$. Therefore,

$$P_1(A_\epsilon) = \Phi(\mu_d/2 - \epsilon/\mu_d),$$
$$P_2(A_\epsilon) = \Phi(-\mu_d/2 - \epsilon/\mu_d).$$

Their difference, with the second term multiplied by e^ϵ, is the Gaussian privacy-profile formula in Proposition 21. Interchanging the laws gives the same profile. Setting $\epsilon = 0$

gives the TV distance. Finally, for $\mu > 0$,

$$\frac{\partial}{\partial \mu}\delta_G(\epsilon;\mu) = \phi(\mu/2 - \epsilon/\mu) > 0. \tag{13}$$

Here one uses $e^{\epsilon}\phi(-\mu/2 - \epsilon/\mu) = \phi(\mu/2 - \epsilon/\mu)$. The Hellinger distance and TV are also increasing and continuous in μ. Taking suprema over neighboring pairs proves the final assertion, even when the sensitivity supremum is approached but not attained. ■

The likelihood-ratio calculation and exact approximate-DP condition are those underlying the analytic Gaussian mechanism of Balle and Wang (2018). In particular, it is the difference of the two event probabilities above, rather than a bound on the first one alone, that gives the exact calibration.

The same likelihood-ratio moments yield the power-divergence identity stated after Proposition 21 in the main paper, including $D_{-1/2} = 2D_{HD}$ and the Gaussian Rényi divergence formula.

The Gaussian testing trade-off is $G_{\mu_d}(a) = \Phi\{\Phi^{-1}(1-a) - \mu_d\}$, where a is the type-I error probability. This follows by projecting the observation onto the direction $w_1 - w_2$ and applying the likelihood-ratio test. Thus the sharp GDP parameter of M_σ is $\mu = \Delta/\sigma$ (Dong et al., 2022). Within this Gaussian family, the GDP parameter, the attained Hellinger level, and the entire approximate-DP profile are determined by the same sensitivity-to-noise ratio.

6.13 Exact scalar Laplace calibration (Proposition 23)

Consider $M_b(D) = W(D) + Z_b$ for a scalar query, where Z_b has density $(2b)^{-1}e^{-|z|/b}$, $b > 0$, and variance $2b^2$. In this paragraph, $\Delta = \sup_{D\sim D'} |W(D) - W(D')| \in (0, \infty)$. For a neighboring pair, translation and reflection allow us to take $P_1 = \text{Laplace}(0, b)$ and $P_2 = \text{Laplace}(d, b)$, with $d \geq 0$. Their log-likelihood ratio is

$$\ell_{12}(y) = \frac{|y-d| - |y|}{b} = \begin{cases} d/b, & y \leq 0, \\ (d-2y)/b, & 0 < y < d, \\ -d/b, & y \geq d. \end{cases} \tag{14}$$

Proposition 23 (Exact scalar Laplace privacy quantities). For the scalar Laplace mechanism, put $r = \Delta/(2b)$. Its exact worst-case Hellinger level and approximate-DP profile are

$$h_L(b) = 2\{1 - (1+r)e^{-r}\}, \qquad \delta_b^L(\epsilon) = [1 - e^{\epsilon/2 - r}]_+, \qquad \epsilon \geq 0. \tag{15}$$

Its exact TV distance is $1 - e^{-r}$, and its smallest pure-DP parameter is $2r = \Delta/b$.

Proof For a pair separated by d, splitting the affinity integral at $0, d$ gives

$$\begin{aligned} \rho(P_1, P_2) &= \frac{1}{2b}\int_{\mathbb{R}} \exp\left\{-\frac{|y| + |y-d|}{2b}\right\} dy \\ &= \frac{1}{2}e^{-d/(2b)} + \frac{d}{2b}e^{-d/(2b)} + \frac{1}{2}e^{-d/(2b)} = \left(1 + \frac{d}{2b}\right)e^{-d/(2b)}. \end{aligned}$$

For $0 \le \epsilon < d/b$, the positive region of $p_1 - e^\epsilon p_2$ is $(-\infty, a_\epsilon)$, where $a_\epsilon = (d - \epsilon b)/2$. Direct integration gives

$$P_1(Y < a_\epsilon) = 1 - \tfrac{1}{2} e^{\epsilon/2 - d/(2b)},$$
$$e^\epsilon P_2(Y < a_\epsilon) = \tfrac{1}{2} e^{\epsilon/2 - d/(2b)}.$$

Subtracting proves the profile formula in this range. For $\epsilon \ge d/b$, the likelihood ratio never exceeds e^ϵ, and the profile is zero. Reflection gives the same reverse-direction profile. Both discrepancies increase continuously with d, so taking the sensitivity supremum replaces d by Δ. The assertions about TV and pure DP follow from the profile. ■

The Laplace privacy profile is also given in Balle et al. (2020, Theorem 3); the affinity calculation supplies its attained Hellinger level. For $\epsilon \ge 0$, $0 \le \delta < 1$, and $(\epsilon, \delta) \ne (0, 0)$, the minimum admissible scale is

$$b_*(\epsilon, \delta) = \frac{\Delta}{\epsilon - 2\log(1-\delta)}, \qquad \mathrm{Var}(Z_{b_*}) = 2b_*^2. \tag{16}$$

Indeed, the exact condition $[1 - e^{\epsilon/2 - r}]_+ \le \delta$ is equivalent to $r \le \epsilon/2 - \log(1-\delta)$. At the minimum scale, put $r_* = \epsilon/2 - \log(1-\delta)$. Keeping b_* fixed, the achieved Hellinger level is

$$h^L_{\text{achieved}}(\epsilon, \delta) = 2\{1 - (1 + r_*)e^{-r_*}\}. \tag{17}$$

Conversely, for $0 < h < 2$, let $r_h > 0$ be the unique solution of $(1 + r_h)e^{-r_h} = 1 - h/2$. Uniqueness follows because the left side has derivative $-re^{-r} < 0$ for $r > 0$ and decreases from 1 to 0. The minimum scale and variance for h-HDP are

$$b_H(h) = \frac{\Delta}{2r_h}, \qquad \mathrm{Var}(Z_{b_H}) = \frac{\Delta^2}{2r_h^2}, \tag{18}$$

and the exact profile at this fixed scale is

$$\delta^L_h(\epsilon) = [1 - e^{\epsilon/2 - r_h}]_+. \tag{19}$$

For $0 \le \epsilon \le 2r_h$, substituting $(\epsilon, \delta^L_h(\epsilon))$ into (16) returns $b_H(h)$. At $\epsilon = 2r_h$, this is the tight pure-DP guarantee. For $\epsilon > 2r_h$, the profile is still zero, but the requirement $(\epsilon, 0)$ permits the smaller minimum scale Δ/ϵ; these looser pure-DP guarantees do not recover the fixed scale. This distinguishes the Laplace reverse correspondence from the strictly positive Gaussian profile at every finite ϵ.

6.14 Gaussian forward and reverse calibration notes

In particular, at $\epsilon = 0$ the exact parameter is

$$\delta^G_h(0) = 2\Phi(\mu_h/2) - 1, \qquad \sigma^2_{\mathrm{BW}}(0, \delta) = \frac{\Delta^2}{4[\Phi^{-1}\{(1+\delta)/2\}]^2}. \tag{20}$$

The corresponding worst-case L_1 distance is $2\delta^G_h(0)$. At any finite ϵ, a nonconstant Gaussian query with finite positive variance has $\delta_G(\epsilon; \Delta/\sigma) > 0$, so it does not satisfy pure ϵ-DP. The

reverse correspondence is not a single pair: it is the curve defining the Gaussian reverse-calibration boundary in Corollary 22. Pairs above this curve are valid but looser descriptions of the same release. The connection with the Gaussian testing profile is given in Dong et al. (2022). If only an upper bound on Δ is used, these formulas provide certified levels; pairwise attained values use the actual separation.

6.15 Additional comparisons for Remark 24

Additional details for Remark 24 (Divergence scale and noise calibration). The dependence of a noise requirement on the divergence scale is not specific to Gaussian mechanisms. For a fixed mechanism family, let $d_j(v)$ be its worst-case discrepancy under criterion j at variance v. If $d_1(v) \le d_2(v)$ throughout the relevant range, then $\{v : d_2(v) \le t\} \subseteq \{v : d_1(v) \le t\}$. Thus the more compressed scale requires no larger minimum variance at the same numerical threshold. This elementary constraint ordering is different from comparison at corresponding operational guarantees.

For example, when $\Delta = 1$ and $h = 0.5$,

$$\sigma_H^2(h) = \frac{1}{8\log(4/3)} = 0.434507\ldots, \qquad \delta_h^G(0) = 0.551865\ldots.$$

These are attained parameters of the same Gaussian release. Imposing TV at most 0.5 instead gives minimum variance $1/\{4[\Phi^{-1}(0.75)]^2\} = 0.549527\ldots$. At the corresponding TV value $0.551865\ldots$, the exact TV calibration returns $0.434507\ldots$. The difference in the numerical coordinates is therefore distinct from a difference in protection at matched parameters. The general scale effect is also visible in the scalar Laplace family, where eliminating r gives $h = 2\{\eta + (1-\eta)\log(1-\eta)\}$ for $\eta = \mathrm{TV} \in [0, 1)$.

6.16 Gaussian calibration under composition

For a fixed neighboring pair, consider independent Gaussian releases with mean separations d_j and noise variances σ_j^2. Their joint log-likelihood ratio is the sum of the individual ones. Writing $\mu_j = d_j/\sigma_j$ and $\mu_{1:K}^2 = \sum_{j=1}^K \mu_j^2$, this sum has distribution $N(\mu_{1:K}^2/2, \mu_{1:K}^2)$ under the first joint law and $N(-\mu_{1:K}^2/2, \mu_{1:K}^2)$ under the second. Hence

$$D_{HD}(P_{1,1:K}, P_{2,1:K}) = 2\left\{1 - \exp\left(-\frac{1}{8}\sum_{j=1}^K \mu_j^2\right)\right\} = 2\left\{1 - \prod_{j=1}^K (1 - h_j/2)\right\}, \tag{21}$$

where $h_j = 2(1 - e^{-\mu_j^2/8})$. The joint profile is exactly $\delta_G(\epsilon; \mu_{1:K})$. These are pairwise identities. For a mechanism-wide product identity, the supremum is over $\sum_j d_j(D, D')^2/\sigma_j^2$; separate sensitivity suprema need not be attained by the same neighboring pair.

For adaptive releases, suppose that at every fixed preceding transcript z, the conditional Gaussian mechanism at step j has standardized separation at most a deterministic $\bar{\mu}_j$ for every neighboring pair. Its conditional affinity is at least $e^{-\bar{\mu}_j^2/8}$. If $R_{1,j}, R_{2,j}$ denote the laws of the first j releases, successive integration gives

$$\rho(R_{1,j}, R_{2,j}) \ge e^{-\bar{\mu}_j^2/8}\rho(R_{1,j-1}, R_{2,j-1}).$$

Thus the transcript is $\bar{h}$-HDP, where

$$\bar{h} = 2\left\{1 - \exp\left(-\frac{1}{8}\sum_{j=1}^{K}\bar{\mu}_j^2\right)\right\}. \tag{22}$$

The GDP composition theorem gives parameter $\bar{\mu}_{1:K} = (\sum_j \bar{\mu}_j^2)^{1/2}$ and hence the approximate-DP guarantees $(\epsilon, \delta_G(\epsilon; \bar{\mu}_{1:K}))$ (Dong et al., 2022). These are transcript guarantees in the adaptive case; the final iterate need not be Gaussian. Releasing only that iterate is post-processing. This is the form used for the iterative estimation procedures below.

To compare equal allocations, fix a reference Gaussian mechanism with Hellinger level $h_0 \in (0, 2)$, variance $\sigma_0^2 = \sigma_H^2(h_0)$, and parameter $\mu_0 = \Delta/\sigma_0$. For K steps with the same sensitivity Δ, HDP and GDP allocations are respectively

$$h_{\text{step}} = 2\{1 - (1 - h_0/2)^{1/K}\}, \qquad \mu_{\text{step}} = \mu_0/\sqrt{K}. \tag{23}$$

Both give the same per-step Gaussian variance:

$$\sigma_{H,K}^2 = \sigma_{\text{GDP},K}^2 = K\sigma_0^2. \tag{24}$$

Indeed, substitute (23) into the reverse Gaussian calibration in Corollary 22, or use $\Delta^2/\mu_{\text{step}}^2$.

For comparison with basic approximate-DP composition, fix $\epsilon_0 \geq 0$ and let $\delta_0 = \delta_G(\epsilon_0; \mu_0)$. Equal allocation uses $(\epsilon_0/K, \delta_0/K)$ at each step, with per-step variance $\sigma_{\text{BW}}^2(\epsilon_0/K, \delta_0/K)$. For the TV rule $\eta_{1:K} \leq 1 - \prod_j(1 - \eta_j)$, let $\eta_0 = 2\Phi(\mu_0/2) - 1$ and use

$$\eta_{\text{step}} = 1 - (1 - \eta_0)^{1/K}, \qquad \sigma_{\text{TV},K}^2 = \frac{\Delta^2}{4[\Phi^{-1}\{(1 + \eta_{\text{step}})/2\}]^2}. \tag{25}$$

The TV composition rule follows by coupling each pair of conditional releases while their histories agree: the probability that all steps agree is at least $\prod_j(1 - \eta_j)$.

Every per-step Gaussian variance in this comparison is calibrated exactly to its stated per-step requirement. Basic approximate-DP and the displayed TV rule provide the specified composition guarantees; they need not equal the attained profile of the joint Gaussian laws. For independent releases with unit separation and a common per-step variance v_K, that attained Hellinger level is $2\{1 - e^{-K/(8v_K)}\}$ by (21). Thus the composition comparison can report both the parameters delivered by the stated rule and the attained quantities when the joint law is available. For $h_0 = 0.5$, the reference constants are, approximately, $\sigma_0^2 = 0.434507$, $\mu_0 = 1.517055$, and $\eta_0 = 0.551865$; choosing $\epsilon_0 = 1$ gives $\delta_0 = 0.327168$. The smallest GDP parameter of the scalar Laplace mechanism is $2\Phi^{-1}\{1 - \frac{1}{2}e^{-\Delta/(2b)}\}$; the full-profile proof and the inverse calibration in the noise scale and Hellinger level are given in Appendix 15.

6.17 Proof of Proposition 30

To show $H_n(\boldsymbol{\theta})$ is $\alpha-$Lipschitz continuous, it is enough to show that the $(i, j)^{th}$ component of $H_n(\boldsymbol{\theta})$ is Lipschitz continuous for all (i, j), where the Lipschitz constant depends only on

m and the upper bounds in assumptions **(U1)** and **(U2)**. That is, we will show that for any $\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)} \in \Theta$,

$$|H_{n,i,j}(\boldsymbol{\theta}^{(1)}) - H_{n,i,j}(\boldsymbol{\theta}^{(2)})| \leq \alpha_{i,j} ||\boldsymbol{\theta}^{(1)} - \boldsymbol{\theta}^{(2)}||_2,$$

where $H_{n,i,j}(\boldsymbol{\theta})$ is the $(i,j)^{th}$ component of $H_n(\boldsymbol{\theta})$. Recall that

$$H_{n,i,j}(\boldsymbol{\theta}) = -\int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},j}(x) u_{\theta,i}(x) \mathrm{d}x - 2\int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i,j}(x) \mathrm{d}x.$$

Then for any $\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)} \in \Theta$,

$$\begin{aligned} |H_{n,i,j}(\boldsymbol{\theta}^{(1)}) - H_{n,i,j}(\boldsymbol{\theta}^{(2)})| \quad \leq \quad & \int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) |T_{1,i,j}(\boldsymbol{\theta}^{(1)}, x) - T_{1,i,j}(\boldsymbol{\theta}^{(2)}, x)| \mathrm{d}x \\ & + 2\int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) |T_{2,i,j}(\boldsymbol{\theta}^{(1)}, x) - T_{2,i,j}(\boldsymbol{\theta}^{(2)}, x)| \mathrm{d}x, \end{aligned} \tag{26}$$

where

$$T_{1,i,j}(\boldsymbol{\theta}, x) = f_{\theta}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},j}(x) u_{\boldsymbol{\theta},i}(x), \quad T_{2,i,j}(\boldsymbol{\theta}, x) = f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i,j}(x).$$

Notice that for $\boldsymbol{\theta} \in \Theta$, $T_{1,i,j}(\boldsymbol{\theta}, x)$ and $T_{2,i,j}(\boldsymbol{\theta}, x)$ are differentiable in $\boldsymbol{\theta}$ by Assumption **(U2)**. Using Cauchy-Schwarz inequality and the upper bounds in Assumption **(U2)**, for $\boldsymbol{\theta} \in \Theta$ we obtain that $g_n^{1/2}(x) ||\nabla T_{1,i,j}(\boldsymbol{\theta}, x)||_2$ and $g_n^{1/2}(x) ||\nabla T_{2,i,j}(\boldsymbol{\theta}, x)||_2$ are integrable with respect to x, where the gradient is taken with respect to $\boldsymbol{\theta}$. By the mean value theorem and Cauchy-Schwarz inequality, there exists $\boldsymbol{\theta}^{(1)*}$ and $\boldsymbol{\theta}^{(2)*}$ on the line between $\boldsymbol{\theta}^{(1)}$ and $\boldsymbol{\theta}^{(2)}$, such that

$$|T_{1,i,j}(\boldsymbol{\theta}^{(1)}, x) - T_{1,i,j}(\boldsymbol{\theta}^{(2)}, x)| \leq ||\nabla T_{1,i,j}(\boldsymbol{\theta}^{(1)*}, x)||_2 \cdot ||\boldsymbol{\theta}^{(1)} - \boldsymbol{\theta}^{(2)}||_2 \tag{27}$$

$$|T_{2,i,j}(\boldsymbol{\theta}^{(1)}, x) - T_{2,i,j}(\boldsymbol{\theta}^{(2)}, x)| \leq ||\nabla T_{2,i,j}(\boldsymbol{\theta}^{(2)*}, x)||_2 \cdot ||\boldsymbol{\theta}^{(1)} - \boldsymbol{\theta}^{(2)}||_2. \tag{28}$$

By the convexity of Θ (see Assumption **(A1)**), we obtain $\boldsymbol{\theta}^{(1)*}, \boldsymbol{\theta}^{(2)*} \in \Theta$. Now, multiplying both sides of (27) and (28) by $g_n^{\frac{1}{2}}(\cdot)$ and using the integrability described above, it follows that

$$\int g_n^{1/2}(x) ||\nabla T_{1,i,j}(\boldsymbol{\theta}, x)||_2 \mathrm{d}x \leq \left(\sup_{\boldsymbol{\theta} \in \Theta} \int g_n^{1/2}(x) ||\nabla T_{1,i,j}(\boldsymbol{\theta}, x)||_2 \mathrm{d}x \right) ||\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2||_2.$$

Using similar arguments for $T_{2,i,j}$ and setting

$$0 < \alpha_{i,j} = \sup_{\boldsymbol{\theta} \in \Theta} \left\{ \int g_n^{1/2}(x) ||\nabla T_{1,i,j}(\boldsymbol{\theta}, x)||_2 \mathrm{d}x + 2\int g_n^{1/2}(x) ||\nabla T_{2,i,j}(\boldsymbol{\theta}, x)||_2 \mathrm{d}x \right\} < \infty.$$

It follows that

$$|H_{n,i,j}(\boldsymbol{\theta}^{(1)}) - H_{n,i,j}(\boldsymbol{\theta}^{(2)})| \leq \alpha_{i,j} ||\boldsymbol{\theta}^{(1)} - \boldsymbol{\theta}^{(2)}||_2.$$

This completes the proof. ∎

Before we prove the theorem, we recall that

$$g_n(x) = \frac{1}{nc_n}\sum_{i=1}^{n} K\left(\frac{x - X_i}{c_n}\right).$$

and for the neighboring i.i.d. observations $\{X_1', X_2, \cdots, X_n\}$, the corresponding density estimator is

$$\tilde{g}_n(x) = \frac{1}{nc_n}\sum_{i=2}^{n} K\left(\frac{x - X_i}{c_n}\right) + \frac{1}{nc_n} K\left(\frac{x - X_1'}{c_n}\right).$$

The corresponding loss functions are given by

$$L_n(\boldsymbol{\theta}) = 2HD^2(g_n, f_{\boldsymbol{\theta}}) \text{ and } \tilde{L}_n(\boldsymbol{\theta}) = 2HD^2(\tilde{g}_n, f_{\boldsymbol{\theta}}),$$

and the Hessian of the loss functions are given by $H_n(\boldsymbol{\theta})$ and $\tilde{H}_n(\boldsymbol{\theta})$.

6.18 Proof of Proposition 31 and Theorem 35

We begin with the proof of Proposition 31. First, notice that for all $1 \le i \le m$

$$\frac{\partial}{\partial \theta_i} L_n(\boldsymbol{\theta}) = -2\int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i}(x)\mathrm{d}x$$

Hence,

$$\bar{\Delta}^{(i)} \coloneqq \frac{\partial}{\partial \theta_i}(L_n(\boldsymbol{\theta}) - \tilde{L}_n(\boldsymbol{\theta})) = -2\int_{\mathbb{R}} (g_n^{\frac{1}{2}}(x) - \tilde{g}_n^{\frac{1}{2}}(x)) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i}(x)\mathrm{d}x$$

where we have suppressed n in the notation $\bar{\Delta}^{(i)}$. Using Cauchy-Schwarz inequality, the $HD^2(g_n, \tilde{g}_n)$ is bounded above by $||g_n - \tilde{g}_n||_1$ and, assumption **(A2)** it follows that

$$|\bar{\Delta}^{(i)}| \le 2HD(g_n, \tilde{g}_n) \cdot \left[\int_{\mathbb{R}} f_{\boldsymbol{\theta}}(x) u_{\boldsymbol{\theta},i}^2(x)\mathrm{d}x\right]^{1/2} \le C_i ||g_n - \tilde{g}_n||_1^{\frac{1}{2}}.$$

Now,

$$||g_n - \tilde{g}_n||_1^{1/2} = \left[\frac{1}{n \cdot c_n}\int_{\mathbb{R}} |K\left(\frac{x - X_1}{c_n}\right) - K\left(\frac{x - X_1'}{c_n}\right)|\mathrm{d}x\right]^{1/2} \le \left(\frac{2}{n}\right)^{1/2}. \tag{29}$$

Hence $|\bar{\Delta}^{(i)}| \le 2C_i \left(\frac{2}{n}\right)^{1/2}$. Let $\bar{\boldsymbol{\Delta}} = [\bar{\Delta}^{(1)}, \cdots, \bar{\Delta}^{(m)}]$. Then $\Delta_{L_1}(\nabla L_n(\boldsymbol{\theta})) \le \sup_{D\sim D'} ||\bar{\boldsymbol{\Delta}}||_1 \le C \cdot m \cdot n^{-\frac{1}{2}}$ and $\Delta_{L_2}(\nabla L_n(\boldsymbol{\theta})) \le \sup_{D\sim D'} ||\bar{\boldsymbol{\Delta}}||_2 \le C \cdot \sqrt{m} \cdot n^{-\frac{1}{2}}$, where $C = \max_i\{2\sqrt{2}C_i\}$.

We now turn to the Hessian. Recall the definition $u_{\boldsymbol{\theta},i}(x) = \frac{1}{f_{\boldsymbol{\theta}}(x)} \cdot \frac{\partial}{\partial \theta_i} f_{\boldsymbol{\theta}}(x)$, $u_{\boldsymbol{\theta},i,j} = \frac{\partial}{\partial \theta_j} u_{\boldsymbol{\theta},i}$, and

$$H_{n,i,j}(\boldsymbol{\theta}) = -\int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x)[u_{\boldsymbol{\theta},j}(x)u_{\boldsymbol{\theta},i}(x) + 2u_{\boldsymbol{\theta},i,j}(x)]\mathrm{d}x.$$

Hence,

$$\bar{\Delta}^{(i,j)} := H_{n,i,j}(\theta) - \tilde{H}_{n,i,j}(\theta) = -\int_{\mathbb{R}} (g_n^{\frac{1}{2}}(x) - \tilde{g}_n^{\frac{1}{2}}(x)) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x)[u_{\boldsymbol{\theta},j}(x)u_{\boldsymbol{\theta},i}(x) + 2u_{\boldsymbol{\theta},i,j}(x)]\mathrm{d}x$$

Using Cauchy-Schwarz inequality, the $HD^2(g_n, \tilde{g}_n)$ is bounded above by $||g_n - \tilde{g}_n||_1$ and, assumptions **(U1)**-**(U2)** it follows that

$$\begin{aligned}|\bar{\Delta}^{(i,j)}| \leq & 2HD(g_n, \tilde{g}_n) \cdot \left[\int_{\mathbb{R}} f_{\boldsymbol{\theta}}(x)[u_{\boldsymbol{\theta},j}(x)u_{\boldsymbol{\theta},i}(x) + 2u_{\boldsymbol{\theta},i,j}(x)]^2\mathrm{d}x\right]^{1/2} \\ \leq & (C_{i,1} + C_{i,2})||g_n - \tilde{g}_n||_1^{\frac{1}{2}},\end{aligned}$$

where $C_{i,1}$ and $C_{i,2}$ are upper bounds given in assumptions **(U1)** and **(U2)**. Using (29), it follows that $|\bar{\Delta}^{(i,j)}| \leq C_{i,j} \cdot n^{-\frac{1}{2}}$. Let $\bar{\boldsymbol{\Delta}}$ be a $m \times m$ matrix with $(i,j)^{\text{th}}$ element $\bar{\Delta}^{(i,j)}$. Then $\Delta_{L_1}(H_n(\boldsymbol{\theta})) \leq \sup_{D\sim D'}||\bar{\boldsymbol{\Delta}}||_1 \leq C \cdot m \cdot n^{-\frac{1}{2}}$ and $\Delta_{L_2}(H_n(\boldsymbol{\theta})) \leq \sup_{D\sim D'}||\bar{\boldsymbol{\Delta}}||_2 \leq C \cdot m \cdot n^{-\frac{1}{2}}$ for some $0 < C < \infty$. This completes the proof of Proposition 31.

We next turn to the Proof of Theorem 35. We first construct the interior region used in assumption **(U3)**. Put

$$\begin{aligned}V_{\boldsymbol{\theta}}(x) = f_{\boldsymbol{\theta}}(x)\Big\{ & \sum_{i=1}^m u_{\boldsymbol{\theta},i}(x)^2 + 2\sum_{i,j=1}^m u_{\boldsymbol{\theta},i}(x)^2 u_{\boldsymbol{\theta},j}(x)^2 \\ & + 8\sum_{i,j=1}^m u_{\boldsymbol{\theta},i,j}(x)^2\Big\}, \qquad T(a) = \sup_{\boldsymbol{\theta}\in\Theta}\int_{|x|>a} V_{\boldsymbol{\theta}}(x)\,\mathrm{d}x.\end{aligned}$$

By **(U1)**–**(U2)**, $V_{\boldsymbol{\theta}}$ is integrable and its integral is continuous in $\boldsymbol{\theta}$. Pointwise continuity and Scheffé's lemma imply L_1-continuity in $\boldsymbol{\theta}$; compactness of Θ gives

$$T(a) \longrightarrow 0 \qquad (a \longrightarrow \infty).$$

For $p \in (1,2)$, choose $a_n \uparrow \infty$ with $T(a_n) \leq n^{1-2/p}$, then choose $b_n > a_n + \beta c_n$ and set

$$A_n = [-a_n, a_n] \subset S_n, \qquad S_n = (-b_n - \beta c_n, b_n + \beta c_n).$$

The construction is independent of the dataset. The bound $\|u_{\boldsymbol{\theta}}u_{\boldsymbol{\theta}}^T + 2\dot{u}_{\boldsymbol{\theta}}\|_F^2 \leq 2\|u_{\boldsymbol{\theta}}u_{\boldsymbol{\theta}}^T\|_F^2 + 8\|\dot{u}_{\boldsymbol{\theta}}\|_F^2$ controls both exterior weights. The KDE lower bound in **(U3)** is imposed only for the given dataset on this A_n; also, $0 < \delta_n \leq (2a_n)^{-1} \to 0$ since $\int \bar{g}_n \leq 1$.

To reduce notational complexity, *redefine* $\tilde{g}_n(x)$ as follows:

$$\tilde{g}_n(x) = \frac{1}{nc_n}\sum_{i=2}^n K\left(\frac{x - X_i}{c_n}\right)\mathbf{1}_{(X_i\in B_n)} + \frac{1}{nc_n}K\left(\frac{x - X_1'}{c_n}\right)\mathbf{1}_{(X_1'\in B_n)},$$

where $X_1, X_1' \in B_n$. Now the $\bar{\Delta}^{(i)}$ is given by

$$\bar{\Delta}^{(i)} := \bar{\Delta}^{(i,1)} + \bar{\Delta}^{(i,2)},$$

where

$$\bar{\Delta}^{(i,1)} = 2\int_{\mathbb{R}} (\bar{g}_n^{\frac{1}{2}}(x) - \tilde{g}_n^{\frac{1}{2}}(x)) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i}(x) \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x, \quad \text{and}$$
$$\bar{\Delta}^{(i,2)} = 2\int_{\mathbb{R}} (\bar{g}_n^{\frac{1}{2}}(x) - \tilde{g}_n^{\frac{1}{2}}(x)) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i}(x) \cdot \mathbf{1}_{(x\notin A_n)} \mathrm{d}x.$$

Using the equation $a^{\frac{1}{2}} - b^{\frac{1}{2}} = \frac{a-b}{2b^{1/2}} - \frac{(a^{1/2}-b^{1/2})^2}{2b^{1/2}}$, and denoting $R_{\boldsymbol{\theta}}(x) = |f_{\boldsymbol{\theta}}^{1/2}(x) u_{\boldsymbol{\theta},i}(x)|$, we obtain

$$|\bar{\Delta}^{(i,1)}| \leq 2(T_1 + T_2), \quad \text{where}$$
$$T_1 = \int_{\mathbb{R}} \left| \frac{\bar{g}_n(x) - \tilde{g}_n(x)}{2\sqrt{\tilde{g}_n(x)}} \right| \cdot R_{\boldsymbol{\theta}}(x) \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x \text{ and}$$
$$T_2 = \int_{\mathbb{R}} \left| \frac{(\sqrt{\bar{g}_n(x)} - \sqrt{\tilde{g}_n(x)})^2}{2\sqrt{\tilde{g}_n(x)}} \right| \cdot R_{\boldsymbol{\theta}}(x) \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x.$$

We first develop the upper bound of T_1 and use the fact that $T_2 \leq T_1$ almost surely to get the final answer. Using Hölder's inequality with $p \in (1,2)$ and integrability of $|R_{\boldsymbol{\theta}}(x)|^q$ in assumption **(U3)** and the boundedness of the kernel function $K(\cdot)$, it follows that

$$\begin{aligned}
T_1 \leq & C_1 \cdot \left[\int_{\mathbb{R}} \left| \frac{\bar{g}_n(x) - \tilde{g}_n(x)}{2\sqrt{\tilde{g}_n(x)}} \right|^p \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x \right]^{\frac{1}{p}} \\
\leq & C_1 \cdot \left[\sup_{x\in A_n} \{|\bar{g}_n(x) - \tilde{g}_n(x)|\} \right]^{\frac{1}{p}} \cdot \left[\int_{\mathbb{R}} \frac{|\bar{g}_n(x) - \tilde{g}_n(x)|^{p-1}}{(2\sqrt{\tilde{g}_n(x)})^p} \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x \right]^{\frac{1}{p}} \\
\leq & C_1 \cdot C_2 \left(\frac{1}{nc_n} \right)^{\frac{1}{p}} \cdot \left[\int_{\mathbb{R}} \frac{|\bar{g}_n(x) - \tilde{g}_n(x)|^{p-1}}{(2\sqrt{\tilde{g}_n(x)})^p} \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x \right]^{\frac{1}{p}},
\end{aligned}$$

where $0 < C_1 < \infty$ is a constant (independent of $\boldsymbol{\theta}$) obtained from assumption **(U3)**. By the one-record replacement bound,

$$\inf_{x\in A_n} \tilde{g}_n(x) \geq \delta_n - \frac{2\|K\|_\infty}{nc_n}.$$

The condition in **(U3)** implies

$$\frac{2\|K\|_\infty}{nc_n\delta_n} \leq 2\|K\|_\infty C_\delta^2 (nc_n)^{1-2/p} \longrightarrow 0,$$

so that, for all sufficiently large n, $\inf_{x\in A_n} \tilde{g}_n(x) \geq \delta_n/2$. Hence

$$\int_{\mathbb{R}} \frac{|\bar{g}_n(x) - \tilde{g}_n(x)|^{p-1}}{(2\sqrt{\tilde{g}_n(x)})^p} \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x \leq \int_{\mathbb{R}} \frac{|\bar{g}_n(x) - \tilde{g}_n(x)|^{p-1}}{(2\sqrt{\delta_n/2})^p} \cdot \mathbf{1}_{(x\in A_n)} \mathrm{d}x.$$

The RHS of the above equation is bounded above by a constant multiple of

$$\frac{1}{\delta_n^{p/2}(nc_n)^{p-1}} \left[\int_{\mathbb{R}} \left| K\left(\frac{x - X_1'}{c_n} \right) - K\left(\frac{x - X_1}{c_n} \right) \right|^{p-1} \mathrm{d}x \right].$$

Next, we establish the upper bound of the term above. By assumption **(A2)**, $K(\cdot)$ has compact support contained in $[-\beta, \beta]$. Fix any $X_1 = x_1$, $X_1' = x_1'$ and $x_1, x_1' \in B_n$, and let

$$S = [x_1 - \beta c_n, x_1 + \beta c_n] \cup [x_1' - \beta c_n, x_1' + \beta c_n].$$

This set contains the supports of the two kernel terms and $\lambda(S) \le 4\beta c_n$. Since $h(x) = x^{p-1}$ is concave for $p \in (1,2)$, Jensen's inequality gives

$$\int_{\mathbb{R}} \left| K\left(\frac{x - x_1'}{c_n}\right) - K\left(\frac{x - x_1}{c_n}\right) \right|^{p-1} \mathrm{d}x \le [\lambda(S)]^{2-p} \left[\int_S \left| K\left(\frac{x - x_1'}{c_n}\right) - K\left(\frac{x - x_1}{c_n}\right) \right| \mathrm{d}x \right]^{p-1} \le C_3 c_n.$$

Equivalently, the intermediate expectation in Jensen's inequality is taken with respect to the uniform distribution on S. Thus, for a constant $C_4 < \infty$,

$$\int_{\mathbb{R}} \left[\frac{|\bar{g}_n(x) - \tilde{g}_n(x)|^{p-1}}{(2\sqrt{\tilde{g}_n(x)})^p} \cdot \mathbf{1}_{(x \in A_n)} \right] \mathrm{d}x \le C_4 \frac{c_n^{2-p}}{\delta_n^{p/2} n^{p-1}} \le C_4 C_\delta^p c_n \longrightarrow 0.$$

Consequently,

$$T_1 \le C_1 C_2 \left(\frac{1}{n c_n}\right)^{1/p} (C_4 C_\delta^p c_n)^{1/p} \le C_5 n^{-1/p}$$

for some $C_5 \in (0, \infty)$. Next, using the same restriction to A_n,

$$T_2 \le \int_{A_n} \frac{|\sqrt{\bar{g}_n(x)} - \sqrt{\tilde{g}_n(x)}|\,|\sqrt{\bar{g}_n(x)} + \sqrt{\tilde{g}_n(x)}|}{2\sqrt{\tilde{g}_n(x)}} R_{\boldsymbol{\theta}}(x)\,\mathrm{d}x = T_1.$$

Therefore $|\bar{\Delta}^{(i,1)}| \le 4C_5 n^{-1/p}$.

We turn to $\bar{\Delta}^{(i,2)}$. The two truncated KDEs vanish outside S_n, so only $S_n \setminus A_n$ contributes. By Cauchy–Schwarz and the inequality $(\sqrt{a} - \sqrt{b})^2 \le |a - b|$,

$$\begin{aligned} |\bar{\Delta}^{(i,2)}|^2 &\le 4 \left[\int_{S_n \setminus A_n} (\sqrt{\bar{g}_n(x)} - \sqrt{\tilde{g}_n(x)})^2 \,\mathrm{d}x \right] \left[\int_{S_n \setminus A_n} f_{\boldsymbol{\theta}}(x) u_{\boldsymbol{\theta},i}(x)^2 \,\mathrm{d}x \right] \\ &\le \frac{8}{n} \int_{S_n \setminus A_n} f_{\boldsymbol{\theta}}(x) u_{\boldsymbol{\theta},i}(x)^2 \,\mathrm{d}x, \end{aligned}$$

where $\|\bar{g}_n - \tilde{g}_n\|_1 \le 2/n$ for every neighboring pair. For the region constructed above, $S_n \setminus A_n \subset \{|x| > a_n\}$. Consequently, by the definition of $V_{\boldsymbol{\theta}}$ and the choice of a_n,

$$\sup_{\boldsymbol{\theta} \in \Theta} \max_i \int_{S_n \setminus A_n} f_{\boldsymbol{\theta}}(x) u_{\boldsymbol{\theta},i}(x)^2 \,\mathrm{d}x \le T(a_n) \le n^{1-2/p}.$$

The same bound applies to every Hessian exterior weight $f_{\boldsymbol{\theta}}(u_{\boldsymbol{\theta},i} u_{\boldsymbol{\theta},j} + 2u_{\boldsymbol{\theta},i,j})^2$, since it is bounded by $V_{\boldsymbol{\theta}}$. It follows that $|\bar{\Delta}^{(i,2)}| \le \sqrt{8C_T}\, n^{-1/p}$. Combining the two parts, for $i = 1, \ldots, m$,

$$|\bar{\Delta}^{(i)}| \le C_6 n^{-1/p}.$$

Setting $\bar{\boldsymbol{\Delta}} = [\bar{\Delta}^{(1)}, \ldots, \bar{\Delta}^{(m)}]^T$, we therefore have, for the neighboring pair under consideration,

$$\|\bar{\boldsymbol{\Delta}}\|_1 \le m C_6 n^{-1/p}, \qquad \|\bar{\boldsymbol{\Delta}}\|_2 \le \sqrt{m}\, C_6 n^{-1/p}.$$

The same argument gives the Hessian bound, completing the proof of Theorem 35. ∎

A random-data extension. The construction is deterministic, but **(U3)** is conditional on the sample. Suppose $K(u) \geq \kappa > 0$ on $|u| \leq \rho$ and cover A_n by M_n intervals $I_{n,j} \subset B_n$ of length ρc_n with

$$\min_{j \leq M_n} \mathbf{P}_g\{X \in I_{n,j}\} \geq c_0 c_n (nc_n)^{-2(1-1/p)}.$$

Set $v_n = (nc_n)^{2/p-1} \to \infty$. Each interval has expected count at least $c_0 v_n$. Chernoff's inequality and a union bound imply that, with probability at least $1 - M_n \exp(-c_0 v_n/8)$, all counts are at least $c_0 v_n/2$. A one-record replacement leaves at least $c_0 v_n/4$ in each interval for large n. Since each $x \in A_n$ belongs to a covering interval, $K \geq \kappa$ yields, simultaneously for the sample and all its one-record replacements,

$$\inf_{x \in A_n} \bar{g}_n(x) \ \wedge \ \inf_{x \in A_n} \tilde{g}_n(x) \ \geq \ \frac{\kappa c_0}{4} \frac{v_n}{nc_n} = \frac{\kappa c_0}{4} (nc_n)^{-2(1-1/p)}.$$

Thus the interior rate requirement in **(U3)** holds on an event $\mathcal{E}_n$ with

$$\mathbf{P}_g(\mathcal{E}_n^c) \leq M_n \exp\{-c_0 (nc_n)^{2/p-1}/8\}.$$

If $\log M_n = o((nc_n)^{2/p-1})$, then $\mathbf{P}_g(\mathcal{E}_n) \to 1$. The $n^{-1/p}$ bound thus holds conditionally and uniformly over the observed dataset's one-record neighbors, as in random differential privacy (Hall et al., 2013), not for every unbounded dataset. The local-mass and covering conditions require verification for the chosen family and sequences.

6.19 Proof of Proposition 34

For PGD, recalling that the mechanism $M_k(w, D) = w - \eta\,(\nabla L_n(w) + \Delta_n \cdot c_{\epsilon'} Z_k)$. M_k satisfies $\epsilon'-$HDP by Proposition 25 and the post processing property. Now starting with the initial estimate $w = \hat{\boldsymbol{\theta}}_n^{(0)}$, we obtain $\hat{\boldsymbol{\theta}}_n^{(k)}$, for $k \geq 1$ using the iteration

$$\hat{\boldsymbol{\theta}}_n^{(k)} = M_k(\hat{\boldsymbol{\theta}}_n^{(k-1)}, D) = \hat{\boldsymbol{\theta}}_n^{(k-1)} - \eta \left(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k-1)}) + \Delta_n \cdot c_{\epsilon'} Z_k \right).$$

Hence, by Corollary 11, $\hat{\boldsymbol{\theta}}_n^{(K)}$ satisfies $h_K(\epsilon')-$HDP. Finally, by the choice of ϵ' satisfying $h_K(\epsilon') = \epsilon$, it follows that $\hat{\boldsymbol{\theta}}_n^{(K)}$ satisfies $\epsilon-$HDP. Next, turning to PNR, the mechanism is $M_k(w, D) = w - (H_n(w) + W_{n,k})^{-1} (\nabla L_n(w) + N_{n,k})$, where $W_{n,k} \in \mathbb{R}^{m \times m}$ and $N_{n,k} \in \mathbb{R}^{m \times 1}$ are the independent random variables added to satisfy the HDP property and

$$N_{n,k} = \Delta_n \cdot c_{\epsilon'/2} \cdot Z_k, \quad W_{n,k} = \Delta_n^{(H)} \cdot c_{\epsilon'/2} \cdot \tilde{Z}_k.$$

Let $M_{k,1}(w, D) = (H_n(w) + W_{n,k})^{-1}$, $M_{k,2}(w, D) = (\nabla L_n(w) + N_{n,k})$. Then $M_{k,1}$ and $M_{k,2}$ satisfies $\frac{\epsilon'}{2}-$HDP by Proposition 25, Proposition 26, and the post processing property. Hence, by Corollary 11, it follows that $M_k(w, D) = w - M_{k,1}(w, D) \cdot M_{k,2}(w, D)$ satisfies $\epsilon'-$HDP. Finally, starting with the initial estimate $w = \hat{\boldsymbol{\theta}}_n^{(0)}$, we obtain $\hat{\boldsymbol{\theta}}_n^{(k)}$ for $k \geq 1$ by iterating

$$\hat{\boldsymbol{\theta}}_n^{(k)} = M_k(\hat{\boldsymbol{\theta}}_n^{(k-1)}, D) = \hat{\boldsymbol{\theta}}_n^{(k-1)} - M_{k,1}(\hat{\boldsymbol{\theta}}_n^{(k-1)}, D) \cdot M_{k,2}(\hat{\boldsymbol{\theta}}_n^{(k-1)}, D)$$

Hence, by Corollary 11, $\hat{\boldsymbol{\theta}}_n^{(K)}$ satisfies $h_K(\epsilon')-$HDP. Also, by the choice of ϵ' satisfying $h_K(\epsilon') = \epsilon$, it follows that $\hat{\boldsymbol{\theta}}_n^{(K)}$ satisfies $\epsilon-$HDP. Finally, to obtain the bounds for $h_K(\epsilon K^{-1})$,

first notice that $h_2(\epsilon K^{-1}) = h_1(\epsilon K^{-1}) + \epsilon' - \frac{1}{2}h_1(\epsilon K^{-1})\epsilon' \leq h_1(\epsilon K^{-1}) + \epsilon' = 2\epsilon K^{-1}$. Iterating, it follows that $h_K(\epsilon K^{-1}) \leq \epsilon$. Now, turning to the lower bound, by iterating Corollary 11 we obtain

$$h_K(\epsilon K^{-1}) = \epsilon' + \epsilon' \left[(K-1) - \frac{1}{2}\sum_{j=1}^{K-1} h_j(\epsilon K^{-1}) \right] = K\epsilon' - \epsilon' \frac{1}{2}\sum_{j=1}^{K-1} h_j(\epsilon K^{-1}).$$

Next, using the upper bound, $h_j(\epsilon K^{-1}) \leq \epsilon \cdot jK^{-1}$, we obtain

$$h_K(\epsilon K^{-1}) \geq \epsilon[1 - \epsilon(K-1)(4K)^{-1}]. \quad \blacksquare$$

6.20 Proof of Theorem 36

The proof of the theorem relies on the behavior of the Hellinger loss function at *private* estimates. Intuitively, we show that under ASLSC and τ_2-smoothness, the closeness of the loss functions implies the closeness of the parameter estimates and vice-versa. This is achieved via Lemma 6.1-Lemma 6.3. We recall that N is defined in Proposition 28. In this proof, for the ease of exposition, we set τ_1 and τ_2 to be $2\tau_1$ and $2\tau_2$.

Lemma 6.1 *Assume that assumptions* **(A1)-(A8)** *in Section 7 and* **(U1)-(U2)** *hold and that* $||\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_g||_2 \leq \frac{1}{2}r$. *Also, assume that for all* $k = 1, \cdots, K$, *and* $n \geq N$, $||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq \frac{1}{2}r$ *with probability* $1 - \frac{k\xi}{K}$ *and* $||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq r$ *with probability* $1 - \frac{(k+1)\xi}{K}$, *where* r *is as defined in Proposition 28. Let* $N_{n,k} = \Delta_n c_{\epsilon'} Z_k$, *where* $Z_k \sim N(\mathbf{0}, \mathbf{I})$. *Then, with probability* $1 - \frac{k\xi}{K}$,

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq (1-\gamma)^k (L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \frac{3r \cdot ||N_{n,k}||_2}{2\gamma}, \tag{30}$$

where η *and* γ *are chosen such that* $0 < \gamma \leq 2\eta\tau_1 \leq 2\eta\tau_2 < 1$.

Proof: Recall that

$$Q_k(\boldsymbol{\theta}) = L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_k, \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle + \frac{1}{2\eta}||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2.$$

Since $\hat{\boldsymbol{\theta}}_n^{(k+1)}$ minimizes $Q_k(\boldsymbol{\theta})$, it follows that by setting $\boldsymbol{\theta}_\gamma = \gamma\hat{\boldsymbol{\theta}}_n + (1-\gamma)\hat{\boldsymbol{\theta}}_n^{(k)}$, that

$$\begin{aligned} Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)}) \leq Q_k(\boldsymbol{\theta}_\gamma) =& L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \gamma\langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}), \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle \\ &+ \frac{\gamma^2}{2\eta}||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 + \gamma\langle N_{n,k}, \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle. \end{aligned} \tag{31}$$

Now using Proposition 29 part (1) we obtain

$$\langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}), \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle \leq L_n(\hat{\boldsymbol{\theta}}_n) - L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - \tau_1||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2.$$

Now, using this bound in the inequality (31), we obtain

$$\begin{aligned} Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)}) \leq& L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - \gamma[L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)] + \left(\frac{\gamma^2}{2\eta} - \gamma\tau_1\right)||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 \\ &+ \gamma\langle N_{n,k}, \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle, \end{aligned} \tag{32}$$

yielding the upper bound of $Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)})$. We next obtain a lower bound for $Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)})$. To this end, we use part (3) of Proposition 29. Specifically, using $L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \geq L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - \langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}), \hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle - \tau_2 ||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n^{(k+1)}||_2^2$, we obtain that

$$Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)}) \geq L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) + (\frac{1}{2\eta} - \tau_2)||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 + \langle N_{n,k}, \hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle.$$

Now since $2\eta\tau_2 \leq 1$, it follows that

$$Q_k(\hat{\boldsymbol{\theta}}_n^{(k+1)}) \geq L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) + \langle N_{n,k}, \hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle. \tag{33}$$

Now using (32) and (33) it follows that

$$\begin{aligned} L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq & (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \left(\frac{\gamma^2}{2\eta} - \gamma\tau_1\right) ||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 \\ & + \gamma \langle N_{n,k}, \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle - \langle N_{n,k}, \hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle. \end{aligned}$$

Now choosing γ so that $0 < \gamma \leq 2\eta\tau_1 < 1$ and applying Cauchy-Schwarz inequality, it follows that

$$\begin{aligned} L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq & (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + ||N_{n,k}||_2 (||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \\ & + ||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2) \\ \leq & (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \frac{3r||N_{n,k}||_2}{2}, \end{aligned} \tag{34}$$

where the last inequality follows from the assumptions $||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq \frac{1}{2}r$ and $||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq r$. Now iterating the above inequality, it follows that

$$\begin{aligned} L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq & (1-\gamma)^k (L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \frac{3r \cdot ||N_{n,k}||_2}{2} \cdot \frac{(1-(1-\gamma)^k)}{\gamma} \\ \leq & (1-\gamma)^k (L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \frac{3r \cdot ||N_{n,k}||_2}{2\gamma}. \quad \blacksquare \end{aligned}$$

Our next key result is Lemma 6.3 below, which verifies that under the assumptions of Lemma 6.3 the private and non-private estimators are close for large n and for every iteration $k = 0, 1, \cdots K$. The proof of this lemma relies on the notion that, under the assumptions in Section 7 and **(U1)**-**(U3)**, if the loss functions are "close", then arguments of the loss functions are also "close". This is the content of our next lemma and the proof is based on almost sure local strong convexity and is provided in Section 10.

Lemma 6.2 *Let assumptions* **(A1)**-**(A8)** *in Section 7 and* **(U1)**-**(U2)** *hold. Then for* $\boldsymbol{\theta} \in B_r(\boldsymbol{\theta}_g)$ *and* $n \geq N$, *if* $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \frac{r^2}{4}\tau_1$ *then* $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq \frac{r}{2}$. *Furthermore, if* $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq \frac{r}{2}$ *for* $\boldsymbol{\theta} \in B_r(\boldsymbol{\theta}_g)$, *then for* $n \geq N$, $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \frac{r^2}{4}\tau_2$.

We next turn to the key result verifying the validity of the conditions in Lemma 6.1 above.

Lemma 6.3 *Under assumptions* **(A1)-(A8)** *and* **(U1)-(U2)**, *for* $\eta \leq \frac{1}{\tau_2}$, *assume that for* $n \geq N$, $\hat{\boldsymbol{\theta}}_n \in B_{r/c}(\boldsymbol{\theta}_g) \subset B_{r/2}(\boldsymbol{\theta}_g)$, *where* $c > 2\left(\frac{\tau_2}{\tau_1}\right)^{\frac{1}{2}}$, *then there exists* $\hat{\boldsymbol{\theta}}_n^{(0)}$, *such that* $L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ *and* $||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$ *hold with probability* $1 - \frac{k\xi}{K}$ *for all* $k = 0, \cdots, K$.

The proof of this lemma is similar to the proof of Lemma 18 in Avella-Medina et al. (2023). A mildly different proof is given in Section 10.

We now turn to the proof of Theorem 36.

Proof of Theorem 36 : Using Proposition 34 with K replaced by K_n, it follows that $\hat{\boldsymbol{\theta}}_n^{(K_n)}$ satisfies $\epsilon-$HDP. We next turn to verification of the utility. The key idea is to use Proposition 29 (1) and Lemma 6.1 and iterate until the required bound is reached. Towards this, using $\nabla L_n(\hat{\boldsymbol{\theta}}_n) = (0, \cdots, 0)$ and taking $\gamma \in (0, 2\eta\tau_1)$, it follows that

$$||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq \frac{(1-\gamma)^k (L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n))}{\tau_1} + \frac{3r \cdot ||N_{n,k}||_2}{2\gamma\tau_1}.$$

Using concentration inequality for L_2-norm of the Gaussian vector (see Rigollet and Hütter (2023)), namely,

$$P\left(||Z_k||_2 \geq 4\sqrt{m} + 2\sqrt{2[\log K - \log \xi]}\right) \leq \frac{\xi}{K},$$

it follows by setting $\epsilon' = \frac{\epsilon}{K}$ that

$$P(||N_{n,k}||_2 \leq \Delta_n c_{\epsilon'} \left(4\sqrt{m} + 2\sqrt{2[\log K - \log \xi]}\right) \coloneqq r_{noi}) > 1 - \frac{\xi}{K}. \tag{35}$$

We emphasize here that r_{noi} depends on n, K_n and ξ. Now, first consider the case $L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) \neq L_n(\hat{\boldsymbol{\theta}}_n)$. By choosing $k > k_0 \coloneqq \max\{0, \frac{-\log(L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \log(3r \cdot r_{noi}) - \log(2\gamma)}{\log(1-\gamma)}\}$, it follows that with probability $(1 - \frac{k\xi}{K})$, and for all $k \geq k_0$

$$||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq \frac{3r \cdot r_{noi}}{\gamma\tau_1} \coloneqq C_0^2 r_{noi}. \tag{36}$$

We notice here that this bound is of order $\Delta_n^{\frac{1}{2}} (K_n \log K_n)^{\frac{1}{4}}$. However, this will not yield efficiency. Our goal is to remove the square root from Δ_n. This suggests one needs larger k in the above bound. This is accomplished by additional iterations (see Theorem 2 in Avella-Medina et al. (2023)). To this end, we need the following claim, whose proof is given below.

Claim: For $k > k_0$, choose n, K such that $r_{noi}^{\frac{1}{2}} < \frac{1}{2\eta} C_0$. Then

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + (2\eta\tau_2 + \frac{3}{2}) C_0 r_{noi}^{\frac{3}{2}}.$$

Using the claim with $k = k_0 + j - 1$ and iterating we get

$$\begin{aligned} L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+j)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq & (1-\gamma)^j (L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \left[\sum_{i=0}^{j} (1-\gamma)^i\right] (2\eta\tau_2 + \frac{3}{2}) C_0 r_{noi}^{\frac{3}{2}} \\ \leq & (1-\gamma)^j (L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \frac{1}{\gamma}(2\eta\tau_2 + \frac{3}{2}) C_0 r_{noi}^{\frac{3}{2}}. \end{aligned}$$

Now, using Proposition 29 (1) and utilizing $\nabla L_n(\hat{\boldsymbol{\theta}}_n) = (0, \cdots, 0)$, it follows that $||\hat{\boldsymbol{\theta}}_n^{(k_0+j)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq \frac{L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+j)}) - L_n(\hat{\boldsymbol{\theta}}_n)}{\tau_1}$ and hence

$$||\hat{\boldsymbol{\theta}}_n^{(k_0+j)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq \frac{(1-\gamma)^j (L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)}) - L_n(\hat{\boldsymbol{\theta}}_n))}{\tau_1} + (2\eta\tau_2 + \frac{3}{2})\frac{C_0}{\gamma\tau_1} r_{noi}^{\frac{3}{2}}.$$

Next, we choose

$$j \geq k_1 \coloneqq \max\left\{0, \frac{-\log(L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + \log\left(\frac{1}{2}\frac{C_0}{\gamma\tau_1} r_{noi}^{\frac{3}{2}}\right)}{\log(1-\gamma)}\right\},$$

and setting $C_1 = (\frac{2\eta\tau_2+2}{\gamma\tau_1} C_0)^{\frac{1}{2}}$, we obtain for $k > k_0 + k_1$,

$$||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq C_1^2 r_{noi}^{\frac{3}{2}}.$$

We notice that the power of r_{noi} is now $3/2$ and is still below the required power of 2. Hence, continuing the iterations and using the Claim with starting value $k_0 + k_1 + \cdots k_i$, we obtain for $k \geq k_0 + k_1 + \cdots k_i$,

$$||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}||_2^2 \leq C_i^2 r_{noi}^{2-\frac{1}{2^i}} \quad \text{if} \quad r_{noi}^{\frac{1}{2^i}} \leq \frac{C_{i-1}}{2\eta},$$

where $C_i = \left(\frac{2\eta\tau_2+2}{\gamma\tau_1} C_{i-1}\right)^{\frac{1}{2}} = \left(\frac{2\eta\tau_2+2}{\gamma\tau_1}\right)^{1-\frac{1}{2^i}} \cdot C_0^{\frac{1}{2^i}}$. Finally, taking $i = \log_2(n)$ we get $k > k_0 + \cdots + k_{\log_2(n)}$

$$||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2^2 \leq C_{\log_2(n)}^2 r_{noi}^{2-\frac{1}{n}}, \quad \text{if} \quad r_{noi}^{\frac{1}{n}} \leq \frac{C_{\log_2(n)-1}}{2\eta},$$

where $C_{\log_2(n)} = \left(\frac{2\eta\tau_2+2}{\gamma\tau_1}\right)^{1-\frac{1}{n}} \cdot C_0^{\frac{1}{n}}$. Now, letting $n \to \infty$, notice that $C_{\log_2(n)}$ converges to $C_\infty(\gamma) \coloneqq 2(\eta\tau_2 + 1)(\gamma\tau_1)^{-1}$. Also, notice that $r_{noi}^{\frac{1}{n}}$ converges to 1. Now choosing $\gamma \in (0, 2\eta\tau_1)$ and $C_\infty > 1$ (such a γ exists) it follows that

$$\limsup_{n\to\infty} r_{noi}^{-2} ||\hat{\boldsymbol{\theta}}_n^{(k_n)} - \hat{\boldsymbol{\theta}}_n||_2 = C_\infty. \tag{37}$$

This requires $K \geq k_0 + k_1 + \cdots + k_{\log_2(n)} \sim (\log n) \cdot (\log r_{noi})$ which implies $K \geq c \log n$, since r_{noi} is bounded by a constant by choice of n and K. Next, we notice that $r_{noi} = \Delta_n c_{\epsilon'} \left(4\sqrt{m} + 2\sqrt{2[\log K - \log \xi]}\right)$ by Theorem 35 and $\Delta_n \sim c \cdot n^{-\frac{1}{p}}$ for $p \in (1, 2]$. Hence, (37) becomes

$$||\hat{\boldsymbol{\theta}}_n^{(K_n)} - \hat{\boldsymbol{\theta}}_n||_2 \leq c \cdot n^{-\frac{1}{p}} (K_n \log(K_n/\xi))^{\frac{1}{2}},$$

for large n with high probability. Thus, to complete the proof of the Theorem, we now establish the claim.

Proof of the Claim: Notice that by Proposition 29 inequality 4 that

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 = ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - \nabla L_n(\hat{\boldsymbol{\theta}}_n)||_2 \leq 2\tau_2 ||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2. \tag{38}$$

Now, first using equation (8) in Definition 32 of the main paper and the expression above and applying (36) it follows that

$$||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq 2\eta\tau_2 C_0 r_{noi}^{\frac{1}{2}} + \eta r_{noi}.$$

From the inequality (34) in the proof of Lemma 6.1, using (36) and from (38) it follows that

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + (2\eta\tau_2 + 1)C_0 r_{noi}^{\frac{3}{2}} + \eta r_{noi}^2.$$

Next, choosing n, K such that $r_{noi}^{\frac{1}{2}} < \frac{1}{2\eta} C_0$ it follows that

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq (1-\gamma)(L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n)) + (2\eta\tau_2 + \frac{3}{2})C_0 r_{noi}^{\frac{3}{2}}.$$

This completes the proof of the claim and the Theorem. ∎

6.21 Proof of Theorem 39

The proof of the Theorem relies on the Lemma 6.4-Lemma 6.6 whose proofs use matrix concentration inequality and is similar to the idea of proof of Theorem 36. We recall that the concentration inequality for L_2-norm of the Gaussian vector and matrix, (see Rigollet and Hütter (2023); Tropp (2015)), is given by

$$P\left(||N_{n,k}||_2 \leq \Delta_n \cdot c_{\epsilon'/2} \cdot [4\sqrt{m} + 2\sqrt{2(\log 2K - \log \xi)}]\right) \geq 1 - \frac{\xi}{2K}, \text{ and}$$
$$P\left(||W_{n,k}||_2 \leq \Delta_n^{(H)} \cdot c_{\epsilon'/2} \cdot \sqrt{2m \log(4Km/\xi)}\right) \geq 1 - \frac{\xi}{2K}.$$

We use these upper bounds on the norms with probability $1 - \frac{\xi}{K}$ in the following lemmas and proofs. In this proof, for the ease of exposition, we set τ_1 and τ_2 to be $2\tau_1$ and $2\tau_2$, and choose $\eta = 1$. Our first lemma provides a useful alternative expression for $\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}$.

Lemma 6.4

$$\left(H_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + W_{n,k}\right)^{-1}\left(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_{n,k}\right) = H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)})\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \tilde{N}_{n,k}.$$

Under assumptions **(A1)-(A8)** *of Section 7 and* **(U1)-(U2)**, *if* $\hat{\boldsymbol{\theta}}_n^{(k)} \in B_r(\boldsymbol{\theta}_g)$, *then for large* n, $||\tilde{N}_{n,k}||_2 \leq \frac{\|N_{n,k}\|_2}{2\tau_1} + \frac{B_1 \cdot ||W_{n,k}||_2}{2\tau_1^2} + \frac{||N_{n,k}||_2 \cdot ||W_{n,k}||_2}{2\tau_1^2}$ *holds with probability* $1 - \frac{\xi}{2K}$. *Additionally,* $\kappa \sim n^{-\frac{1}{p}}(K\log(K/\xi))^{\frac{1}{2}}$, *there exists* N_κ *such that for all* $n > N_\kappa$ *and* $k = 1, 2, \cdots K$ $P(||\tilde{N}_{n,k}||_2 \leq \kappa) > 1 - \frac{\xi}{K}$.

Proof: Using Neumann series formula, note that

$$\begin{aligned}
&\left(H_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + W_{n,k}\right)^{-1}\left(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_{n,k}\right) \\
=&H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)})\left[\mathbf{I} + \sum_{j=1}^{\infty}(-W_{n,k}H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)}))^j\right]\left(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_{n,k}\right) \\
:=&H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)})\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \tilde{N}_{n,k},
\end{aligned}$$

where

$$\tilde{N}_{n,k} = H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)}) \left\{ N_{n,k} + \left[\sum_{j=1}^{\infty} (-W_{n,k} H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)}))^j \right] \left(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_{n,k} \right) \right\}.$$

Now, applying the properties of matrix norms, Proposition 28, and Proposition 25, we obtain

$$||\tilde{N}_{n,k}||_2 \leq \frac{1}{2\tau_1} \cdot \left[\|N_{n,k}\|_2 + \left[\sum_{j=1}^{\infty} \left(\frac{||W_{n,k}||_2}{2\tau_1} \right)^j \right] (B_1 + ||N_{n,k}||_2) \right].$$

Let n be large enough such that $||W_{n,k}||_2 \leq \tau_1$ with probability $1 - \frac{\xi}{2K}$. Then it follows that

$$||\tilde{N}_{n,k}||_2 \leq \frac{\|N_{n,k}\|_2}{2\tau_1} + \frac{B_1 \cdot ||W_{n,k}||_2}{2\tau_1^2} + \frac{||N_{n,k}||_2 \cdot ||W_{n,k}||_2}{2\tau_1^2}.$$

Notice that as $n \to \infty$, both $||N_{n,k}||_2$ and $||W_{n,k}||_2$ converge to 0 in probability at rate $n^{-\frac{1}{p}}(K \log(K/\xi))^{\frac{1}{2}}$. ∎

Lemma 6.5 *Under assumptions* **(A1)**-**(A8)** *and* **(U1)**-**(U2)**, *if* $\hat{\boldsymbol{\theta}}_n^{(k)} \in B_r(\boldsymbol{\theta}_g)$, *then*

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)})||_2 \leq \frac{\alpha}{2\tau_1^2} ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + C||\tilde{N}_{n,k}||_2.$$

This bound holds with probability $1 - \frac{\xi}{K}$.

Proof: Recall that from the PNR iteration, namely, $\hat{\boldsymbol{\theta}}_n^{(k+1)} = \hat{\boldsymbol{\theta}}_n^{(k)} - H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)}) \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \tilde{N}_{n,k}$, that $\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + H_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \cdot [\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n^{(k+1)} - \tilde{N}_{n,k}] = 0$. We now rewrite $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)})||_2$ as

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)})||_2 = ||T_1 - T_2 + H_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \tilde{N}_{n,K}||_2, \quad \text{where}$$
$$T_1 = \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \quad \text{and} \quad T_2 = H_n(\hat{\boldsymbol{\theta}}_n^{(k)})(\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}).$$

Notice that $T_1 - T_2$ can be written as

$$\begin{aligned} T_1 - T_2 &= \int_0^1 H_n(\hat{\boldsymbol{\theta}}_n^{(k)} + t(\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)})) \cdot (\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}) \mathrm{d}t - \int_0^1 H_n(\hat{\boldsymbol{\theta}}_n^{(k)})(\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}) \mathrm{d}t \\ &= (\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}) \int_0^1 \left[H_n(\hat{\boldsymbol{\theta}}_n^{(k)} + t(\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)})) - H_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \right] \mathrm{d}t. \end{aligned}$$

Using Proposition 30 (namely the Lipschitz property of the Hessian), it follows that with probability 1,

$$||T_1 - T_2||_2 \leq ||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \cdot \int_0^1 \alpha \cdot t \cdot ||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \mathrm{d}t = \frac{\alpha}{2} ||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2.$$

Using the upper bound of Proposition 25, $\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)} = -H_n^{-1}(\hat{\boldsymbol{\theta}}_n^{(k)})\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \tilde{N}_{n,k}$, Proposition 28, and for large n that $||\tilde{N}_{n,k}||_2 \leq 1$ with probability $1 - \frac{\xi}{K}$, we obtain

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)})||_2 \leq \frac{\alpha}{2}||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 + B_2||\tilde{N}_{n,k}||_2 \leq \frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + C||\tilde{N}_{n,k}||_2,$$

where the constant $C \in (0, \infty)$ only depends on α, τ_1, B_1, B_2. ∎

The next lemma concerns the "distance" between the private and non-private estimators at every iteration, and the proof is based on induction. The choice of $\hat{\boldsymbol{\theta}}_n^{(0)}$, verifies the assumption that the assumptions in Lemma 6.5 hold; that's is, for all $k = 1, 2 \cdots K$, $\hat{\boldsymbol{\theta}}_n^{(k)} \in B_r(\boldsymbol{\theta}_g)$.

Lemma 6.6 *Under assumption* **(A1)-(A8)** *and* **(U1)-(U2)**, *if* $\hat{\boldsymbol{\theta}}_n \in B_{r/2}(\boldsymbol{\theta}_g)$, *and*

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2 \leq \min\{\frac{\tau_1 r}{2}, \frac{\tau_1^2}{\alpha}\},$$

then for $k = 0, 1, \cdots, K$, $||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$ *holds with probability* $1 - \frac{k\xi}{K}$.

Proof: We prove the lemma using the following claim:
Claim: If $\hat{\boldsymbol{\theta}}_n \in B_{r/2}(\boldsymbol{\theta}_g)$ and $||\nabla L_n(\boldsymbol{\theta})||_2 \leq 2\tau_1 r$, then $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n||_2 \leq r$.

First, we finish the proof of the lemma using the Claim and then prove the Claim. We prove the lemma by induction. First notice that by assumption $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2 \leq \min\{\frac{\tau_1 r}{2}, \frac{\tau_1^2}{\alpha}\} \leq \tau_1 r$ and hence from the claim it follows, with $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_n^{(0)}$ and r replaced by $\frac{r}{2}$, that $||\hat{\boldsymbol{\theta}}_n^{(0)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$. We start the inductive hypothesis with $k = k_0$. That is, assume for $k = k_0$, $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)})||_2 \leq \min\{\tau_1 r, \frac{\tau_1^2}{\alpha}\}$ and $||\hat{\boldsymbol{\theta}}_n^{(k_0)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$, and $\hat{\boldsymbol{\theta}}_n^{(k_0)} \in B_r(\boldsymbol{\theta}_g)$. Also from Lemma 6.5, and for large n such that $||\tilde{N}_{n,k}||_2 \leq \min\{\tau_1 r, \frac{\tau_1^2}{\alpha}\}$ with probability $1 - \frac{\xi}{K}$, we obtain

$$\begin{aligned}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+1)})||_2 \leq& \frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)})||_2^2 + C||\tilde{N}_{n,k}||_2 \\ \leq& \frac{\alpha}{2\tau_1^2} \cdot \left(\frac{\tau_1^2}{\alpha}\right)^2 + C||\tilde{N}_{n,k}||_2 \leq \min\{\frac{\tau_1^2}{\alpha}, \tau_1 r\}.\end{aligned}$$

Now, applying the claim with $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_n^{(k_0+1)}$ and replacing r by $\frac{r}{2}$, it follows that $||\hat{\boldsymbol{\theta}}_n^{(k_0+1)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$. This completes the induction. Now, we turn to the proof of the claim.
Proof of the claim: The proof uses the ASLSC property and is similar to the one used in Avella-Medina et al. (2023). Specifically, we establish the proof using contradiction. To this end, suppose $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n||_2 > r$; let $\tilde{\boldsymbol{\theta}}$ denote the point on the boundary of $\mathcal{B}_r(\hat{\boldsymbol{\theta}})$. By Proposition 29,

$$\nabla L_n(\tilde{\boldsymbol{\theta}})^T \cdot (\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) \geq 2\tau_1||\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}||_2^2.$$

Define $\boldsymbol{v} = \frac{\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}}{||\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}||_2}$; then we have

$$\nabla L_n(\tilde{\boldsymbol{\theta}})^T \cdot \boldsymbol{v} \geq 2\tau_1||\tilde{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}||_2 = 2\tau_1 r.$$

Set $f(t) = \nabla L_n(\hat{\boldsymbol{\theta}} + t \cdot v)^T \cdot v$ for $t \geq 0$, then $f'(t) = \boldsymbol{v}^T H_n(\hat{\boldsymbol{\theta}} + t \cdot \boldsymbol{v}) \cdot \boldsymbol{v} \geq 0$, since Hessian matrix is positive definite by Proposition 28. Hence $f(t)$ is increasing in t and this implies that

$$||\nabla L_n(\boldsymbol{\theta})||_2 \geq \nabla L_n(\boldsymbol{\theta})^T \cdot \boldsymbol{v} \geq \nabla L_n(\tilde{\boldsymbol{\theta}})^T \cdot \boldsymbol{v} \geq 2\tau_1 r$$

which is a contradiction since $||\nabla L_n(\boldsymbol{\theta})||_2 \leq 2\tau_1 r$. Therefore, it follows that

$$||\nabla L_n(\boldsymbol{\theta})||_2 \leq 2\tau_1 r \Longrightarrow ||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq r.$$

This completes the proof of the claim and the lemma. ∎

We now turn to the proof of the Theorem.

Proof of Theorem 39 : Using Proposition 34 with K replaced by K_n, it follows that $\hat{\boldsymbol{\theta}}_n^{(K_n)}$ satisfies $\epsilon-$HDP. We next turn to verification of the utility. We assume N is large enough to satisfy the conditions in Lemma 6.4. That is for $n > N$ such that $P(||\tilde{N}_{n,k}||_2 \leq r_{noi}) \geq 1 - \frac{\xi}{K}$ for $r_{noi} \sim n^{-\frac{1}{p}}(K \log(K/\xi))^{\frac{1}{2}}$. We will use Lemma 6.5 and Lemma 6.6 to obtain the following claim:

Claim: For $\hat{\boldsymbol{\theta}}_n \in B_{r/2}(\boldsymbol{\theta}_g)$ and $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2 \leq \min\{\frac{\tau_1 r}{2}, \frac{\tau_1^2}{\alpha}\}$, the inequality

$$\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\theta}_n^{(K)})||_2 \leq \left(\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\theta}_n^{(0)})||_2\right)^{2^K} + 3C \cdot r_{noi}$$

holds for some constant $C \in (0, \infty)$ with probability $1 - \xi$.

Using Proposition 29 (2), and multiplying both side by $\frac{\alpha}{\tau_1}$, we obtain

$$\frac{\alpha}{\tau_1}||\hat{\boldsymbol{\theta}}_n^{(K)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(K)}) - \nabla L_n(\hat{\boldsymbol{\theta}}_n)||_2.$$

Now using the fact that $\nabla L_n(\hat{\boldsymbol{\theta}}_n) = 0$, the claim, we obtain (since $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2 \leq \min\{\frac{\tau_1 r}{2}, \frac{\tau_1^2}{\alpha}\}$)

$$\frac{\alpha}{\tau_1}||\hat{\boldsymbol{\theta}}_n^{(K)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \left(\frac{1}{2}\right)^{2^K} + 3Cr_{noi}.$$

Choose K large such that $\left(\frac{1}{2}\right)^{2^K} \leq Cr_{noi}$, that is $K \geq \frac{1}{\log 2} \log \frac{\log Cr_{noi}}{\log(1/2)}$, then

$$||\hat{\boldsymbol{\theta}}_n^{(K)} - \hat{\boldsymbol{\theta}}_n||_2 \leq 4Cr_{noi}.$$

By Lemma 6.4, $r_{noi} \sim n^{-\frac{1}{p}}(K \log(K/\xi))^{\frac{1}{2}}$. Using the sharp bound of $\Delta_n^{(H)}$ in Theorem 35, we obtain the utility bound. This also implies that $K \geq C' \log\log n$ for some $C' \in (0, \infty)$. We complete the proof by establishing the claim.

Proof the the claim: We prove the claim by induction. Notice that for $k = 1$, the claim is true by Lemma 6.5 and Lemma 6.6. Assume that the claim holds for $k = k_0$. Then for $k = k_0 + 1$, using Lemma 6.5 and the choice of $\hat{\boldsymbol{\theta}}_n^{(0)}$ such that $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2 \leq \min\{\frac{\tau_1 r}{2}, \frac{\tau_1^2}{\alpha}\}$, it follows that

$$\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+1)})||_2 \leq \left(\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0)})||_2\right)^2 + Cr_{noi}.$$

Now by inductive hypothesis, it follows that

$$\begin{aligned}\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+1)})||_2 &\leq \left[\left(\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2\right)^{2^{k_0}} + 3Cr_{noi}\right]^2 + Cr_{noi} \\ &\leq \left(\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2\right)^{2^{k_0+1}} + \frac{3}{2}Cr_{noi} + 9C^2r_{noi}^2 + Cr_{noi}.\end{aligned}$$

Let n be large such that $9C^2r_{noi}^2 \leq \frac{1}{2}Cr_{noi}$. It then follows that

$$\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k_0+1)})||_2 \leq \left(\frac{\alpha}{2\tau_1^2}||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(0)})||_2\right)^{2^{k_0+1}} + 3Cr_{noi}.$$

This completes the induction and the proof of the Claim and the Theorem. ∎

We next turn to the proof of Theorem 42. First, we recall that Q is the distribution associated with the mechanism, representing the noise distribution.

6.22 Proof of Theorem 42

We begin with part (1). Suppose $\boldsymbol{\theta}_n^{(K_n)}$ is obtained using the PGD or PNR algorithm. Then, using Theorem 36 or Theorem 39 with $p \in (1,2)$ it follows that $n^{\frac{1}{2}}||\hat{\boldsymbol{\theta}}_n^{(K_n)} - \hat{\boldsymbol{\theta}}_n||_2$ converges to zero in probability (with respect to the joint distribution of $P_g \times Q$) since $K_n \sim \log n$ for PGD algorithm and $K_n \sim \log\log n$ for the PNR algorithm. Turning to part (2), observe that

$$(\hat{\boldsymbol{\theta}}_n^{(K_n)} - \boldsymbol{\theta}_g) = (\hat{\boldsymbol{\theta}}_n^{(K_n)} - \hat{\boldsymbol{\theta}}_n) + (\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_g). \tag{39}$$

Now, taking the norm, the first term on the RHS of the above equation converges to 0 in probability by part (1), and the second term converges to zero almost surely under the assumptions **(A1)-(A8)** in Section 7. Finally, turning to part (3), by multiplying both sides of (39) by $\sqrt{n}$, the first term converges to zero in $P_g \times Q$ probability by part (1). The second term converges to a normal distribution under the assumptions **(A1)-(A8)** in Section 7 under P_g, by Theorem 7.1 in Section 7. Hence, $n^{\frac{1}{2}}(\hat{\boldsymbol{\theta}}_n^{(K_n)} - \boldsymbol{\theta}_g)$ converges in distribution (under $P_g \times Q$) to a multivariate normal distribution; that is,

$$\lim_{n\to\infty} P_g \times Q\left(n^{\frac{1}{2}}(\hat{\boldsymbol{\theta}}_n^{(K_n)} - \boldsymbol{\theta}_g) \leq \boldsymbol{x}\right) = P(\mathbf{Z} \leq \boldsymbol{x}),$$

where $\mathbf{Z} \sim N(0, \Sigma_g)$.

7 Regularity conditions and asymptotic properties of MHDE

Let $f(x)$ and $g(x)$ be any two probability density functions. The Hellinger distance between $f(x)$ and $g(x)$ is defined as the L_2-norm of the difference between the square root of density functions, that is,

$$HD^2(f,g) = ||f^{\frac{1}{2}}(x) - g^{\frac{1}{2}}(x)||_2^2 = \int \left[f^{\frac{1}{2}}(x) - g^{\frac{1}{2}}(x)\right]^2 \mathrm{d}x.$$

Let $\{X_1, X_2, \cdots, X_n\}$ be i.i.d. real-valued random variables with density $g(\cdot)$, and postulated to belong to a parametric family $\{f_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta \subset \mathbb{R}^m\}$. The minimum Hellinger distance estimator in the population, $\boldsymbol{\theta}_g$, if it exists, is the minimizer of the $||f_{\boldsymbol{\theta}}^{\frac{1}{2}} - g^{\frac{1}{2}}||_2$; that is,

$$\boldsymbol{\theta}_g = \underset{\boldsymbol{\theta}\in\Theta}{\operatorname{argmin}} ||f_{\boldsymbol{\theta}}^{\frac{1}{2}} - g^{\frac{1}{2}}||_2 = \underset{\boldsymbol{\theta}\in\Theta}{\operatorname{argmin}} HD(f_{\boldsymbol{\theta}}, g).$$

When $g(\cdot) = f_{\boldsymbol{\theta}_0}(\cdot)$, $\boldsymbol{\theta}_g = \boldsymbol{\theta}_0$. We also assume that $\boldsymbol{\theta}_g$ and $\boldsymbol{\theta}_0$ belong to the interior of Θ. Beran (1977) and Cheng and Vidyashankar (2006) establish that under the assumption,

(A1) *$\Theta \subset \mathbb{R}^m$ is compact and convex and the family $\{f_{\boldsymbol{\theta}}(\cdot) : \boldsymbol{\theta} \in \Theta\}$ is identifiable; that is, if $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$ then $f_{\boldsymbol{\theta}_1}(\cdot) \neq f_{\boldsymbol{\theta}_2}(\cdot)$ on a set of positive Lebesgue measure.*

that $\boldsymbol{\theta}_g$ exists and is unique. We will assume this condition holds. In practice, one replaces $g(\cdot)$ by $g_n(\cdot)$, where $g_n(\cdot)$ is a nonparametric estimate of of $g(\cdot)$; specifically, a kernel density estimator, defined below.

$$g_n(x) = \frac{1}{n \cdot c_n} \sum_{i=1}^{n} K\left(\frac{x - X_i}{c_n}\right).$$

The MHDE is obtained by minimizing the loss function

$$\hat{\boldsymbol{\theta}}_n = \arg\min_{\boldsymbol{\theta}\in\Theta} L_n(\boldsymbol{\theta}), \quad \text{where } L_n(\boldsymbol{\theta}) = \int_{\mathbb{R}} (\sqrt{f_{\boldsymbol{\theta}}(x)} - \sqrt{g_n(x)})^2 \mathrm{d}x.$$

Asymptotic properties of $\hat{\boldsymbol{\theta}}_n$ rely on the bandwidth c_n and additional regularity assumptions on the parametric family. We provide the assumptions below:

(A2) *The kernel function $K(\cdot)$ is symmetric (about 0) density with compact support. The bandwidth c_n satisfies $c_n \to 0$, $n^{\frac{1}{2}} c_n^2 \to 0$, $n^{\frac{1}{2}} c_n \to \infty$.*

(A3) *$f_{\boldsymbol{\theta}}(x)$ is twice continuously differentiable in $\boldsymbol{\theta}$. Also, the Fisher information matrix $I(\boldsymbol{\theta})$ is positive definite and continuous in $\boldsymbol{\theta}$ with finite maximum eigenvalue.*

(A4) *$||\mathbf{u}_{\boldsymbol{\theta}}(\cdot) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(\cdot)||_2$, $||\dot{\mathbf{u}}_{\boldsymbol{\theta}}(\cdot) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(\cdot)||_2$, $||\mathbf{u}_{\boldsymbol{\theta}}(\cdot)\mathbf{u}_{\boldsymbol{\theta}}^T(\cdot) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(\cdot)||_2$ exist and are continuous in $\boldsymbol{\theta}$.*

(A5) *Let $\{a_n, n \geq 1\}$ be a sequence diverging to infinity. Assume $\lim_{n\to\infty} n \sup_{t\in supp(K)} \mathbf{P}(|X - c_n t| > a_n) = 0$, where $supp(K)$ is the support of the kernel density $K(\cdot)$ and X is a generic random variable with density $f_{\boldsymbol{\theta}_g}(\cdot)$.*

(A6) *Let* $M(n) = \sup_{|x|\le a_n} \sup_{t\in supp(K)} |f_{\boldsymbol{\theta}_g}^{-1}(x) f_{\boldsymbol{\theta}_g}(x + tc_n)|$. *Assume* $\sup_{n\ge 1} M(n) < \infty$.

(A7) *The score function has a regular central behavior,*

$$\lim_{n\to\infty} (n^{\frac{1}{2}} c_n)^{-1} \int_{-a_n}^{a_n} \mathbf{u}_{\boldsymbol{\theta}_g}(x)\mathrm{d}x = \mathbf{0};\ \textit{also, assume that}\ \lim_{n\to\infty} (n^{\frac{1}{2}} c_n^4) \int_{-a_n}^{a_n} \mathbf{u}_{\boldsymbol{\theta}_g}(x)\mathrm{d}x = \mathbf{0}.$$

(A8) *The score function is smooth in an* L_2 *sense; i.e.*

$$\lim_{n\to\infty} \sup_{t\in supp(K)} \int_{\mathbb{R}} [\mathbf{u}_{\boldsymbol{\theta}_g}(x + tc_n) - \mathbf{u}_{\boldsymbol{\theta}_g}(x)]^2 f_{\boldsymbol{\theta}_g}(x)\mathrm{d}x = \mathbf{0}.$$

It is known that, under the above conditions, $\hat{\boldsymbol{\theta}}_n$ is known to be unique, consistent, and asymptotically efficient (see Beran (1977), Cheng and Vidyashankar (2006)). Write $g(x) = \lim_{n\to\infty} g_n(x)$ (which exists by **(A2)**) and set

$$\rho_{\boldsymbol{\theta}}(x) = -4 \left[\int g^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [\mathbf{u}_{\boldsymbol{\theta}}(x)\mathbf{u}_{\boldsymbol{\theta}}^T(x) + 2\dot{\mathbf{u}}_{\boldsymbol{\theta}}(x)]\mathrm{d}x \right]^{-1} \cdot \nabla f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \quad \text{and}$$
$$\Sigma_g = 4^{-1} \int \rho_{\boldsymbol{\theta}_g}(x)\rho_{\boldsymbol{\theta}_g}^T(x)\mathrm{d}x.$$

The next theorem is concerned with the limit distribution of MHDE and is similar to the proof in Cheng and Vidyashankar (2006) when the true model is $g(\cdot)$.

Theorem 7.1 *Under the assumptions* **(A1)**-**(A8)**, $\rho_{\boldsymbol{\theta}}(\cdot)$ *is continuous at* $\boldsymbol{\theta}_g$. *Furthermore,*

1. $||\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_g||_2 \xrightarrow{P} 0$,
2. $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_g) \xrightarrow{d} N(0, \Sigma_g)$.
3. *In particular, if* $g(\cdot) = f_{\boldsymbol{\theta}_0}(\cdot)$, *then* $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(0, I^{-1}(\boldsymbol{\theta}_0))$.

8 Calculations for additive mechanisms

8.1 Gaussian mechanism

Lemma 8.1 *For two m dimensional random variable $\mathbf{X} \sim N(\mathbf{w}_1, \sigma^2 \cdot \mathbf{I})$ and $\mathbf{Y} \sim N(\mathbf{w}_2, \sigma^2 \cdot \mathbf{I})$, the power divergence with parameter λ is given by*

$$D_\lambda(\mathbf{X}, \mathbf{Y}) = \begin{cases} \frac{1}{\lambda(\lambda+1)} \left[e^{\frac{\lambda(\lambda+1)||\mathbf{v}||_2^2}{2\sigma^2}} - 1 \right], & \lambda(\lambda+1) \neq 0 \\ \frac{||\mathbf{v}||_2^2}{2\sigma^2}, & \lambda(\lambda+1) = 0, \end{cases}$$

where $\mathbf{v} = \mathbf{w}_1 - \mathbf{w}_2$. In particular, if $\lambda = -\frac{1}{2}$ then $D_\lambda(\mathbf{X}, \mathbf{Y}) = -4 \left[e^{-\frac{||\mathbf{v}||_2^2}{8\sigma^2}} - 1 \right]$.

Proof: Denote the density function for $\mathbf{X}$ and $\mathbf{Y}$ by $p(\cdot)$ and $q(\cdot)$ correspondingly, that is

$$p(\mathbf{x}) = \frac{1}{(\sqrt{2\pi}\sigma)^m} e^{-\frac{||\mathbf{x}-\mathbf{w}_1||_2^2}{2\sigma^2}} \quad \text{and} \quad q(\mathbf{x}) = \frac{1}{(\sqrt{2\pi}\sigma)^m} e^{-\frac{||\mathbf{x}-\mathbf{w}_2||_2^2}{2\sigma^2}}.$$

Let $\mathbf{y} = \mathbf{x} - \mathbf{w}_2$, $\mathbf{v} = \mathbf{w}_1 - \mathbf{w}_2$, and denote by y_i, v_i the ith element of $\mathbf{y}$ and $\mathbf{v}$. For the case $\lambda(\lambda+1) \neq 0$, the power divergence with parameter λ between $\mathbf{X}$ and $\mathbf{Y}$ is given by

$$\begin{aligned} D_\lambda(\mathbf{X}, \mathbf{Y}) =& \frac{1}{\lambda(\lambda+1)} \int_{\mathbb{R}^m} \left[\frac{p^{\lambda+1}(\mathbf{x})}{q^{\lambda+1}(\mathbf{x})} \cdot q(\mathbf{x}) - q(\mathbf{x}) \right] \mathrm{d}\mathbf{x} \\ =& \frac{1}{\lambda(\lambda+1)} \left[\int_{\mathbb{R}^m} \frac{1}{(\sqrt{2\pi}\sigma)^m} e^{-\frac{(\lambda+1)||\mathbf{x}-\mathbf{w}_1||_2^2 - \lambda||\mathbf{x}-\mathbf{w}_2||_2^2}{2\sigma^2}} \mathrm{d}\mathbf{x} - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[\int_{\mathbb{R}^m} \frac{1}{(\sqrt{2\pi}\sigma)^m} e^{-\frac{(\lambda+1)||\mathbf{y}-\mathbf{v}||_2^2 - \lambda||\mathbf{y}||_2^2}{2\sigma^2}} \mathrm{d}\mathbf{y} - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\lambda+1)(y_i-v_i)^2 - \lambda y_i^2}{2\sigma^2}} \mathrm{d}y_i \right) - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[\left(\prod_{i=1}^m e^{\frac{\lambda(\lambda+1)v_i^2}{2\sigma^2}} \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y_i-(1+\lambda)v_i)^2}{2\sigma^2}} \mathrm{d}y_i \right) - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[e^{\frac{\lambda(\lambda+1)||\mathbf{v}||_2^2}{2\sigma^2}} - 1 \right]. \end{aligned}$$

Next consider the case $\lambda(\lambda+1)=0$. Denote the ith element of $\mathbf{x}$, $\mathbf{w}_1$, and $\mathbf{w}_2$ by $x_i, w_{1,i}, w_{2,i}$ correspondingly. First we study the case $\lambda=0$.

$$\begin{aligned}
D_0(\mathbf{X},\mathbf{Y}) =& \int_{\mathbb{R}^m} p(\mathbf{x})\log(\frac{p(\mathbf{x})}{q(\mathbf{x})})\mathrm{d}x \\
=& \int_{\mathbb{R}^m} p(\mathbf{x})\frac{-(||\mathbf{x}-\mathbf{w}_1||_2^2-||\mathbf{x}-\mathbf{w}_2||_2^2)}{2\sigma^2}\mathrm{d}\mathbf{x} \\
=& \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-w_{1,i})^2}{2\sigma^2}} \cdot \frac{(w_{1,i}-w_{2,i})(2x-w_{1,i}-w_{2,i})}{2\sigma^2}\mathrm{d}x_i \\
=& \sum_{i=1}^m \left(\frac{w_{1,i}-w_{2,i}}{2\sigma^2}\mathbf{E}_{X\sim N(w_{1,i},\sigma^2)}[2X-w_{1,i}-w_{2,i}] \right) \\
=& \frac{||\mathbf{w}_1-\mathbf{w}_2||_2^2}{2\sigma^2} = \frac{||\mathbf{v}||_2^2}{2\sigma^2}
\end{aligned}$$

The case $\lambda=-1$ is similar and this completes the proof. ∎

8.2 Laplace mechanism

Lemma 8.2 *For two m dimensional random variable $\mathbf{X}$ and $\mathbf{Y}$, where $X_i \sim Lap(w_{1,i}, b)$ and $Y_i \sim Lap(w_{2,i}, b)$, the power divergence between them, with parameter λ is given by*

$$D_\lambda(\mathbf{X},\mathbf{Y}) = \begin{cases} \frac{1}{\lambda(\lambda+1)}\left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i\right)-1\right], & \lambda(\lambda+1)\neq 0 \\ \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|y_j|}{b}} \cdot \frac{|y_i-v_i|-|y_i|}{b}\mathrm{d}y_i, & \lambda(\lambda+1)=0, \end{cases}$$

where $v_i = w_{1,i}-w_{2,i}$. Furthermore

$$D_\lambda(\mathbf{X},\mathbf{Y}) \leq \begin{cases} \frac{1}{\lambda(\lambda+1)}\left[e^{\frac{sign(\lambda)(\lambda+1)||\mathbf{v}||_1}{b}}-1\right], & \lambda(\lambda+1)\neq 0 \\ \frac{||\mathbf{v}||_1}{b}, & \lambda(\lambda+1)=0, \end{cases}$$

In particular, if $\lambda=-\frac{1}{2}$, then

$$\begin{aligned}
D_\lambda(\mathbf{X},\mathbf{Y}) =& -4\left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|y_i-v_i|+|y_i|}{2b}}\mathrm{d}y_i\right)-1\right] \\
\leq& -4\left[e^{\frac{-||\mathbf{v}||_1}{2b}}-1\right].
\end{aligned}$$

Proof: Denote the density function for $\mathbf{X}$ and $\mathbf{Y}$ by $p(\cdot)$ and $q(\cdot)$ correspondingly, that is

$$p(\mathbf{x}) = \frac{1}{(2b)^m} e^{-\frac{||\mathbf{x}-\mathbf{w}_1||_1}{b}} \quad \text{and} \quad q(\mathbf{x}) = \frac{1}{(2b)^m} e^{-\frac{||\mathbf{x}-\mathbf{w}_2||_1}{b}}.$$

Let $\mathbf{y} = \mathbf{x} - \mathbf{w}_2$, $\mathbf{v} = \mathbf{w}_1 - \mathbf{w}_2$, and denote y_i, v_i the ith element of $\mathbf{y}$ and $\mathbf{v}$. For the case $\lambda(\lambda+1) \neq 0$, the power divergence with parameter λ between $\mathbf{X}$ and $\mathbf{Y}$ is given by

$$\begin{aligned} D_\lambda(\mathbf{X},\mathbf{Y}) =& \frac{1}{\lambda(\lambda+1)} \int_{\mathbb{R}^m} \frac{p^{\lambda+1}(\mathbf{x})}{q^{\lambda+1}(\mathbf{x})} \cdot q(\mathbf{x}) - q(\mathbf{x}) \mathrm{d}\mathbf{x} \\ =& \frac{1}{\lambda(\lambda+1)} \left[\int_{\mathbb{R}^m} \frac{1}{(2b)^m} e^{-\frac{(\lambda+1)||\mathbf{y}-\mathbf{v}||_1 - \lambda||\mathbf{y}||_1}{b}} \mathrm{d}\mathbf{y} - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i - v_i| - \lambda|y_i|}{b}} \mathrm{d}y_i \right) - 1 \right]. \end{aligned}$$

Furthermore,

$$\begin{aligned} D_\lambda(\mathbf{X},\mathbf{Y}) \leq& \frac{1}{\lambda(\lambda+1)} \left[\left(\prod_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i| - sign(\lambda)(\lambda+1)|v_i| - \lambda|y_i|}{b}} \mathrm{d}y_i \right) - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[\left(\prod_{i=1}^m e^{\frac{sign(\lambda)(\lambda+1)|v_i|}{b}} \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|y_i|}{b}} \mathrm{d}y_i \right) - 1 \right] \\ =& \frac{1}{\lambda(\lambda+1)} \left[e^{\frac{sign(\lambda)(\lambda+1)||\mathbf{v}||_1}{b}} - 1 \right]. \end{aligned}$$

Next we consider the case $\lambda(\lambda+1) = 0$. Denote the ith element of $\mathbf{x}$, $\mathbf{w}_1$, and $\mathbf{w}_2$ by $x_i, w_{1,i}, w_{2,i}$ correspondingly. We first study the case $\lambda = 0$.To this end,

$$\begin{aligned} D_0(\mathbf{X},\mathbf{Y}) =& \int_{\mathbb{R}^m} p(\mathbf{x}) \log(\frac{p(\mathbf{x})}{q(\mathbf{x})}) \mathrm{d}\mathbf{x} \\ =& \int_{\mathbb{R}^m} p(\mathbf{x}) \frac{-(||\mathbf{x}-\mathbf{w}_1||_1 - ||\mathbf{x}-\mathbf{w}_2||_1)}{b} \mathrm{d}\mathbf{x} \\ =& \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|x_i - w_{1,i}|}{b}} \cdot \frac{|x_i - w_{2,i}| - |x_i - w_{1,i}|}{b} \mathrm{d}x_i \\ =& \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|y_i|}{b}} \cdot \frac{|y_i - v_i| - |y_i|}{b} \mathrm{d}y_i. \end{aligned}$$

Furthermore,

$$\begin{aligned} D_0(\mathbf{X},\mathbf{Y}) \leq& \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|x_i - w_{1,i}|}{b}} \cdot \left| \frac{|x_i - w_{2,i}| - |x_i - w_{1,i}|}{b} \right| \mathrm{d}x_i \\ \leq& \sum_{i=1}^m \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|x_i - w_{1,i}|}{b}} \cdot \frac{|w_{1,i} - w_{2,i}|}{b} \mathrm{d}x_i = \frac{||\mathbf{w}_1 - \mathbf{w}_2||_1}{b}. \end{aligned}$$

The case $\lambda = -1$ is similar, and this completes the proof. ∎

8.3 Exact Laplace mechanism

Lemma 8.3 *For two m dimensional random variable $\mathbf{X}$ and $\mathbf{Y}$, where $X_i \sim Lap(w_{1,i}, b)$ and $Y_i \sim Lap(w_{2,i}, b)$, the $\lambda-$power divergence between, $D_\lambda(\mathbf{X},\mathbf{Y})$, for $\lambda(\lambda+1) \neq 0$ and*

$\lambda \neq -\frac{1}{2}$ *given by*

$$\frac{1}{\lambda(\lambda+1)}\left[\left(\prod_{i=1}^{m}\frac{1}{2b}\left[e^{\frac{\lambda|v_i|}{b}}\left(b+\frac{b}{2\lambda+1}\right)+e^{-\frac{(\lambda+1)|v_i|}{b}}\left(b-\frac{b}{2\lambda+1}\right)\right]\right)-1\right].$$

If $\lambda(\lambda+1)=0$ *or* $\lambda=-\frac{1}{2}$, *then*

$$D_\lambda(\mathbf{X},\mathbf{Y})=\begin{cases}-4\left[\left(\prod_{i=1}^{m}\left(e^{\frac{-|v_i|}{2b}}+\frac{|v_i|}{2b}e^{-\frac{|v_i|}{2b}}\right)\right)-1\right], & \lambda=-\frac{1}{2}\\ \frac{1}{2b}\sum_{i=1}^{m}\left[2|v_i|-2b+2be^{-\frac{|v_i|}{b}}\right], & \lambda(\lambda+1)=0.\end{cases}$$

Proof: In case $\lambda(\lambda+1)\neq 0, \lambda\neq -\frac{1}{2}$, using Lemma 8.2, it follows that

$$D_\lambda(\mathbf{X},\mathbf{Y})=\frac{1}{\lambda(\lambda+1)}\left[\left(\prod_{i=1}^{m}\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i\right)-1\right].$$

For each i, we remove the absolute sign by studying case $v_i<0$ and $v_i\geq 0$. If $v_i<0$,

$$\begin{aligned}&\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i\\ =&\frac{1}{2b}\left[\int_{-\infty}^{v_i}e^{-\frac{-(\lambda+1)(y_i-v_i)+\lambda y_i}{b}}\mathrm{d}y_i+\int_{v_i}^{0}e^{-\frac{(\lambda+1)(y_i-v_i)+\lambda y_i}{b}}\mathrm{d}y_i+\int_{0}^{\infty}e^{-\frac{(\lambda+1)(y_i-v_i)-\lambda y_i}{b}}\mathrm{d}y_i\right]\\ =&\frac{1}{2b}\left[e^{\frac{-\lambda v_i}{b}}\left(b+\frac{b}{2\lambda+1}\right)+e^{\frac{(\lambda+1)v_i}{b}}\left(b-\frac{b}{2\lambda+1}\right)\right].\end{aligned}$$

If $v_i\geq 0$,

$$\begin{aligned}&\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i\\ =&\frac{1}{2b}\left[\int_{-\infty}^{0}e^{-\frac{-(\lambda+1)(y_i-v_i)+\lambda y_i}{b}}\mathrm{d}y_i+\int_{0}^{v_i}e^{-\frac{-(\lambda+1)(y_i-v_i)-\lambda y_i}{b}}\mathrm{d}y_i+\int_{v_i}^{\infty}e^{-\frac{(\lambda+1)(y_i-v_i)-\lambda y_i}{b}}\mathrm{d}y_i\right]\\ =&\frac{1}{2b}\left[e^{\frac{\lambda v_i}{b}}\left(b+\frac{b}{2\lambda+1}\right)+e^{\frac{-(\lambda+1)v_i}{b}}\left(b-\frac{b}{2\lambda+1}\right)\right].\end{aligned}$$

Combining the cases $v_i<0$ and $v_i\geq 0$, we get

$$\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i=\frac{1}{2b}\left[e^{\frac{\lambda|v_i|}{b}}\left(b+\frac{b}{2\lambda+1}\right)+e^{-\frac{(\lambda+1)|v_i|}{b}}\left(b-\frac{b}{2\lambda+1}\right)\right].$$

Therefore,

$$D_\lambda(\mathbf{X},\mathbf{Y})=\frac{1}{\lambda(\lambda+1)}\left[\left(\prod_{i=1}^{m}\frac{1}{2b}\left[e^{\frac{\lambda|v_i|}{b}}\left(b+\frac{b}{2\lambda+1}\right)+e^{-\frac{(\lambda+1)|v_i|}{b}}\left(b-\frac{b}{2\lambda+1}\right)\right]\right)-1\right].$$

We now turn to the case $\lambda=-\frac{1}{2}$. If $v_i<0$, by the same calculation of the integral, it follows that

$$\int_{\mathbb{R}}\frac{1}{2b}e^{-\frac{(\lambda+1)|y_i-v_i|-\lambda|y_i|}{b}}\mathrm{d}y_i=e^{\frac{v_i}{2b}}-\frac{v_i}{2b}e^{\frac{v_i}{2b}}.$$

If $v_i \geq 0$,

$$\int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i - v_i| - \lambda|y_i|}{b}} \mathrm{d}y_i = e^{\frac{-v_i}{2b}} + \frac{v_i}{2b} e^{-\frac{v_i}{2b}}.$$

Combining $v_i < 0$ and $v_i \geq 0$, we get

$$\int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{(\lambda+1)|y_i - v_i| - \lambda|y_i|}{b}} \mathrm{d}y_i = e^{\frac{-|v_i|}{2b}} + \frac{|v_i|}{2b} e^{-\frac{|v_i|}{2b}}.$$

Therefore

$$D_\lambda(\mathbf{X}, \mathbf{Y}) = -4 \left[\left(\prod_{i=1}^{m} e^{\frac{-|v_i|}{2b}} + \frac{|v_i|}{2b} e^{-\frac{|v_i|}{2b}} \right) - 1 \right].$$

Finally, we turn to the case $\lambda(\lambda + 1) = 0$. We study the case $\lambda = 0$. Using Lemma 8.2, it follows that

$$D_0(\mathbf{X}, \mathbf{Y}) = \sum_{i=1}^{m} \int_{\mathbb{R}} \frac{1}{2b} e^{-\frac{|y_j|}{b}} \cdot \frac{|y_i - v_i| - |y_i|}{b} \mathrm{d}y_i.$$

If $v_i \geq 0$,

$$\begin{aligned}
&\int_{\mathbb{R}} e^{-\frac{|y_i|}{b}} \cdot \frac{|y_i - v_i| - |y_i|}{b} \mathrm{d}y_i \\
=&\int_{-\infty}^{0} e^{-\frac{-y_i}{b}} \frac{-(y_i - v_i) + y_i}{b} \mathrm{d}y_i + \int_{0}^{v_i} e^{-\frac{y_i}{b}} \frac{-(y_i - v_i) - y_i}{b} \mathrm{d}y_i + \int_{v_i}^{\infty} e^{-\frac{y_i}{b}} \frac{(y_i - v_i) - y_i}{b} \mathrm{d}y_i \\
=&2v_i - 2b + 2be^{-\frac{v_i}{b}}.
\end{aligned}$$

If $v_i < 0$,

$$\begin{aligned}
&\int_{\mathbb{R}} e^{-\frac{|y_i|}{b}} \cdot \frac{|y_i - v_i| - |y_i|}{b} \mathrm{d}y_i \\
=&\int_{-\infty}^{v_i} e^{-\frac{-y_i}{b}} \frac{-(y_i - v_i) + y_i}{b} \mathrm{d}y_i + \int_{v_i}^{0} e^{-\frac{-y_i}{b}} \frac{(y_i - v_i) + y_i}{b} \mathrm{d}y_i + \int_{0}^{\infty} e^{-\frac{y_i}{b}} \frac{(y_i - v_i) - y_i}{b} \mathrm{d}y_i \\
=&-2v_i - 2b + 2be^{\frac{v_i}{b}}.
\end{aligned}$$

Combining the cases $v_i < 0$ and $v_i \geq 0$, we get

$$\int_{\mathbb{R}} e^{-\frac{|y_i|}{b}} \cdot \frac{|y_i - v_i| - |y_i|}{b} \mathrm{d}y_i = 2|v_i| - 2b + 2be^{-\frac{|v_i|}{b}}.$$

Therefore,

$$D_0(\mathbf{X}, \mathbf{Y}) = \frac{1}{2b} \sum_{i=1}^{m} \left[2|v_i| - 2b + 2be^{-\frac{|v_i|}{b}} \right].$$

The case $\lambda = -1$ is similar, and this completes the proof. ∎

9 Relationships between privacy frameworks

9.1 Proof of Remark 5

Link to $\rho-$zCDP: Suppose $M(D) = W(D) + \mathbf{Y}$, where $\mathbf{Y} \sim N_m(0, \sigma^2 I_m)$ with a dataset-independent covariance, satisfies $(\lambda, \epsilon)-$PDP for $\lambda > 0$. Put $\Delta = \Delta_{L_2} W$ and $\rho = \Delta^2/(2\sigma^2)$. For every order $a > 1$ and every adjacent D, D', the Gaussian Rényi divergence satisfies

$$D_a^{(R)}(P_D, P_{D'}) = \frac{a\|W(D) - W(D')\|_2^2}{2\sigma^2} \leq a\rho.$$

At the given PDP order, the Gaussian power-divergence identity yields

$$\frac{e^{\lambda(\lambda+1)\rho} - 1}{\lambda(\lambda+1)} \leq \epsilon, \qquad \rho \leq \frac{\log\{1 + \lambda(\lambda+1)\epsilon\}}{\lambda(\lambda+1)} \leq \epsilon.$$

The Gaussian bound holds at *every* Rényi order $a > 1$, and hence M satisfies $\rho-$zCDP, in particular $\epsilon-$zCDP.

Link to $(\epsilon, \delta)-$differential privacy: Suppose a mechanism M satisfies $(\lambda, \epsilon)-$PDP, then by definition, $D_\lambda(f_1, f_2) \leq \epsilon$, where f_1 is the density of $M(w, D)$ and f_2 is the density of $M(w, D')$. We now determine the relationship to $(\epsilon, \delta)-$DP.

If $\lambda > 0$, then

$$\begin{aligned} D_\lambda(f_1, f_2) = \frac{1}{\lambda(\lambda+1)} \left[\int_{\mathbb{R}^m} \frac{f_1^{\lambda+1}(x)}{f_2^\lambda(x)} \mathrm{d}x - 1 \right] \leq \epsilon \\ \iff \int_{\mathbb{R}^m} \frac{f_1^{\lambda+1}(x)}{f_2^\lambda(x)} \mathrm{d}x \leq \lambda(\lambda+1)\epsilon + 1 = e^{\log(\lambda(\lambda+1)\epsilon+1)}. \end{aligned}$$

For any set $A \subset \mathbb{R}^m$, applying Holder inequality for $p = \lambda + 1$ and $q = \frac{\lambda+1}{\lambda}$, it follows that

$$\begin{aligned} \mathbf{P}_{X\sim f_1}(X \in A) &= \int_A f_1(x)\mathrm{d}x = \int_A \frac{f_1(x)}{[f_2(x)]^{\frac{\lambda}{\lambda+1}}} \cdot [f_2(x)]^{\frac{\lambda}{\lambda+1}} \mathrm{d}x \\ &\leq \left(\int_A \left(\frac{f_1(x)}{[f_2(x)]^{\frac{\lambda}{\lambda+1}}} \right)^p \mathrm{d}x \right)^{\frac{1}{p}} \cdot \left(\int_A \left([f_2(x)]^{\frac{\lambda}{\lambda+1}} \right)^q \mathrm{d}x \right)^{\frac{1}{q}} \\ &= \left(\int_A \frac{[f_1(x)]^{\lambda+1}}{[f_2(x)]^\lambda} \mathrm{d}x \right)^{\frac{1}{\lambda+1}} \cdot \left(\int_A f_2(x)\mathrm{d}x \right)^{\frac{\lambda}{\lambda+1}} \\ &\leq \left(\int_{\mathbb{R}^m} \frac{[f_1(x)]^{\lambda+1}}{[f_2(x)]^\lambda} \mathrm{d}x \right)^{\frac{1}{\lambda+1}} \cdot \left(\int_A f_2(x)\mathrm{d}x \right)^{\frac{\lambda}{\lambda+1}} \\ &\leq e^{\frac{1}{\lambda+1}\log(\lambda(\lambda+1)\epsilon+1)} \cdot [\mathbf{P}_{X\sim f_2}(X \in A)]^{\frac{\lambda}{\lambda+1}} \\ &= \left[e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)} \cdot \mathbf{P}_{X\sim f_2}(X \in A) \right]^{\frac{\lambda}{\lambda+1}}. \end{aligned}$$

If $e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)>\delta^{\frac{\lambda+1}{\lambda}}$, then

$$\begin{aligned}\mathbf{P}_{X\sim f_1}(X\in A)\leq&\left[e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\right]^{\frac{\lambda}{\lambda+1}}\\=&e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\cdot\left[e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\right]^{\frac{-1}{\lambda+1}}\\\leq&e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\cdot\delta^{\frac{-1}{\lambda}}\\=&e^{\frac{1}{\lambda}\log(\frac{\lambda(\lambda+1)\epsilon+1}{\delta})}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\end{aligned}\tag{40}$$

If $e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\leq\delta^{\frac{\lambda+1}{\lambda}}$, then

$$\mathbf{P}_{X\sim f_1}(X\in A)\leq\left[e^{\frac{1}{\lambda}\log(\lambda(\lambda+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\right]^{\frac{\lambda}{\lambda+1}}=\delta.$$

Therefore

$$\mathbf{P}_{X\sim f_1}(X\in A)\leq e^{\frac{1}{\lambda}\log(\frac{\lambda(\lambda+1)\epsilon+1}{\delta})}\cdot\mathbf{P}_{X\sim f_2}(X\in A)+\delta.$$

This implies that M satisfies $(\frac{1}{\lambda}\log(\frac{\lambda(\lambda+1)\epsilon+1}{\delta}),\delta)-$DP.

If $\lambda<-1$, write $\lambda'=-\lambda-1>0$,

$$\begin{aligned}&D_\lambda(f_2,f_1)=\frac{1}{\lambda(\lambda+1)}\left[\int_{\mathbb{R}^m}\frac{f_2^{\lambda+1}(x)}{f_1^{\lambda}(x)}\mathrm{d}x-1\right]\leq\epsilon\\\Longleftrightarrow&\int_{\mathbb{R}^m}\frac{f_2^{\lambda+1}(x)}{f_1^{\lambda}(x)}\mathrm{d}x\leq\lambda(\lambda+1)\epsilon+1=e^{\log(\lambda(\lambda+1)\epsilon+1)}\\\Longleftrightarrow&\int_{\mathbb{R}^m}\frac{f_1^{\lambda'+1}(x)}{f_2^{\lambda'}(x)}\mathrm{d}x\leq\lambda'(\lambda'+1)\epsilon+1=e^{\log(\lambda'(\lambda'+1)\epsilon+1)}.\end{aligned}$$

Applying Holder inequality for $p=\lambda'+1>1$, $q=\frac{\lambda'+1}{\lambda'}>1$, by the same method, we obtain

$$\mathbf{P}_{X\sim f_1}(X\in A)\leq\left[e^{\frac{1}{\lambda'}\log(\lambda'(\lambda'+1)\epsilon+1)}\cdot\mathbf{P}_{X\sim f_2}(X\in A)\right]^{\frac{\lambda'}{\lambda'+1}}.\tag{41}$$

Using the same δ, it follows that

$$\mathbf{P}_{X\sim f_1}(X\in A)\leq e^{\frac{-1}{\lambda+1}\log(\frac{\lambda(\lambda+1)\epsilon+1}{\delta})}\cdot\mathbf{P}_{X\sim f_2}(X\in A)+\delta.$$

This implies that M satisfies $(\frac{-1}{\lambda+1}\log(\frac{\lambda(\lambda+1)\epsilon+1}{\delta}),\delta)-$DP.

Link to $\mu-$GDP for Gaussian and scalar Laplace mechanisms: The GDP conclusions of Remark 5 of the main paper concern the additive mechanisms with common dataset-independent noise scales, rather than arbitrary PDP mechanisms. For the Gaussian mechanism $M(D)=W(D)+\mathbf{Y}$, with $\mathbf{Y}\sim N_m(0,\sigma^2I_m)$, let $\Delta_2=\Delta_{L_2}W$. The exact

Gaussian testing trade-off is G_{μ_G}, where $\mu_G = \Delta_2/\sigma$. For $\lambda(\lambda+1) \neq 0$, the Gaussian power divergence of an adjacent pair with shift v satisfies

$$D_\lambda\{N_m(0,\sigma^2 I_m), N_m(v,\sigma^2 I_m)\} = \frac{\exp\{\lambda(\lambda+1)\|v\|_2^2/(2\sigma^2)\} - 1}{\lambda(\lambda+1)}.$$

Thus $(\lambda,\epsilon)-$PDP implies

$$\mu_G^2 \leq \frac{2}{\lambda(\lambda+1)} \log\{1 + \lambda(\lambda+1)\epsilon\},$$

with the continuous bound $\mu_G^2 \leq 2\epsilon$ at $\lambda = 0, -1$.

For the scalar Laplace mechanism $M(D) = W(D) + Z_b$, where $Z_b \sim \text{Lap}(0,b)$ and b is common to all datasets, let $\Delta_1 = \Delta_{L_1} W$ and $r = \Delta_1/(2b)$. Proposition 15.1 establishes from the complete testing profile that its smallest GDP parameter is

$$\mu_L(r) = 2\Phi^{-1}\left(1 - \frac{1}{2}e^{-r}\right).$$

In particular, the scalar Laplace PDP bound restricts r through the exact power-divergence formula in Lemma 8.3. To make the dependence on (λ,ϵ) precise, put

$$d_\lambda(t) = D_\lambda\{\text{Lap}(0,1), \text{Lap}(t,1)\}, \qquad t \geq 0.$$

The formulas in Lemma 8.3 show that d_λ is strictly increasing for $t > 0$. In the nontrivial admissible PDP range, $\Delta_1/b \leq d_\lambda^{-1}(\epsilon)$, so the mechanism is also μ-GDP with

$$\mu = 2\Phi^{-1}\left(1 - \frac{1}{2}\exp\left\{-\frac{d_\lambda^{-1}(\epsilon)}{2}\right\}\right).$$

This uses the full family-specific GDP comparison, not a general conversion from a single power-divergence order. ∎

10 Supporting results for Hessian convergence and utility

10.1 Details on the convergence of $H_n(\cdot)$ to $H_\infty(\cdot)$ in Proposition 26

We write $H_{n,i,j}(\boldsymbol{\theta})$ as the $i-th$ row and $j-th$ column element of $H_n(\boldsymbol{\theta})$, and $I_{i,j}(\boldsymbol{\theta})$ as the $i-th$ row and $j-th$ column element of $I(\boldsymbol{\theta})$. Then we only need to show $H_{n,i,j}(\boldsymbol{\theta}_0) \to I_{i,j}(\boldsymbol{\theta}_0)$ for any $i, j = 1, \cdots, m$. Recall that

$$H_{n,i,j}(\boldsymbol{\theta}) = \frac{\partial^2}{\partial \boldsymbol{\theta}_i \partial \boldsymbol{\theta}_j} L_n(\boldsymbol{\theta}) = -T_{1,n,i,j}(\boldsymbol{\theta}) - 2T_{2,n,i,j}(\boldsymbol{\theta}),$$

where

$$T_{1,n,i,j}(\boldsymbol{\theta}) = \int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},j}(x) u_{\boldsymbol{\theta},i}(x) \mathrm{d}x, \quad T_{2,n,i,j}(\boldsymbol{\theta}) = \int_{\mathbb{R}} g_n^{\frac{1}{2}}(x) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i,j}(x) \mathrm{d}x.$$

We decompose $T_{1,n,i,j}(\boldsymbol{\theta})$ and $T_{2,n,i,j}(\boldsymbol{\theta})$ as follows:

$$T_{1,n,i,j}(\boldsymbol{\theta}) = T^{(1)}_{1,n,i,j}(\boldsymbol{\theta}) + I_{i,j}(\boldsymbol{\theta}), \quad \text{and} \quad T_{2,n,i,j}(\boldsymbol{\theta}) = T^{(1)}_{2,n,i,j}(\boldsymbol{\theta}) - I_{i,j}(\boldsymbol{\theta}),$$

where

$$T^{(1)}_{1,n,i,j}(\boldsymbol{\theta}) = \int_{\mathbb{R}} \left(g_n^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},j}(x) u_{\boldsymbol{\theta},i}(x) \mathrm{d}x, \quad \text{and}$$
$$T^{(1)}_{2,n,i,j}(\boldsymbol{\theta}) = \int_{\mathbb{R}} \left(g_n^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i,j}(x) \mathrm{d}x.$$

Then $H_{n,i,j}(\boldsymbol{\theta})$ can be written as follows:

$$H_{n,i,j}(\boldsymbol{\theta}) = I_{i,j}(\boldsymbol{\theta}) - D_{n,i,j}(\boldsymbol{\theta}), \tag{42}$$

where $D_{n,i,j}(\boldsymbol{\theta}) = T^{(1)}_{1,n,i,j}(\boldsymbol{\theta}) + 2T^{(1)}_{2,n,i,j}(\boldsymbol{\theta})$. We are going to show $D_{n,i,j}(\boldsymbol{\theta}) \to D_{i,j}(\boldsymbol{\theta})$ almost surely as $n \to \infty$, where $D_{i,j}(\boldsymbol{\theta}) = \int_{\mathbb{R}} \left(g^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) + 2u_{\boldsymbol{\theta},i,j}(x)] \mathrm{d}x$. Notice that

$$D_{n,i,j}(\boldsymbol{\theta}) = D_{i,j}(\boldsymbol{\theta}) + \int_{\mathbb{R}} \left(g_n^{\frac{1}{2}}(x) - g^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) + 2u_{\boldsymbol{\theta},i,j}(x)] \mathrm{d}x.$$

We are going to show $\int_{\mathbb{R}} \left(g_n^{\frac{1}{2}}(x) - g^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) + 2u_{\boldsymbol{\theta},i,j}(x)] \mathrm{d}x \to 0$ almost surely as $n \to \infty$. Using the Cauchy–Schwarz inequality and the upper bounds in assumptions **(U1)-(U2)**, it follows that as $n \to \infty$,

$$\begin{aligned}
&\left| \int_{\mathbb{R}} \left(g_n^{\frac{1}{2}}(x) - g^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) + 2u_{\boldsymbol{\theta},i,j}(x)] \mathrm{d}x \right| \\
\leq & HD(g_n, g) \cdot \left(\mathbf{E}_{\boldsymbol{\theta}} \left[u^2_{\boldsymbol{\theta},i}(X) u^2_{\boldsymbol{\theta},j}(X) \right] \right)^{\frac{1}{2}} + 2HD(g_n, g) \cdot \left(\mathbf{E}_{\boldsymbol{\theta}} \left[u^2_{\boldsymbol{\theta},i,j}(X) \right] \right)^{\frac{1}{2}} \\
\leq & c \cdot HD(g_n, g) \xrightarrow{a.s.} 0.
\end{aligned}$$

The convergence follows from $HD(g_n, g) \xrightarrow{a.s.} 0$ when $n \to \infty$, since by assumption **(A2)**, $||g_n - g||_1$ converges to zero almost surely. Thus, $H_\infty(\boldsymbol{\theta}) = I(\boldsymbol{\theta}) - D(\boldsymbol{\theta})$. This completes the proof.

10.2 Establish the upper bound for $D_{i,j}(\cdot)$ in Proposition 27

Using Cauchy- Schwarz inequality and **(U1)-(U2)** the result follows. To see this, notice that

$$|D_{i,j}(\boldsymbol{\theta})| = \left| \int_{\mathbb{R}} \left(g^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) [u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) + 2u_{\boldsymbol{\theta},i,j}(x)] \mathrm{d}x \right|.$$

Now, splitting the RHS of the above equation, we see that it is bounded above by

$$\left| \int_{\mathbb{R}} \left(g^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i}(x) u_{\boldsymbol{\theta},j}(x) \mathrm{d}x \right| + 2 \left| \int_{\mathbb{R}} \left(g^{\frac{1}{2}}(x) - f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) \right) f_{\boldsymbol{\theta}}^{\frac{1}{2}}(x) u_{\boldsymbol{\theta},i,j}(x) \mathrm{d}x \right|.$$

Now, applying Cauchy-Schwarz inequality, we get

$$\begin{aligned} |D_{i,j}^2(\boldsymbol{\theta})| \leq & 2HD^2(g, f_{\boldsymbol{\theta}}) \cdot \mathbf{E}_{\boldsymbol{\theta}} \left[u_{\boldsymbol{\theta},i}^2(X) u_{\boldsymbol{\theta},j}^2(X) \right] + 8HD^2(g, f_{\boldsymbol{\theta}}) \cdot \mathbf{E}_{\boldsymbol{\theta}} \left[u_{\boldsymbol{\theta},i,j}^2(X) \right] \\ \leq & c \cdot HD^2(g, f_{\boldsymbol{\theta}}), \end{aligned} \tag{43}$$

where $0 < c = 1 + \sup_{\boldsymbol{\theta} \in \Theta} \left\{ 2\mathbf{E}_{\boldsymbol{\theta}} \left[u_{\boldsymbol{\theta},i}^2(X) u_{\boldsymbol{\theta},j}^2(X) \right] + 8\mathbf{E}_{\boldsymbol{\theta}} \left[u_{\boldsymbol{\theta},i,j}^2(X) \right] \right\} < \infty$.

10.3 Proof of Lemma 6.2

Statement: Let assumptions **(A1)-(A8)** and **(U1)-(U2)** hold. Then for $\boldsymbol{\theta} \in B_r(\boldsymbol{\theta}_g)$ and $n \geq N$, if $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \frac{r^2}{4}\tau_1$ then $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq \frac{r}{2}$. Furthermore, if $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq \frac{r}{2}$ for $\boldsymbol{\theta} \in B_r(\boldsymbol{\theta}_g)$, then for $n \geq N$, $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \frac{r^2}{4}\tau_2$.

Proof: Let $n \geq N$ and $\boldsymbol{\theta}, \hat{\boldsymbol{\theta}} \in B_r(\boldsymbol{\theta}_g)$. Suppose $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \frac{r^2}{4}\tau_1$. Then using Proposition 29 (1), it follows that $L_n(\boldsymbol{\theta}) \geq L_n(\hat{\boldsymbol{\theta}}) + \langle \nabla L_n(\hat{\boldsymbol{\theta}}), \boldsymbol{\theta} - \hat{\boldsymbol{\theta}} \rangle + \tau_1 ||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2^2$. Since $\nabla L_n(\hat{\boldsymbol{\theta}}) = (0, \cdots, 0)$ it follows that $\frac{r^2\tau_1}{4} \geq L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \geq \tau_1 ||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2^2$, the result follows. The rest of the proof follows similarly, using Proposition 29 (3); that is, if $||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2 \leq \frac{r}{2}$, then $L_n(\boldsymbol{\theta}) - L_n(\hat{\boldsymbol{\theta}}) \leq \langle \nabla L_n(\hat{\boldsymbol{\theta}}), \boldsymbol{\theta} - \hat{\boldsymbol{\theta}} \rangle + \tau_{2,n} ||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2^2 = \tau_2 ||\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}||_2^2 \leq \frac{r^2}{4}\tau_2$. ∎

10.4 Proof of Lemma 6.3

Statement: Under assumptions **(A1)-(A8)** and **(U1)-(U2)**, for $\eta \leq \frac{1}{\tau_2}$, assume that for $n \geq N$, $\hat{\boldsymbol{\theta}}_n \in B_{r/c}(\boldsymbol{\theta}_g) \subset B_{r/2}(\boldsymbol{\theta}_g)$, where $c > 2\left(\frac{\tau_2}{\tau_1}\right)^{\frac{1}{2}}$, then there exists $\hat{\boldsymbol{\theta}}_n^{(0)}$, such that $L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ and $||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$ hold with probability $1 - \frac{k\xi}{K}$ for all $k = 0, \cdots, K$.

Proof: The Lemma states that under the stated conditions, the estimators from each iteration, $\hat{\boldsymbol{\theta}}_n^{(k)}$, belong to the ball $B_{r/2}(\hat{\boldsymbol{\theta}}_n) \subset B_r(\boldsymbol{\theta}_g)$. We prove the result by induction. First, for $k = 0$, we choose the initial estimator to be a consistent estimator of $\boldsymbol{\theta}_g$. Hence for large n, $||\hat{\boldsymbol{\theta}}_n^{(0)} - \boldsymbol{\theta}_g|| \leq \left(\frac{\tau_1}{\tau_2}\right)^{\frac{1}{2}} \frac{r}{2} - \frac{r}{c}$. Hence, for large n, $||\hat{\boldsymbol{\theta}}_n^{(0)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \left(\frac{\tau_1}{\tau_2}\right)^{\frac{1}{2}} \frac{r}{2}$. By Lemma 6.2, $L_n(\hat{\boldsymbol{\theta}}_n^{(0)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ and $||\hat{\boldsymbol{\theta}}_n^{(0)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$ hold. Hence, by induction hypothesis, let $L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ and $||\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$ hold. We will establish that $L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ and $||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$. The proof of this relies on the behavior

of $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2$, $||N_{n,k}||_2$, and their relationships which is described in the following claim whose proof is relegated to the end.

Claim: If $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \geq \sqrt{\frac{(1+2\eta\tau_2)B||N_{n,k}||_2+\eta\tau_2||N_{n,k}||_2^2}{1-\eta\tau_2}}$, where B is the upper bound of $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2$ from Proposition 25, then the following inequality holds.

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n).$$

Using the claim for k, and the assumption $L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$, we obtain that $L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$. Next, applying Lemma 6.2 we get $||\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n||_2 \leq \frac{r}{2}$. This completes the proof under the condition of the claim.

Next, we turn to the case $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 < \sqrt{\frac{(1+2\eta\tau_2)B||N_{n,k}||_2+\eta\tau_2||N_{n,k}||_2^2}{1-\eta\tau_2}}$. By (35), it follows that with probability $1 - \frac{k\xi}{K}$ (since we have k iterations here)

$$||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 < \sqrt{\frac{(1+2\eta\tau_2)Br_{noi} + \eta\tau_2 r_{noi}^2}{1-\eta\tau_2}} := \bar{r}_{noi}.$$

Using Proposition 29 (1), $\tau_1||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 \leq L_n(\hat{\boldsymbol{\theta}}_n) - L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - \langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}), \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle$. Since $\hat{\boldsymbol{\theta}}_n$ is the minimizer of $L_n(\boldsymbol{\theta})$, it follows that $\tau_1||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 \leq |\langle \nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}), \hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)} \rangle|$. Now, applying the Cauchy-Schwarz inequality, it follows that $\tau_1||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2^2 \leq ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \cdot ||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2$. Hence, we obtain

$$||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k)}||_2 \leq \frac{||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2}{\tau_1} \leq \frac{\bar{r}_{noi}}{\tau_1}.$$

Now using $\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n^{(k+1)} = \eta(\nabla L_n(\hat{\boldsymbol{\theta}}_n^k) + N_k)$, it follows that

$$\begin{aligned} ||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k+1)}||_2 \leq & ||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^k||_2 + \eta||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 + \eta||N_k||_2 \leq \frac{\bar{r}_{noi}}{\tau_1} + \eta\bar{r}_{noi} + \eta r_{noi} \\ & \leq \left(\frac{\tau_1}{\tau_2}\right)^{\frac{1}{2}} \frac{r}{2} \leq \frac{r}{2}, \end{aligned}$$

where the last inequality follows by taking n large. This is equivalent to choosing n such that

$$\Delta_n \frac{\left(4\sqrt{m} + 2\sqrt{2\log(\frac{K}{\xi})}\right)}{\sqrt{-8\log(1 - 0.5\frac{\epsilon}{K})}} \leq r_u.$$

Finally, the inequality $L_n(\hat{\boldsymbol{\theta}}^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq \tau_1 \frac{r^2}{4}$ follows using Lemma 6.2 and $||\hat{\boldsymbol{\theta}}_n - \hat{\boldsymbol{\theta}}_n^{(k+1)}||_2 \leq \left(\frac{\tau_1}{\tau_2}\right)^{\frac{1}{2}} \frac{r}{2}$. This completes the induction. To complete the proof of the Lemma, we now establish the claim.

Proof of the claim: Using equation (8) in Definition 21 of the main paper, and let $\boldsymbol{\theta}^* = \gamma\hat{\boldsymbol{\theta}}_n^{(k+1)} + (1-\gamma)\hat{\boldsymbol{\theta}}_n^{(k)}$ for some $\gamma \in [0,1]$ in the Taylor expansion of $L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)})$ up-to

second order, and apply Cauchy–Schwarz inequality to get

$$\begin{aligned}L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) =& L_n(\hat{\boldsymbol{\theta}}_n^{(k)} - \eta(\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + N_k)) - L_n(\hat{\boldsymbol{\theta}}_n) \\ \leq& L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) - \eta||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + \eta||N_{n,k}||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \\ &+ \frac{\eta^2}{2}\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})^T H_n(\boldsymbol{\theta}^*)\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) + \frac{\eta^2}{2}N_{n,k}^T H_n(\boldsymbol{\theta}^*)N_{n,k} \\ &+ \eta^2 N_{n,k}^T H_n(\boldsymbol{\theta}^*)\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}).\end{aligned}$$

Furthermore, use Proposition 29 (3) and Cauchy–Schwarz inequality to get

$$\begin{aligned}&\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})^T H_n(\theta^*)\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) \leq 2\tau_2||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2, \\ &N_{n,k}^T H_n(\boldsymbol{\theta}^*)N_k \leq 2\tau_2||N_{n,k}||_2^2, \text{ and} \\ &|N_{n,k}^T H_n(\boldsymbol{\theta}^*)\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})| \leq ||N_{n,k}^T||_2 \cdot ||H_n(\boldsymbol{\theta}^*)\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \leq 2\tau_2||N_{n,k}^T||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2.\end{aligned}$$

These give the upper bound of $L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n)$ as follows:

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n) + Term_{n,k},$$

where $Term_{n,k}$ is given by

$$\begin{aligned}&- \eta||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + \eta||N_{n,k}||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 + \eta^2\tau_2||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + \eta^2\tau_2||N_{n,k}||_2^2 \\ &+ 2\eta^2\tau_2||N_{n,k}^T||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2.\end{aligned}$$

Hence the condition $||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \geq \sqrt{\frac{(1+2\eta\tau_2)B||N_{n,k}||_2 + \eta\tau_2||N_{n,k}||_2^2}{1-\eta\tau_2}}$ implies that $Term_{n,k} \leq 0$, that is

$$\begin{aligned}&- \eta||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 + \eta||N_{n,k}||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 + \eta^2\tau_2||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2^2 \\ &+ \eta^2\tau_2||N_{n,k}||_2^2 + 2\eta^2\tau_2||N_{n,k}^T||_2 \cdot ||\nabla L_n(\hat{\boldsymbol{\theta}}_n^{(k)})||_2 \leq 0,\end{aligned}$$

and furthermore

$$L_n(\hat{\boldsymbol{\theta}}_n^{(k+1)}) - L_n(\hat{\boldsymbol{\theta}}_n) \leq L_n(\hat{\boldsymbol{\theta}}_n^{(k)}) - L_n(\hat{\boldsymbol{\theta}}_n).$$

This completes the proof. ∎

11 Numerical calculations and additional experiments

In this section, we provide a Monte-Carlo approximation to the loss function and give calculation details for the Normal distribution used in numerical experiments.

$$\begin{aligned}\tilde{L}_n(\boldsymbol{\theta}) =& 2\int_{\mathbb{R}}(\sqrt{f_{\boldsymbol{\theta}}(x)}-\sqrt{g_n(x)})^2\mathrm{d}x = 2\left[2-2\int_{\mathbb{R}} g_n(x)\left(\frac{f_{\boldsymbol{\theta}}(x)}{g_n(x)}\right)^{\frac{1}{2}}\mathrm{d}x\right]\\ \approx& 2\left[2-\frac{2}{r_n}\sum_{i=1}^{r_n}\left(\frac{f_{\boldsymbol{\theta}}(X_{n,i})}{g_n(X_{n,i})}\right)^{\frac{1}{2}}\right],\end{aligned}$$

where $\{X_{n,1},\cdots X_{n,r_n}\}|(X_1,\cdots X_n)$ are i.i.d. $g_n(\cdot)$. Next, the gradient is given by

$$\nabla\tilde{L}_n(\boldsymbol{\theta}) = -\frac{2}{r_n}\sum_{i=1}^{r_n}\left(\frac{f_{\boldsymbol{\theta}}(X_{n,i})}{g_n(X_{n,i})}\right)^{\frac{1}{2}} u_{\boldsymbol{\theta}}(X_{n,i}),$$

where $u_{\boldsymbol{\theta}}(x) = \frac{\nabla f_{\boldsymbol{\theta}}(x)}{f_{\boldsymbol{\theta}}(x)}$, while the Hessian is given by

$$\begin{aligned}\tilde{H}_n(\boldsymbol{\theta}) =& \frac{1}{r_n}\sum_{i=1}^{r_n}\left(\frac{f_{\boldsymbol{\theta}}(X_{n,i})}{g_n(X_{n,i})}\right)^{\frac{1}{2}} u_{\boldsymbol{\theta}}(X_{n,i})\cdot u_{\boldsymbol{\theta}}^T(X_{n,i})\\ &-\frac{2}{r_n}\sum_{i=1}^{r_n}\left(\frac{f_{\boldsymbol{\theta}}(X_{n,i})}{g_n(X_{n,i})}\right)^{\frac{1}{2}}\frac{1}{f_{\boldsymbol{\theta}}(X_{n,i})}H_f(X_{n,i}),\end{aligned}$$

where H_f is Hessian of $f_{\boldsymbol{\theta}}(\cdot)$. For the normal distribution, the gradient is given by

$$\nabla f_{\boldsymbol{\theta}}(x) = f_{\boldsymbol{\theta}}(x)\cdot\begin{bmatrix}\frac{x-\mu}{\sigma^2}\\ \frac{(x-\mu)^2-\sigma^2}{\sigma^3}\end{bmatrix},\quad u_{\boldsymbol{\theta}}(x) = \begin{bmatrix}\frac{x-\mu}{\sigma^2}\\ \frac{(x-\mu)^2-\sigma^2}{\sigma^3}\end{bmatrix},$$

and the Hessian of $f_{\boldsymbol{\theta}}(\cdot)$ is given by

$$H_f(x) = f_{\boldsymbol{\theta}}(x)\cdot\begin{bmatrix}\frac{(x-\mu)^2-\sigma^2}{\sigma^4} & \frac{(x-\mu)((x-\mu)^2-3\sigma^2)}{\sigma^5}\\ \frac{(x-\mu)((x-\mu)^2-3\sigma^2)}{\sigma^5} & \frac{(x-\mu)^4-5\sigma^2(x-\mu)^2+2\sigma^4}{\sigma^6}\end{bmatrix}.$$

11.1 Additional figures and tables for PGD and PNR

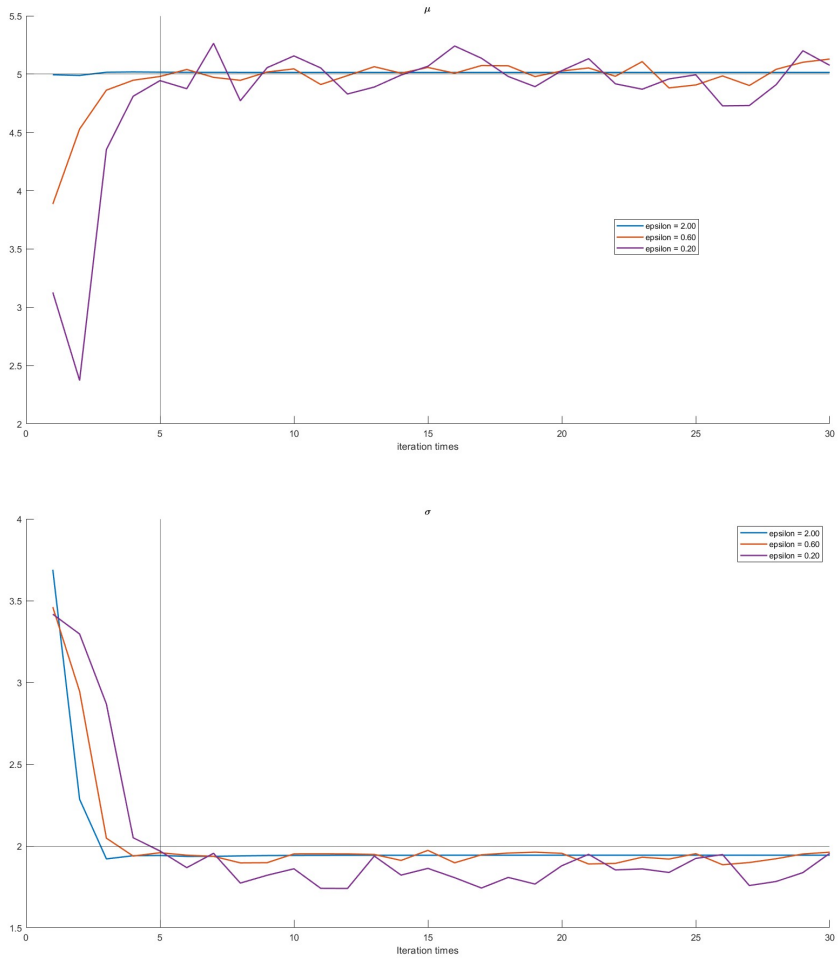


Figure 11.1: Private Newton's method path

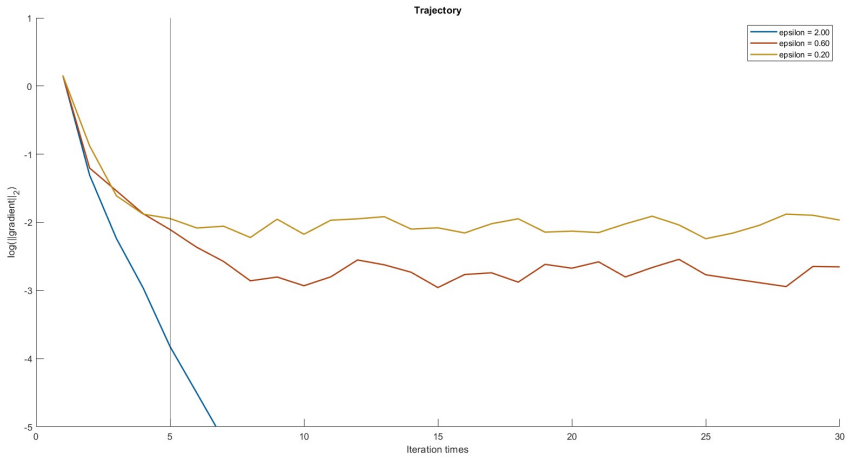


Figure 11.2: Private Newton's method trajectory

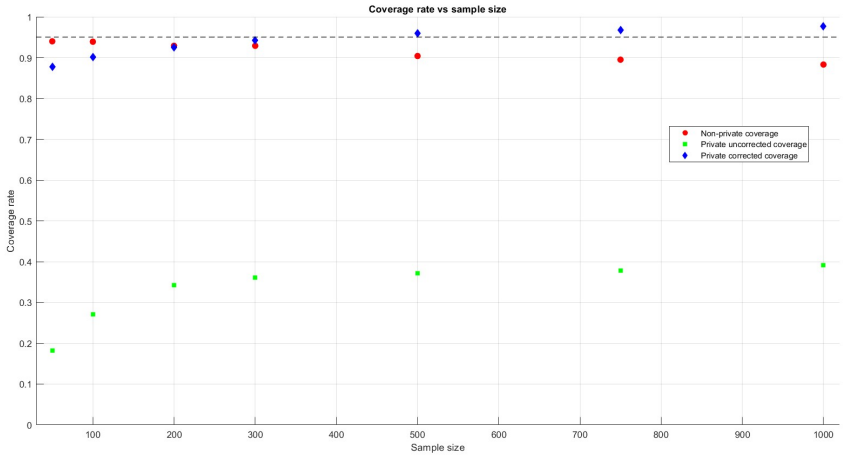


Figure 11.3: Private and non-private Newton's method 95% confidence interval coverage

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	5 (0.08)	4.948 (0.332)	4.868 (1.756)
	σ: Mean (Std. Error)	1.975 (0.076)	1.987 (0.349)	2.196 (1.99)
CI coverage for μ	Corrected	0.883	0.977	0.95
	Uncorrected	0.883	0.391	0.247
CI coverage for σ	Corrected	0.739	0.913	0.904
	Uncorrected	0.739	0.442	0.264

Table 11.1: Results for different values of ϵ (Newton-Raphson). Sample size is 1000, $K = 5$.

		$\lambda = 1$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	4.991 (0.083)	4.986 (0.295)	4.96 (2.009)
	σ: Mean (Std. Error)	1.984 (0.058)	2.023 (0.21)	2.038 (1.809)
CI coverage for μ	Corrected	0.861	0.823	0.817
	Uncorrected	0.861	0.371	0.292
CI coverage for σ	Corrected	0.819	0.93	0.913
	Uncorrected	0.819	0.332	0.266

Table 11.2: Results for different values of ϵ (Gradient descent). Sample size is 1000, $K = 50$.

		$\lambda = 1$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	5 (0.08)	4.927 (0.727)	4.76 (5.962)
	σ: Mean (Std. Error)	1.975 (0.076)	2.107 (1.265)	2.513 (7.652)
CI coverage for μ	Corrected	0.883	0.965	0.934
	Uncorrected	0.883	0.288	0.208
CI coverage for σ	Corrected	0.739	0.918	0.894
	Uncorrected	0.739	0.309	0.235

Table 11.3: Results for different values of epsilon (Newton-Raphson). Sample size is 1000, $K = 5$.

	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	5.001 (0.063)	5.241 (0.061)	5.476 (0.059)	5.952 (0.056)	6.422 (0.052)
PMHDE $\epsilon = 2$ (Std. Error)	5 (0.08)	5.174 (0.085)	5.309 (0.087)	5.555 (0.091)	5.778 (0.096)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.952 (0.326)	5.119 (0.333)	5.252 (0.341)	5.472 (0.38)	5.646 (0.42)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.942 (13.391)	4.905 (5.895)	5.054 (2.78)	5.349 (4.023)	5.169 (15.035)

Table 11.4: Contamination results, Newton-Raphson, sample size is 1000, $\mu = 5$

11.2 Estimation and coverage rate for $(\lambda, \epsilon)-$PDP

In this section, we provide PMHDE and coverage rates for PGD and PNR algorithms for different values of λ.

		$\lambda = -0.1$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	4.991 (0.083)	4.992 (0.212)	4.979 (0.454)
	σ: Mean (Std. Error)	1.984 (0.058)	2.001 (0.152)	2.045 (0.291)
CI coverage for μ	Corrected	0.861	0.836	0.82
	Uncorrected	0.861	0.468	0.33
CI coverage for σ	Corrected	0.819	0.931	0.927
	Uncorrected	0.819	0.428	0.296

Table 11.5: Results for different values of ϵ (Gradient descent). Sample size is 1000, $K = 50$.

		$\lambda = -0.1$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	5 (0.08)	4.955 (0.348)	4.827 (3.731)
	σ: Mean (Std. Error)	1.975 (0.076)	1.992 (0.353)	2.223 (1.554)
CI coverage for μ	Corrected	0.883	0.977	0.948
	Uncorrected	0.883	0.382	0.234
CI coverage for σ	Corrected	0.739	0.917	0.902
	Uncorrected	0.739	0.41	0.264

Table 11.6: Results for different values of ϵ (Newton). Sample size is 1000, $K = 5$.

		$\lambda = 0.5$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	4.991 (0.083)	4.991 (0.256)	4.976 (0.549)
	σ: Mean (Std. Error)	1.984 (0.058)	2.013 (0.18)	2.056 (0.322)
CI coverage for μ	Corrected	0.861	0.826	0.817
	Uncorrected	0.861	0.396	0.306
CI coverage for σ	Corrected	0.819	0.931	0.922
	Uncorrected	0.819	0.379	0.277

Table 11.7: Results for different values of ϵ (Gradient descent). Sample size is 1000, $K = 50$.

		$\lambda = 0.5$, ϵ		
		Non-private	1.20	0.40
Estimator	μ: Mean (Std. Error)	5 (0.08)	4.942 (0.483)	4.808 (2.064)
	σ: Mean (Std. Error)	1.975 (0.076)	2.03 (0.523)	2.297 (2.313)
CI coverage for μ	Corrected	0.883	0.972	0.94
	Uncorrected	0.883	0.322	0.222
CI coverage for σ	Corrected	0.739	0.915	0.899
	Uncorrected	0.739	0.355	0.254

Table 11.8: Results for different values of ϵ (Newton). Sample size is 1000, $K = 5$.

11.3 Additional results for HDP and robustness evaluation

In this section, we provide additional results for PMHDE for sample sizes 200, 300, and 500 for both PGD and PNR algorithms. As explained in Section 4 of the main paper, when n and ϵ are both small, the algorithms can produce aberrant values, reducing their usefulness. For this reason, we use only estimates within the lower 0.7% and upper 99.5% percentiles of a Gaussian distribution with non-private $\hat{\mu}_n$ and $\hat{\sigma}_n$. All the Tables in this section are based on such a thresholding strategy. Since the confidence intervals are unaffected by thresholding, we retain all the simulation experiments for constructing the confidence intervals.

Tables 11.9 and 11.10 provide the estimators and the coverage rates for sample size 200, while Tables 11.11 and 11.12 provide the behavior of PMHDE under contamination for the sample size 200. The corresponding Tables for sample size 300 and 500 are given in Tables 11.13, 11.14, 11.15, 11.16, 11.17, 11.18, 11.19, 11.20 respectively.

Sample size 200:

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.992 (0.153)	4.978 (0.538)	4.844 (1.22)
	σ: Mean (Std. Error)	1.952 (0.104)	2.036 (0.588)	1.744 (6.164)
CI coverage for μ	Corrected	0.921	0.839	0.626
	Uncorrected	0.921	0.51	0.27
CI coverage for σ	Corrected	0.846	0.921	0.6
	Uncorrected	0.846	0.484	0.216

Table 11.9: Results for different values of ϵ (Gradient descent). Sample size is 200, $K = 50$.

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.993 (0.148)	4.819 (1.242)	4.703 (1.972)
	σ: Mean (Std. Error)	1.93 (0.139)	2.465 (2.452)	3.118 (3.792)
CI coverage for μ	Corrected	0.929	0.925	0.888
	Uncorrected	0.929	0.343	0.157
CI coverage for σ	Corrected	0.771	0.879	0.829
	Uncorrected	0.771	0.359	0.169

Table 11.10: Results for different values of ϵ (Newton). Sample size is 200, $K = 5$.

	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	5 (0.142)	5.238 (0.138)	5.483 (0.134)	5.948 (0.127)	6.418 (0.119)
PMHDE $\epsilon = 2$ (Std. Error)	4.99 (0.153)	5.169 (0.155)	5.301 (0.158)	5.562 (0.165)	5.783 (0.172)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.977 (0.521)	5.141 (0.534)	5.283 (0.54)	5.508 (0.548)	5.721 (0.576)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.856 (1.228)	5.024 (1.16)	5.13 (1.22)	5.313 (1.215)	5.463 (1.306)

Table 11.11: Contamination results, gradient descent, sample size is 200, $\mu = 5$.

	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	5 (0.142)	5.238 (0.138)	5.483 (0.134)	5.948 (0.127)	6.418 (0.119)
PMHDE $\epsilon = 2$ (Std. Error)	4.991 (0.148)	5.174 (0.15)	5.308 (0.151)	5.578 (0.155)	5.823 (0.156)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.809 (1.275)	4.974 (1.221)	5.021 (1.216)	5.17 (1.308)	5.304 (1.375)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.744 (1.964)	4.815 (2.071)	4.866 (2.116)	4.954 (2.21)	5.002 (2.206)

Table 11.12: Contamination results, Newton, sample size is 200, $\mu = 5$.

Sample size 300:

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.989 (0.128)	4.979 (0.405)	4.926 (0.866)
	σ: Mean (Std. Error)	1.962 (0.086)	2.023 (0.338)	1.952 (1.971)
CI coverage for μ	Corrected	0.918	0.837	0.733
	Uncorrected	0.918	0.493	0.317
CI coverage for σ	Corrected	0.85	0.944	0.763
	Uncorrected	0.85	0.477	0.29

Table 11.13: Results for different values of ϵ (Gradient descent). Sample size is 300, $K = 50$.

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.993 (0.123)	4.863 (0.934)	4.697 (1.722)
	σ: Mean (Std. Error)	1.946 (0.113)	2.233 (1.442)	2.785 (6.688)
CI coverage for μ	Corrected	0.929	0.943	0.897
	Uncorrected	0.929	0.361	0.182
CI coverage for σ	Corrected	0.777	0.9	0.836
	Uncorrected	0.777	0.377	0.204

Table 11.14: Results for different values of ϵ (Newton). The Sample size is 300, $K = 5$.

	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (std.error)	4.999 (0.115)	5.238 (0.113)	5.474 (0.11)	5.938 (0.103)	6.433 (0.096)
PMHDE $\epsilon = 2$ (Std. Error)	4.989 (0.128)	5.164 (0.132)	5.309 (0.135)	5.542 (0.14)	5.756 (0.149)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.992 (0.408)	5.161 (0.398)	5.291 (0.408)	5.514 (0.416)	5.717 (0.434)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.925 (0.858)	5.095 (0.868)	5.241 (0.844)	5.396 (0.905)	5.562 (0.932)

Table 11.15: Contamination results, gradient descent, sample size is 300, $\mu = 5$.

	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	4.999 (0.115)	5.238 (0.113)	5.474 (0.11)	5.938 (0.103)	6.433 (0.096)
PMHDE $\epsilon = 2$ (Std. Error)	4.993 (0.124)	5.173 (0.127)	5.32 (0.129)	5.562 (0.132)	5.806 (0.135)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.856 (0.947)	5.009 (0.922)	5.114 (0.882)	5.296 (0.991)	5.42 (1.055)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.724 (1.724)	4.863 (1.729)	4.904 (1.782)	5.023 (1.849)	5.063 (1.894)

Table 11.16: Contamination results, Newton, sample size is 300, $\mu = 5$.

Sample size 500:

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.989 (0.106)	4.986 (0.296)	4.951 (0.559)
	σ: Mean (Std. Error)	1.973 (0.072)	2.016 (0.213)	2.058 (0.799)
CI coverage for μ	Corrected	0.892	0.842	0.798
	Uncorrected	0.892	0.489	0.326
CI coverage for σ	Corrected	0.848	0.941	0.888
	Uncorrected	0.848	0.44	0.301

Table 11.17: Results for different values of ϵ (gradient descent). Sample size is 500, $K = 50$.

		ϵ		
		2.00	0.60	0.20
Estimator	μ: Mean (Std. Error)	4.995 (0.102)	4.929 (0.631)	4.785 (1.322)
	σ: Mean (Std. Error)	1.959 (0.094)	2.089 (1.514)	2.554 (2.009)
CI coverage for μ	Corrected	0.904	0.96	0.916
	Uncorrected	0.904	0.371	0.213
CI coverage for σ	Corrected	0.767	0.908	0.874
	Uncorrected	0.767	0.391	0.243

Table 11.18: Results for different values of ϵ. Sample size is 500 (Newton), $K = 5$.

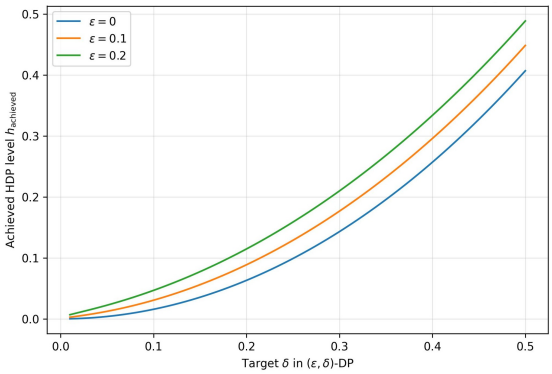
	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	4.998 (0.09)	5.24 (0.087)	5.473 (0.086)	5.946 (0.08)	6.424 (0.076)
PMHDE $\epsilon = 2$ (Std. Error)	4.989 (0.107)	5.157 (0.109)	5.293 (0.115)	5.526 (0.119)	5.738 (0.126)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.986 (0.293)	5.157 (0.294)	5.291 (0.299)	5.514 (0.302)	5.719 (0.312)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.987 (0.556)	5.146 (0.553)	5.256 (0.571)	5.464 (0.583)	5.65 (0.636)

Table 11.19: Contamination results, gradient descent, sample size is 500, $\mu = 5$.

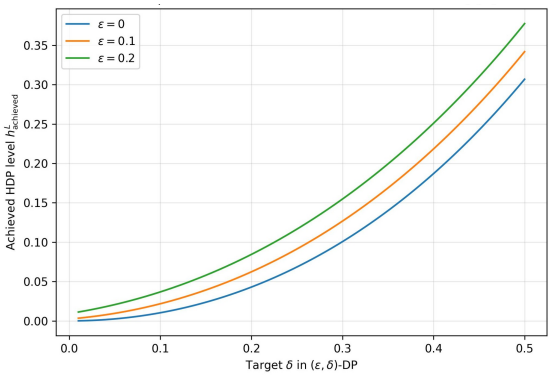
	Contamination percentage α				
	0%	5%	10%	20%	30%
MLE (Std. Error)	4.998 (0.09)	5.24 (0.087)	5.473 (0.086)	5.946 (0.08)	6.424 (0.076)
PMHDE $\epsilon = 2$ (Std. Error)	4.995 (0.103)	5.169 (0.105)	5.309 (0.11)	5.553 (0.113)	5.793 (0.117)
PMHDE $\epsilon = 0.6$ (Std. Error)	4.915 (0.631)	5.069 (0.606)	5.2 (0.629)	5.383 (0.659)	5.546 (0.716)
PMHDE $\epsilon = 0.2$ (Std. Error)	4.777 (1.316)	4.923 (1.306)	4.995 (1.384)	5.151 (1.386)	5.259 (1.419)

Table 11.20: Contamination results, Newton, sample size is 500, $\mu = 5$.

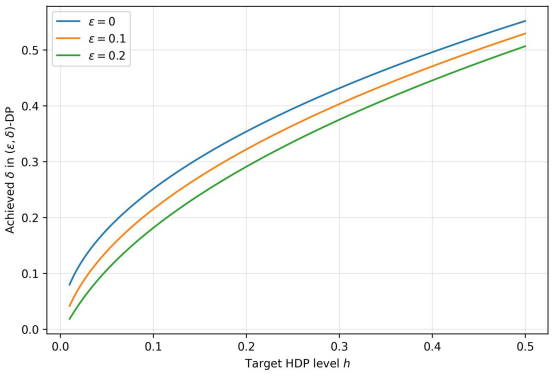
11.4 Forward and reverse privacy calibrations



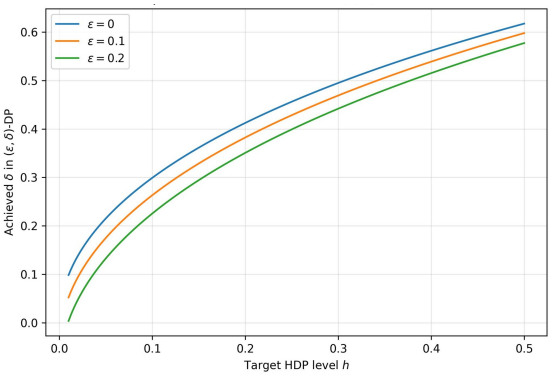
(a) Gaussian forward: attained h vs. target δ



(b) Laplace forward: attained h vs. target δ



(c) Gaussian reverse: attained δ vs. target h



(d) Laplace reverse: attained δ vs. target h

Figure 11.4: Exact forward and reverse privacy calibrations for the additive Gaussian and scalar Laplace families, using the noise level calibrated to the target guarantee. Forward panels show the achieved Hellinger level at target (ϵ, δ)-DP; reverse panels show the achieved δ for target h-HDP at a fixed ϵ. The four supplied numerical curves are unchanged.

12 Uniform rare-event bounds and large-deviation consequences

We derive the consequences of the Hellinger constraint for sequences of neighboring output laws. These statements concern the actual output probabilities. They are not applications of an empirical-divergence large-deviation theorem. The partition viewpoint is related to Beirlant et al. (2001) and Biau and Györfi (2005), whose sampling experiments involve empirical distributions on partitions.

For each n, let M_n satisfy h_n-HDP, with $0 \le h_n \le 2$, and let $\mathcal{I}_n$ be a nonempty collection of triples $\iota = (D, D', A)$ consisting of an ordered neighboring pair and a measurable output event. Write

$$p_{n,\iota} = P_{n,D}(A), \qquad q_{n,\iota} = P_{n,D'}(A).$$

The event classes and neighboring pairs may vary with n. No restriction on the cardinality of $\mathcal{I}_n$ is needed.

Proposition 12.1 (Uniform probability comparison) *For every n and $\iota \in \mathcal{I}_n$,*

$$(\sqrt{q_{n,\iota}} - \sqrt{h_n})_+^2 \le p_{n,\iota} \le \min\{1, (\sqrt{q_{n,\iota}} + \sqrt{h_n})^2\}. \tag{44}$$

If $\inf_{\iota\in\mathcal{I}_n} q_{n,\iota} \ge a_n > 0$, then

$$\sup_{\iota\in\mathcal{I}_n} \left| \frac{p_{n,\iota}}{q_{n,\iota}} - 1 \right| \le 2\sqrt{\frac{h_n}{a_n}} + \frac{h_n}{a_n}. \tag{45}$$

In particular, $h_n/a_n \to 0$ implies uniform convergence of these probability ratios to one.

Proof Apply the binary-partition inequality (4) to each triple and drop the complementary-event term. This yields $|\sqrt{p_{n,\iota}} - \sqrt{q_{n,\iota}}| \le \sqrt{h_n}$ and proves (44). Multiplying by $\sqrt{p_{n,\iota}} + \sqrt{q_{n,\iota}}$ gives

$$|p_{n,\iota} - q_{n,\iota}| \le 2\sqrt{h_n q_{n,\iota}} + h_n.$$

Division by $q_{n,\iota}$ and the assumed lower bound prove (45). These are simultaneous inequalities between the output laws; no union bound over the event class is involved. ∎

Corollary 12.2 (Exponential bounds) *Let $v_n \to \infty$ and $a, b > 0$. If $\sup_{\iota\in\mathcal{I}_n} q_{n,\iota} \le e^{-v_n a}$ and $h_n \le e^{-v_n b}$, then*

$$\sup_{\iota\in\mathcal{I}_n} p_{n,\iota} \le 4e^{-v_n \min\{a,b\}}. \tag{46}$$

More generally, if the corresponding logarithmic upper limits are at most $-a$ and $-b$, then

$$\limsup_{n\to\infty} \frac{1}{v_n} \log \sup_{\iota\in\mathcal{I}_n} p_{n,\iota} \le -\min\{a, b\},$$

where $\log 0 = -\infty$. If instead $\inf_{\iota\in\mathcal{I}_n} q_{n,\iota} \ge e^{-v_n a}$ and $h_n \le e^{-v_n b}$ with $b > a$, then

$$\sup_{\iota\in\mathcal{I}_n} \left| \frac{p_{n,\iota}}{q_{n,\iota}} - 1 \right| \le 2e^{-v_n(b-a)/2} + e^{-v_n(b-a)}. \tag{47}$$

Proof By (44), the first supremum is bounded by $(e^{-v_n a/2} + e^{-v_n b/2})^2$, which is at most the right side of (46). For logarithmic upper limits, apply this argument with $a-\eta, b-\eta$ for $0 < \eta < \min(a,b)$ and then let $\eta \downarrow 0$. The last assertion follows by substituting $a_n = e^{-v_n a}$ into (45). ■

Proposition 12.3 (Transfer of relative approximations) *Suppose positive reference quantities $r_{n,\iota}$ satisfy*

$$\eta_n := \sup_{\iota \in \mathcal{I}_n} \left| \frac{q_{n,\iota}}{r_{n,\iota}} - 1 \right| \longrightarrow 0, \qquad r_{n,*} := \inf_{\iota \in \mathcal{I}_n} r_{n,\iota} > 0.$$

Then

$$\sup_{\iota \in \mathcal{I}_n} \left| \frac{p_{n,\iota}}{r_{n,\iota}} - 1 \right| \leq \eta_n + 2\sqrt{\frac{(1+\eta_n)h_n}{r_{n,*}}} + \frac{h_n}{r_{n,*}}. \tag{48}$$

Hence $h_n/r_{n,*} \to 0$ transfers the reference approximation uniformly, including any prefactor present in $r_{n,\iota}$.

Proof Use $|p_{n,\iota} - q_{n,\iota}| \leq 2\sqrt{h_n q_{n,\iota}} + h_n$, divide by $r_{n,\iota}$, and apply $q_{n,\iota}/r_{n,\iota} \leq 1 + \eta_n$. The triangle inequality proves (48). ■

For example, suppose $r_{n,\iota} = C_{n,\iota} v_n^{-\kappa_{n,\iota}} e^{-v_n I_{n,\iota}}$, where $C_{n,\iota} \geq c > 0$, $0 \leq \kappa_{n,\iota} \leq K$, and $I_{n,\iota} \leq I_* < b$. If $h_n \leq e^{-v_n b}$, then, for $v_n \geq 1$,

$$\frac{h_n}{r_{n,*}} \leq c^{-1} v_n^K e^{-v_n(b - I_*)} \longrightarrow 0.$$

Thus a separately established sharp reference approximation transfers to the neighboring output laws. This conclusion does not assume that the reference probabilities arise from an empirical partition statistic.

Remark 12.4 (Sharpness of the exponential scale) *Define $\mathcal{R}(h,q) = \sup\{P_1(A) : D_{HD}(P_1,P_2) \leq h,\ P_2(A) \leq q\}$, where the supremum is over probability laws and events. Since U_h is nondecreasing, Proposition 16 of the main paper and its binary attaining examples give $\mathcal{R}(h,q) = U_h(q)$. In particular,*

$$\max\{q, h - h^2/4\} \leq \mathcal{R}(h,q) \leq (\sqrt{q} + \sqrt{h})^2.$$

The lower bounds follow by taking $P_1 = P_2$ and by the distinguishing-event example, respectively. If $q_n, h_n \to 0$ and $-v_n^{-1} \log q_n \to a > 0$, $-v_n^{-1} \log h_n \to b > 0$, then

$$-\frac{1}{v_n} \log \mathcal{R}(h_n, q_n) \longrightarrow \min\{a,b\}.$$

Indeed, for $0 < h \leq 2$ the lower bound is at least $\max\{q, h/2\}$, while the upper bound is at most $4\max\{q,h\}$. For fixed $h > 0$, instead, $\lim_{q \downarrow 0} \mathcal{R}(h,q) = h - h^2/4$. Thus relative preservation of arbitrarily rare events requires a corresponding restriction on the budget scale.

13 An example of crossing privacy profiles

The following two neighboring-output experiments have the same attained Hellinger distance but different orderings at different privacy-profile thresholds. Consider

$$P_1^R = (3/4, 1/4, 0), \qquad P_2^R = (1/4, 3/4, 0).$$

Their likelihood ratio takes the values 3 and $1/3$ on the common support, so this mechanism is pure $\log 3$-DP. Direct calculation gives $D_{HD}(P_1^R, P_2^R) = 2 - \sqrt{3}$. Now set $r = 1 - \sqrt{3}/2$ and take

$$P_1^H = (1 - r, r, 0), \qquad P_2^H = (1 - r, 0, r).$$

Here $D_{HD}(P_1^H, P_2^H) = 2r = 2 - \sqrt{3}$ as well.

For $t \geq 0$, summing the positive cell contributions in (5) gives the two directional profiles, which coincide within each experiment:

$$\delta_R(t) = \frac{(3 - e^t)_+}{4}, \qquad \delta_H(t) = r. \tag{49}$$

Indeed, in the first experiment only the cell with likelihood ratio 3 can have a positive contribution, and it contributes $(3-e^t)_+/4$. In the second, the common cell contributes zero, while the distinguishing cell contributes r. Reversing the laws gives the same quantities.

Thus $\delta_H(0) = r < 1/2 = \delta_R(0)$, whereas $\delta_R(t) = 0 < r = \delta_H(t)$ for $t \geq \log 3$. The crossing occurs at $t = \log(2\sqrt{3} - 1)$. Consequently, equality of attained Hellinger distances does not impose a pointwise ordering of privacy profiles. The uniform likelihood-ratio cutoff of the first experiment describes one aspect of its protection, not an ordering at every disclosure threshold.

Both experiments can be realized as mechanisms on a binary database domain. They also admit versions with Lebesgue densities by replacing the three points by uniform laws on three disjoint unit intervals, using the same weights. The example compares these specified mechanisms and does not assert a universal ordering of privacy classes.

14 Influence functions and breakdown of PMHDE

Write $T(g) = \boldsymbol{\theta}_g$, let $\hat{\boldsymbol{\theta}}_n$ denote MHDE, and let $\widetilde{\boldsymbol{\theta}}_{A,n} = \hat{\boldsymbol{\theta}}_n^{(K_{A,n})}$ be its private version, where $A \in \{\mathrm{PGD}, \mathrm{PNR}\}$. The estimators are computed from the same sample and use the same density estimate, bandwidth, truncation, and criterion. Set $R_{A,n} = \widetilde{\boldsymbol{\theta}}_{A,n} - \hat{\boldsymbol{\theta}}_n$. The probability $\mathbf{P}_g$ includes sampling from g and the algorithmic randomization.

The inputs are the asymptotic-equivalence and consistency conclusions of Theorem 42, and the approximation bounds in Theorems 36 and 39. Each is applied under its hypotheses at the stated sampling law and with the stipulated initialization and tuning. In particular, whenever Theorem 42 applies under g,

$$\sqrt{n} R_{A,n} \longrightarrow 0, \qquad \widetilde{\boldsymbol{\theta}}_{A,n} \longrightarrow T(g) \quad \text{in probability.} \tag{50}$$

The results below are consequences of these inputs. Uniform statements and full-parameter-space breakdown specify their additional domain conditions separately.

14.1 Influence and the finite-α response

Proposition 14.1 (Asymptotic linear influence of PMHDE) *Assume the hypotheses of Theorem 42 under g, and suppose the ordinary MHDE has the representation*

$$\sqrt{n}\{\hat{\boldsymbol{\theta}}_n - T(g)\} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_g(X_i) + r_{n,g}, \qquad r_{n,g} \longrightarrow 0 \quad \textit{in probability,}$$

where $\mathbf{E}_g\psi_g(X) = 0$ and $\mathbf{E}_g\|\psi_g(X)\|_2^2 < \infty$. Then PGD and PNR PMHDE have the same asymptotic linear influence function ψ_g.

Proof By Theorem 42(i), $\sqrt{n}R_{A,n} \to 0$ in probability. Hence

$$\sqrt{n}\{\widetilde{\boldsymbol{\theta}}_{A,n} - T(g)\} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\psi_g(X_i) + r_{n,g} + \sqrt{n}R_{A,n}.$$

The last two terms tend to zero, proving the assertion. ■

Let $g_0 = f_{\boldsymbol{\theta}_0}$ and $g_{\alpha,z} = (1-\alpha)g_0 + \alpha q_z$, where q_z is a density. Following the finite-contamination formulation of Beran (1977), write

$$\mathrm{IF}_\alpha(z; T, g_0) = \frac{T(g_{\alpha,z}) - T(g_0)}{\alpha}, \qquad 0 < \alpha < 1. \tag{51}$$

For each sampling law on which the private estimator has a deterministic probability limit, denote this limit by $T_A(g)$. Define $\mathrm{IF}_\alpha(z; T_A, g_0)$ by the same expression with T_A in place of T.

For a sample-level version, let superscripts (α, z) and 0 denote samples from $g_{\alpha,z}$ and g_0, respectively. Use any coupling with these prescribed marginal sampling and randomization laws, and put

$$\begin{aligned}\widehat{\mathrm{IF}}_{\alpha,A,n}(z) &= \frac{\widetilde{\boldsymbol{\theta}}^{\alpha,z}_{A,n} - \widetilde{\boldsymbol{\theta}}^{0}_{A,n}}{\alpha},\\ \widehat{\mathrm{IF}}_{\alpha,H,n}(z) &= \frac{\hat{\boldsymbol{\theta}}^{\alpha,z}_{n} - \hat{\boldsymbol{\theta}}^{0}_{n}}{\alpha}.\end{aligned} \tag{52}$$

Here H labels the ordinary MHDE, not another algorithm.

Proposition 14.2 (α-influence function of PMHDE) *Fix $0 < \alpha < 1$ and z. Assume the hypotheses of Theorem 42 at both g_0 and $g_{\alpha,z}$, with the same Hellinger minimizer selection used for the private and non-private procedures. Then, for $A \in \{\mathrm{PGD}, \mathrm{PNR}\}$,*

$$\mathrm{IF}_\alpha(z; T_A, g_0) = \mathrm{IF}_\alpha(z; T, g_0), \qquad \widehat{\mathrm{IF}}_{\alpha,A,n}(z) \longrightarrow \mathrm{IF}_\alpha(z; T, g_0) \quad \textit{in probability.} \tag{53}$$

Moreover,

$$\sqrt{n}\,\big\|\widehat{\mathrm{IF}}_{\alpha,A,n}(z) - \widehat{\mathrm{IF}}_{\alpha,H,n}(z)\big\|_2 \longrightarrow 0 \quad \textit{in probability.} \tag{54}$$

For any set of locations on which the hypotheses hold pointwise, the two limiting α-influence curves therefore have identical suprema of their norms, allowing the value $+\infty$.

Proof Apply Theorem 42(ii) at the two sampling laws. It gives $T_A(g_0) = T(g_0)$ and $T_A(g_{\alpha,z}) = T(g_{\alpha,z})$, which prove the functional identity. The sample response differs from that common limit in norm by at most

$$\frac{\|\widetilde{\boldsymbol{\theta}}^{\alpha,z}_{A,n} - T(g_{\alpha,z})\|_2 + \|\widetilde{\boldsymbol{\theta}}^{0}_{A,n} - T(g_0)\|_2}{\alpha}.$$

Both terms in the numerator tend to zero in probability, proving the sample convergence for fixed α, z. Finally, exactly,

$$\left\|\widehat{\mathrm{IF}}_{\alpha,A,n}(z) - \widehat{\mathrm{IF}}_{\alpha,H,n}(z)\right\|_2 \le \frac{\|R^{\alpha,z}_{A,n}\|_2 + \|R^{0}_{A,n}\|_2}{\alpha}. \tag{55}$$

Multiply by $\sqrt{n}$ and apply Theorem 42(i) at both laws. A union bound suffices; independence is not required. The statement about suprema follows from equality of the limiting functions at every location in the specified set. ∎

The convergence statements above refer to the random sample response, not its expectation. The functional identity requires only the consistency conclusion of the efficiency theorem; the root-n comparison uses its asymptotic-equivalence conclusion. If the clean and contaminated samples and the two algorithmic randomizations are independent, Theorem 42(iii) also gives, for fixed α, z,

$$\sqrt{n}\left\{\widehat{\mathrm{IF}}_{\alpha,A,n}(z) - \mathrm{IF}_\alpha(z; T, g_0)\right\} \implies N\left(0, \frac{\Sigma_{g_{\alpha,z}} + \Sigma_{g_0}}{\alpha^2}\right). \tag{56}$$

Indeed, apply the two central limit statements jointly using independence and take their difference divided by α. Equation (54) transfers the same limit to the ordinary sample response. Equality of the limiting curves does not assert uniform convergence of the random curves over all z.

The local functional derivative. For the manuscript's criterion $2HD^2(f_{\boldsymbol{\theta}}, g)$, the population estimating function is $\Psi(\boldsymbol{\theta}, g) = -2\int \sqrt{g f_{\boldsymbol{\theta}}}\, \mathbf{u}_{\boldsymbol{\theta}}$. Consider a dominated density path $g_t = g_0 + ta$ with $\int a = 0$, and a differentiable local minimizer $T(g_t)$. Suppose the required derivatives may be taken under the integral, $I(\boldsymbol{\theta}_0)$ is nonsingular, and $\int \|\mathbf{u}_{\boldsymbol{\theta}_0}\||a| < \infty$. The gradient and Hessian formulas in the manuscript give

$$\partial_{\boldsymbol{\theta}}\Psi(\boldsymbol{\theta}_0, g_0) = I(\boldsymbol{\theta}_0), \qquad \partial_g \Psi(\boldsymbol{\theta}_0, g_0)[a] = -\int \mathbf{u}_{\boldsymbol{\theta}_0}(x) a(x)\, dx.$$

Differentiation of $\Psi\{T(g_t), g_t\} = 0$ yields

$$\left.\frac{d}{dt} T(g_t)\right|_{t=0} = I(\boldsymbol{\theta}_0)^{-1} \int \mathbf{u}_{\boldsymbol{\theta}_0}(x) a(x)\, dx. \tag{57}$$

Thus the influence at the model is $\psi_{g_0} = I(\boldsymbol{\theta}_0)^{-1}\mathbf{u}_{\boldsymbol{\theta}_0}$ on these regular paths. For $a = q_z - g_0$, the derivative is $\int \psi_{g_0}(x) q_z(x)\, dx$. The asymptotic linear representation also includes the density-estimation remainder, as in Proposition 14.1. The finite-α identity in Proposition 14.2 requires no interchange between $n \to \infty$ and $\alpha \downarrow 0$.

14.2 The limiting-functional breakdown point

Let $\Theta_* \subseteq \mathbb{R}^m$ be the full open parameter space, and let $\mathcal{Q}$ be a nonempty class of contaminating densities. For $0 < \alpha < 1$, set $\mathcal{G}_\alpha = \{(1-\alpha)g_0 + \alpha q : q \in \mathcal{Q}\}$. For a functional S on these laws, define

$$b^*(S; \mathcal{Q}, g_0) = \inf\{\alpha \in (0,1) : \{S(g) : g \in \mathcal{G}_\alpha\} \text{ is not contained in a compact subset of } \Theta_*\}, \tag{58}$$

with value one if the set in braces is empty. Leaving compact subsets includes location escape, scale explosion, and scale collapse for $\Theta_* = \mathbb{R} \times (0, \infty)$.

For the private sequence, the limiting-functional breakdown point is $b^*(T_A; \mathcal{Q}, g_0)$, where $T_A(g)$ is its probability limit at each fixed sampling law. Thus this definition takes $n \to \infty$ first. For these breakdown statements the estimators and targets are considered on Θ_*. The efficiency conclusion must apply at each specified target through its local assumptions; a fixed compact constraint is not used to prevent escape.

Proposition 14.3 (Limiting-functional breakdown of PMHDE) *Suppose MHDE is consistent for $T(g)$ and the consistency conclusion of Theorem 42(ii) applies to PGD or PNR PMHDE at every fixed $g \in \{g_0\} \cup \bigcup_{0<\alpha<1} \mathcal{G}_\alpha$, with the same global minimizer selection on Θ_*. Then*

$$b^*(T_A; \mathcal{Q}, g_0) = b^*(T; \mathcal{Q}, g_0). \tag{59}$$

In particular, whenever the ordinary MHDE functional has breakdown point $1/2$ on this class, both private procedures have the same limiting-functional breakdown point $1/2$.

Proof Theorem 42(ii) gives $T_A(g) = T(g)$ at every stated sampling law. Fix α. If the ordinary targets lie in a compact set $K \subset \Theta_*$, all the private probability limits lie in that same K. Conversely, if no compact subset contains the ordinary targets, then for every compact K there is $g \in \mathcal{G}_\alpha$ with $T(g) \notin K$; its private probability limit $T_A(g)$ is also outside K. The sets of contamination fractions in (58) are therefore identical, which proves (59). ■

Lemma 14.4 (A classical Hellinger benchmark) *Let $g_0 = f_{\boldsymbol{\theta}_0}$ and $\rho(f, g) = \int \sqrt{fg}$. Assume global maximizers $T(g) \in \arg\max_{\boldsymbol{\theta} \in \Theta_*} \rho(f_{\boldsymbol{\theta}}, g)$ exist for the mixtures in the specified class, with a fixed selection. Suppose that, for each $c > 0$, $\{\boldsymbol{\theta} : \rho(f_{\boldsymbol{\theta}}, g_0) \geq c\}$ is contained in a compact subset of Θ_*, and that $\mathcal{Q}$ contains a sequence $q_j = f_{\boldsymbol{\theta}_j}$ satisfying*

$$\sup_{\boldsymbol{\theta} \in K} \rho(f_{\boldsymbol{\theta}}, q_j) \longrightarrow 0 \quad \text{for every compact } K \subset \Theta_*. \tag{60}$$

Then $b^*(T; \mathcal{Q}, g_0) = 1/2$.

Proof For $g_\alpha = (1-\alpha)g_0 + \alpha q$, comparison with $f_{\boldsymbol{\theta}_0}$ and $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ give

$$\sqrt{1-\alpha} \leq \rho(f_{T(g_\alpha)}, g_\alpha) \leq \sqrt{1-\alpha}\rho(f_{T(g_\alpha)}, g_0) + \sqrt{\alpha}.$$

For $\alpha < 1/2$, it follows that

$$\rho(f_{T(g_\alpha)}, g_0) \geq 1 - \sqrt{\frac{\alpha}{1-\alpha}} > 0. \tag{61}$$

All the targets therefore lie in a compact set independent of q. For $\alpha > 1/2$, take $q = q_j$. Fitting q_j itself gives affinity at least $\sqrt{\alpha}$. On any fixed compact K,

$$\sup_{\boldsymbol{\theta}\in K} \rho(f_{\boldsymbol{\theta}}, g_{\alpha,j}) \le \sqrt{1-\alpha} + \sqrt{\alpha} \sup_{\boldsymbol{\theta}\in K} \rho(f_{\boldsymbol{\theta}}, q_j).$$

For sufficiently large j, the right side is strictly less than $\sqrt{\alpha}$. Every global maximizer must then lie outside K. Thus all fractions below $1/2$ preserve compact containment and every fraction above $1/2$ admits an escape sequence. Taking the infimum in (58) proves the result. ■

This is a population affinity argument of the classical minimum-distance type (Beran, 1977; Basu et al., 2011); its sufficient conditions are stated here to identify the benchmark used in the private transfer.

Corollary 14.5 (Half breakdown for the limiting PMHDE functional) *Suppose the conditions of Lemma 14.4 hold, and the consistency conditions of Proposition 14.3 hold for both PGD and PNR. Then*

$$b^*(T_{\mathrm{PGD}}; \mathcal{Q}, g_0) = b^*(T_{\mathrm{PNR}}; \mathcal{Q}, g_0) = b^*(T; \mathcal{Q}, g_0) = \tfrac{1}{2}.$$

For every fixed $0 < \alpha < 1/2$, the limiting α-influence functions of the two private procedures are uniformly bounded over $q_z \in \mathcal{Q}$ and equal the MHDE curve.

Proof The breakdown identity follows by applying Proposition 14.3 to Lemma 14.4. For fixed $\alpha < 1/2$, (61) puts all $T(g_{\alpha,z})$ in one compact set. Their Euclidean distances from $T(g_0)$ are uniformly bounded, as are those distances divided by the fixed positive α. The identity $T_A(g) = T(g)$ from Proposition 14.3 identifies the same limiting curves for PMHDE. ■

14.3 Probability bounds and simultaneous breakdown transfer

For a fixed privacy budget, the approximation bounds of Theorems 36 and 39 have the form

$$\mathbf{P}_g\{\|R_{A,n}\|_2 > e_{A,n}(g;\xi)\} \le \xi, \qquad e_{A,n}(g;\xi) = C_A(g) n^{-1/p} \{K_{A,n} \log(K_{A,n}/\xi)\}^{1/2}, \quad (62)$$

for $0 < \xi < 1$ on the domains and sample sizes covered by those results. The endpoint $p = 2$ refers to the consistency-scale use of the weak sensitivity bound. Local constants, initialization, and sample-size cutoffs are retained when the sampling laws vary.

Lemma 14.6 (Finite-sample contamination comparison) *Put $B_n = \|\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0\|_2$ and $\widetilde{B}_{A,n} = \|\widetilde{\boldsymbol{\theta}}_{A,n} - \boldsymbol{\theta}_0\|_2$. Whenever (62) holds, let $e = e_{A,n}(g;\xi)$. For $t > e$,*

$$\mathbf{P}_g\{B_n > t + e\} - \xi \le \mathbf{P}_g\{\widetilde{B}_{A,n} > t\} \le \mathbf{P}_g\{B_n > t - e\} + \xi. \quad (63)$$

If the bound is applied at both $g_{\alpha,z}$ and g_0, then with probability at least $1 - 2\xi$,

$$\left\|\widehat{\mathrm{IF}}_{\alpha,A,n}(z) - \widehat{\mathrm{IF}}_{\alpha,H,n}(z)\right\|_2 \le \frac{e_{A,n}(g_{\alpha,z};\xi) + e_{A,n}(g_0;\xi)}{\alpha}. \quad (64)$$

Proof The reverse triangle inequality gives $|\widetilde{B}_{A,n} - B_n| \le \|R_{A,n}\|_2$. On $\{\|R_{A,n}\|_2 \le e\}$, $\{B_n > t + e\} \subseteq \{\widetilde{B}_{A,n} > t\} \subseteq \{B_n > t - e\}$. Adding the probability of the complementary event proves (63). Equation (55) and a union bound prove (64). ∎

If displacement is measured from a clean estimate instead of $\boldsymbol{\theta}_0$, put $V_{H,n} = \hat{\boldsymbol{\theta}}_n^{\alpha,z} - \hat{\boldsymbol{\theta}}_n^0$ and $V_{A,n} = \widetilde{\boldsymbol{\theta}}_{A,n}^{\alpha,z} - \widetilde{\boldsymbol{\theta}}_{A,n}^0$. Then $|\|V_{A,n}\|_2 - \|V_{H,n}\|_2| \le \|R_{A,n}^{\alpha,z}\|_2 + \|R_{A,n}^0\|_2$. Thus (63) holds with radius $e_{A,n}(g_{\alpha,z};\xi) + e_{A,n}(g_0;\xi)$ and failure probability 2ξ. This comparison does not require independent clean and contaminated samples.

For a simultaneous sample-size and contamination limit, use the compact exhaustion of the full open parameter space

$$\mathcal{K}_R = \left\{\boldsymbol{\theta} \in \Theta_* : \|\boldsymbol{\theta}\|_2 \le R, \quad \operatorname{dist}(\boldsymbol{\theta}, \mathbb{R}^m \setminus \Theta_*) \ge R^{-1}\right\}, \qquad R \ge 1.$$

The distance to an empty complement is infinite. These compact sets increase to Θ_* and contain every compact subset for large R. An estimator sequence S_n is asymptotically tight uniformly over $\mathcal{G}_\alpha$ when

$$\lim_{R\to\infty} \limsup_{n\to\infty} \sup_{g\in\mathcal{G}_\alpha} \mathbf{P}_g\{S_n \notin \mathcal{K}_R\} = 0. \tag{65}$$

Its simultaneous asymptotic breakdown threshold is the infimum of those $\alpha \in (0,1)$ for which this limit is positive, with value one if there are none. An escape witness is a sequence $g_{\alpha,n} \in \mathcal{G}_\alpha$ for which $\mathbf{P}_{g_{\alpha,n}}\{S_n \in K\} \to 0$ for every compact $K \subset \Theta_*$. A failed or inadmissible output is counted as outside every compact set; the approximation error is assigned value $+\infty$ on failures.

Proposition 14.7 (Simultaneous asymptotic breakdown transfer) *Fix a threshold $b \in (0,1)$. For every $\alpha < b$, suppose MHDE is asymptotically tight uniformly over $\mathcal{G}_\alpha$ and there are $r_{A,n,\alpha} \to 0$, $\xi_{A,n,\alpha} \to 0$ such that*

$$\sup_{g\in\mathcal{G}_\alpha} \mathbf{P}_g\{\|R_{A,n}\|_2 > r_{A,n,\alpha}\} \le \xi_{A,n,\alpha}. \tag{66}$$

For every $\alpha > b$, suppose there is an MHDE escape witness $g_{\alpha,n} \in \mathcal{G}_\alpha$ along which $R_{A,n} \to 0$ in probability. Then PMHDE has simultaneous asymptotic breakdown threshold b. In particular, the threshold is $1/2$ for PGD and PNR whenever these hypotheses hold with $b = 1/2$.

Proof Fix $R \ge 1$. If $\boldsymbol{\theta} \in \mathcal{K}_R$ and $\|\boldsymbol{\theta}' - \boldsymbol{\theta}\|_2 \le (2R)^{-1}$, then $\boldsymbol{\theta}' \in \mathcal{K}_{2R+1}$: its norm is at most $R + (2R)^{-1}$ and its distance from the parameter-space complement is at least $(2R)^{-1}$. For $\alpha < b$, the error radius is eventually at most $(2R)^{-1}$. Hence

$$\sup_{g\in\mathcal{G}_\alpha} \mathbf{P}_g\{\widetilde{\boldsymbol{\theta}}_{A,n} \notin \mathcal{K}_{2R+1}\} \le \sup_{g\in\mathcal{G}_\alpha} \mathbf{P}_g\{\hat{\boldsymbol{\theta}}_n \notin \mathcal{K}_R\} + \xi_{A,n,\alpha}.$$

Take $\limsup_n$, then $R \to \infty$, to prove tightness below b. For the escape witness above b, reverse the roles of the estimates:

$$\mathbf{P}_{g_{\alpha,n}}\{\widetilde{\boldsymbol{\theta}}_{A,n} \in \mathcal{K}_R\} \le \mathbf{P}_{g_{\alpha,n}}\{\hat{\boldsymbol{\theta}}_n \in \mathcal{K}_{2R+1}\} + \mathbf{P}_{g_{\alpha,n}}\{\|R_{A,n}\|_2 > (2R)^{-1}\}.$$

Both terms tend to zero. Every $\alpha > b$ therefore admits escape and the threshold equals b. No claim at $\alpha = b$ is needed. ∎

The same compact-set argument also transfers uniform tightness from bounded population targets whenever an ordinary estimation error bound holds uniformly over the contamination class. Along a population escape sequence it transfers escape if both the ordinary estimation error and the private approximation error vanish. This is the extra sampling step needed to pass from Corollary 14.5 to the simultaneous statement.

For location–scale parameters, a lower bound $\hat{\sigma}_n \geq c > 0$ and private error at most $c/2$ give $\widetilde{\sigma}_{A,n} \geq c/2$. The compact-set proof therefore includes scale collapse. For an exploding Euclidean displacement, the weaker requirement that the private error be negligible relative to that displacement suffices by (63).

14.4 PGD and PNR error radii

The following local calculations display the constants in the probability comparisons. They use the manuscript's Hessian bounds and perturbation structure without changing either algorithm. They are stated on a good-data event $\mathcal{E}_{A,n}(g)$ on which an interior MHDE root exists, the starting point is in the stated neighborhood, and the local derivative bounds hold. Write $\ell_{A,n}(g) = \mathbf{P}_g\{\mathcal{E}_{A,n}(g)^c\}$. If the original approximation theorem is used as an unconditional input, its failure probability already accounts for the required localization; no additional $\ell_{A,n}$ is then added.

Use a closed ball $\overline{B}_r(\hat{\boldsymbol{\theta}}_n) \subset \Theta_*$ on which $2\tau_1 I \preceq H_n(\boldsymbol{\theta}) \preceq 2\tau_2 I$, with $0 < \tau_1 \leq \tau_2 < \infty$. Constants and radii below may depend on g, but are deterministic bounds on the specified good-data event. Let $e_{0,n}$ be a deterministic upper bound for $\|\hat{\boldsymbol{\theta}}_n^{(0)} - \hat{\boldsymbol{\theta}}_n\|_2$ on this event. Let $c_u = \{-8\log(1-u/2)\}^{-1/2}$, $0 < u < 2$, as in the manuscript. For total privacy budget $0 < \epsilon < 2$ and K iterations, the existing allocation uses $\epsilon' = \epsilon/K$.

PGD. The Gaussian vector radius used in the PGD proof is

$$u_{G,n}(\xi) = \Delta_n c_{\epsilon'}\{4\sqrt{m} + 2\sqrt{2\log(K/\xi)}\}. \tag{67}$$

The corresponding Gaussian norm estimate and a union bound give $\max_{k<K} \|N_{n,k}\|_2 \leq u_{G,n}(\xi)$ with probability at least $1-\xi$. For $0 < \eta \leq (2\tau_2)^{-1}$, put $q_G = 1 - 2\eta\tau_1 \in [0,1)$. The integral Hessian formula along the line from $\hat{\boldsymbol{\theta}}_n$ to an iterate gives

$$\|\hat{\boldsymbol{\theta}}_n^{(k+1)} - \hat{\boldsymbol{\theta}}_n\|_2 \leq q_G\|\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n\|_2 + \eta u_{G,n}(\xi).$$

If $e_{0,n} \leq r$ and $u_{G,n}(\xi)/(2\tau_1) \leq r$, induction keeps the iterates in this ball and yields

$$e_{G,n}^{\mathrm{loc}}(\xi) = q_G^K e_{0,n} + \frac{u_{G,n}(\xi)}{2\tau_1}(1 - q_G^K). \tag{68}$$

The probability that the actual PGD error exceeds this radius is at most $\ell_{G,n}(g) + \xi$. This expression separates finite iteration and privacy perturbation.

PNR. For the unit-step local Newton phase used in the PNR proof, let

$$u_{N,n}(\xi) = \Delta_n c_{\epsilon'/2}\{4\sqrt{m} + 2\sqrt{2\log(2K/\xi)}\},$$
$$v_{N,n}(\xi) = \Delta_n^{(H)} c_{\epsilon'/2}\{2m\log(4Km/\xi)\}^{1/2}. \tag{69}$$

These are the gradient and symmetric-matrix Gaussian norm radii used in the supplementary PNR proof; the concentration tools are cited there as Rigollet and Hütter (2023); Tropp (2015). The independent entries of the upper triangle are reflected into the lower triangle. A union bound over the two noise sequences gives simultaneous bounds $\|N_{n,k}\|_2 \le u_{N,n}$, $\|W_{n,k}\|_2 \le v_{N,n}$ for all $k < K$, with probability at least $1 - \xi$. Suppose $v_{N,n} \le \tau_1$ and $\|\nabla L_n(\boldsymbol{\theta})\|_2 \le B_1$ on the local ball. The inverse identity $(H+W)^{-1} - H^{-1} = -(H+W)^{-1}WH^{-1}$ gives

$$\|(H+W)^{-1} - H^{-1}\|_2 \le \frac{v_{N,n}}{2\tau_1^2}.$$

Writing the noisy direction as the exact Newton direction plus an error, as in the manuscript's PNR decomposition, gives the bound

$$a_{N,n}(\xi) = \frac{u_{N,n}}{2\tau_1} + \frac{B_1 v_{N,n}}{2\tau_1^2} + \frac{u_{N,n}v_{N,n}}{2\tau_1^2}. \tag{70}$$

In particular, both the gradient and Hessian perturbations and their interaction remain in the calculation.

If H_n is Lipschitz with constant L_H on the local ball, set $c_N = L_H/(4\tau_1)$. Integration of the Hessian between an iterate and the MHDE root proves

$$E_{k+1} \le c_N E_k^2 + a_{N,n}(\xi), \qquad E_k = \|\hat{\boldsymbol{\theta}}_n^{(k)} - \hat{\boldsymbol{\theta}}_n\|_2. \tag{71}$$

Indeed, the exact Newton remainder has norm at most $\|H_n^{-1}\|(L_H/2)E_k^2 \le c_N E_k^2$. For a concrete sufficient local choice, suppose $c_N r \le 1/4$, $e_{0,n} \le r/2$, and $a_{N,n} \le r/8$. If $c_N > 0$, induction gives

$$e_{N,n}^{\mathrm{loc}}(\xi) = c_N^{-1}(c_N e_{0,n})^{2^K} + 2a_{N,n}(\xi). \tag{72}$$

To check the induction, write $x_k = c_N E_k$ and $b = c_N a_{N,n}$. Then $x_0 \le 1/8$, $b \le 1/32$, and $x_{k+1} \le x_k^2 + b$. If $x_k \le x_0^{2^k} + 2b$, putting $t = x_0^{2^k} \le 1/8$ yields $(t+2b)^2 + b \le t^2 + 2b$ because $4t + 4b \le 1$. Also the resulting radius is at most $e_{0,n} + 2a_{N,n} \le 3r/4$, so the iterates remain in the ball. If $c_N = 0$, one Newton step leaves only the direction error and $E_K \le a_{N,n}$ for $K \ge 1$. The failure probability is at most $\ell_{N,n}(g) + \xi$. For a fixed damped step $0 < \eta < 1$, the corresponding verified recursion is $E_{k+1} \le (1-\eta)E_k + \eta c_N E_k^2 + \eta a_{N,n}$; the displayed quadratic-iteration bound pertains to $\eta = 1$.

14.5 Varying contamination and privacy levels

Proposition 14.8 (Private response at a changing contamination fraction) *Let $\alpha_n > 0$, allow $z = z_n$, and suppose the two approximation bounds* (62) *hold under g_{α_n,z_n} and g_0 with failure probabilities at most ξ_n. Put*

$$d_{A,n} = \frac{e_{A,n}(g_{\alpha_n,z_n};\xi_n) + e_{A,n}(g_0;\xi_n)}{\alpha_n}. \tag{73}$$

Then

$$\mathbf{P}\Big\{\big\|\widehat{\mathrm{IF}}_{\alpha_n,A,n}(z_n)-\widehat{\mathrm{IF}}_{\alpha_n,H,n}(z_n)\big\|_2>d_{A,n}\Big\}\le 2\xi_n.$$

If $d_{A,n}\to 0$ and $\xi_n\to 0$, the private and non-private sample responses differ by a term tending to zero in probability.

Proof Apply (55) with $(\alpha,z)=(\alpha_n,z_n)$ and bound the two approximation errors by a union bound. ■

This result compares the sample responses at the stated contamination scale. At $\alpha_n = c/\sqrt{n}$, $c>0$, root-n negligible errors along the two sampling sequences suffice. For fixed positive α, consistency-scale errors suffice. In all cases the exact decomposition is

$$\widetilde{\boldsymbol{\theta}}_{A,n}-\boldsymbol{\theta}_0=\{T(g_{\alpha_n,z_n})-\boldsymbol{\theta}_0\}+\{\hat{\boldsymbol{\theta}}_n-T(g_{\alpha_n,z_n})\}+R_{A,n}. \tag{74}$$

Take $\xi_n=n^{-2}$ and a fixed privacy budget. On a family where the sensitivity and local constants have the required uniform bounds, choose $K_{G,n}$ between fixed positive multiples of $\log n$ and $K_{N,n}$ between fixed positive multiples of $\log\log n$ for large n, with the lower constants large enough that the initialization terms are no larger than the noise terms. The existing approximation rates, or the local bounds above, give

$$\begin{aligned} e_{G,n}(g;\xi_n)&\le C_G(g)n^{-1/p}\log n,\\ e_{N,n}(g;\xi_n)&\le C_N(g)n^{-1/p}\{\log n\,\log\log n\}^{1/2}. \end{aligned} \tag{75}$$

Both vanish even at $p=2$ when the stated constants stay bounded and localization failure tends to zero. Thus consistency-scale sensitivity can suffice for breakdown; the root-n statements require the stronger rate. Along moving contamination laws, the displayed constants and sample-size cutoffs must satisfy the stated bounds along that sequence.

The noise-scale dependence on a varying privacy budget can be retained directly. Since $-\log(1-x)\ge x$ for $0<x<1$,

$$c_{\epsilon_n/K}\le\sqrt{\frac{K}{4\epsilon_n}},\qquad c_{\epsilon_n/(2K)}\le\sqrt{\frac{K}{2\epsilon_n}}.$$

If both sensitivities are bounded by constants times $n^{-1/p}$ and $0<\epsilon_n<2$ satisfies $\epsilon_n\ge c_\epsilon n^{-\gamma}$, with $c_\epsilon>0$ and $0\le\gamma<2/p$, the local error bounds give, once the initialization terms are smaller,

$$e_{A,n}(g;\xi_n)\le C_A(g)n^{-1/p+\gamma/2}L_A(n),\quad L_G(n)=\log n,\quad L_N(n)=\{\log n\,\log\log n\}^{1/2}. \tag{76}$$

The PNR gradient–Hessian interaction is a product of vanishing radii and is absorbed for sufficiently large n. With bounded constants, $\gamma<2/p$ gives vanishing error, and $\gamma<2/p-1$ gives root-n negligible error. Uniform versions also require uniform localization and a common prescribed tuning rule.

If $\alpha_n\ge c_\alpha n^{-r}$, where $c_\alpha>0$ and $r\ge 0$, substitution into (73) gives

$$d_{A,n}\le\frac{C_A(g_{\alpha_n,z_n})+C_A(g_0)}{c_\alpha}n^{r-1/p+\gamma/2}L_A(n).$$

For bounded constants and vanishing failure probabilities, $r < 1/p - \gamma/2$ is sufficient for equality of the normalized sample responses in probability. This bound retains the interaction of sample size, privacy budget, and contamination fraction.

Finally, suppose for fixed q_z that $\{T(g_{\alpha,z}) - \boldsymbol{\theta}_0\}/\alpha \to b_z$ as $\alpha \downarrow 0$, and that under $g_{\alpha_n,z}$ the non-private centered estimator $\sqrt{n}\{\hat{\boldsymbol{\theta}}_n - T(g_{\alpha_n,z})\}$ converges to $N(0, V_0)$. If $\sqrt{n}\alpha_n \to c < \infty$ and $\sqrt{n}R_{A,n} \to 0$ along those laws, then

$$\sqrt{n}(\widetilde{\boldsymbol{\theta}}_{A,n} - \boldsymbol{\theta}_0) \implies N(cb_z, V_0).$$

Multiply (74) by $\sqrt{n}$ and use Slutsky's theorem. Here the non-private triangular-array limit and the private approximation along that sequence are the stated premises.

15 GDP calibration for the scalar Laplace mechanism

Consider $M_b(D) = W(D) + Z_b$, where W is scalar, Z_b has density $(2b)^{-1}e^{-|z|/b}$, and the common scale $b > 0$ does not depend on the dataset. Let $\Delta = \sup_{D \sim D'} |W(D) - W(D')| \in (0, \infty)$ be its exact sensitivity and put $r = \Delta/(2b)$. The exact Laplace calculation gives

$$\delta_L(t; r) = [1 - e^{t/2-r}]_+, \qquad \eta_L(r) = 1 - e^{-r}, \qquad h_L(b) = 2\{1 - (1 + r)e^{-r}\}, \tag{77}$$

where δ_L is the worst-case profile over ordered neighboring pairs, η_L is the worst-case TV distance, and h_L is the attained HDP level. The profile formula is also given by Balle et al. (2020, Theorem 3). The affinity formula follows by splitting the integral at the two neighboring query values. All suprema can be interpreted as limits along pairs approaching Δ; attainment of the sensitivity is not required.

For $\mu > 0$, write

$$\delta_G(t; \mu) = \Phi\left(\frac{\mu}{2} - \frac{t}{\mu}\right) - e^t \Phi\left(-\frac{\mu}{2} - \frac{t}{\mu}\right), \qquad t \geq 0. \tag{78}$$

The full-profile characterization of Dong et al. (2022) states that M_b is μ-GDP if and only if $\delta_L(t; r) \leq \delta_G(t; \mu)$ for every $t \geq 0$. The following argument verifies this inequality over the entire profile, not only at its TV value $t = 0$.

Proposition 15.1 (Smallest GDP parameter for scalar Laplace noise) *The scalar Laplace mechanism is μ-GDP if and only if*

$$\mu \geq \mu_L(r) := 2\Phi^{-1}\left(1 - \frac{1}{2}e^{-r}\right) = 2\Phi^{-1}\left(\frac{1 + \eta_L(r)}{2}\right). \tag{79}$$

Here $r = \Delta/(2b) > 0$. Thus $\mu_L(r)$ is the smallest admissible GDP parameter of the mechanism.

Proof Necessity follows at $t = 0$: any GDP parameter must satisfy

$$1 - e^{-r} = \delta_L(0; r) \leq \delta_G(0; \mu) = 2\Phi(\mu/2) - 1,$$

which is equivalent to (79).

For sufficiency, put $k = \Phi^{-1}(1 - e^{-r}/2) > 0$ and $q = \Phi(-k) = e^{-r}/2$. The profiles $\delta_G(t; 2k)$ and $\delta_L(t; r)$ agree at $t = 0$. For $0 < t < 2r$, differentiation gives

$$\frac{d}{dt}\{\delta_G(t; 2k) - \delta_L(t; r)\} = e^{t/2}\left\{q - e^{t/2}\Phi\left(-k - \frac{t}{2k}\right)\right\}. \tag{80}$$

Indeed, the derivative of the Gaussian profile is $-e^t\Phi(-k - t/(2k))$; the normal-density terms cancel since $\phi(k - t/(2k)) = e^t\phi(k + t/(2k))$.

To determine the sign in (80), define $v(t) = e^{t/2}\Phi(-k-t/(2k))$. With $x = k+t/(2k) > 0$,

$$\frac{v'(t)}{v(t)} = \frac{1}{2} - \frac{\phi(x)}{2k\Phi(-x)} < \frac{1}{2} - \frac{x}{2k} \le 0.$$

The strict inequality follows from

$$\phi(x) = \int_x^\infty u\phi(u)\, du > x\int_x^\infty \phi(u)\, du = x\Phi(-x).$$

Thus $v(t) \le v(0) = q$, and (80) is nonnegative. It follows that $\delta_L(t; r) \le \delta_G(t; 2k)$ on $[0, 2r]$. For $t \ge 2r$ the Laplace profile is zero, so the inequality continues to hold. This proves $2k$-GDP by the full-profile characterization. Since $\partial_\mu\delta_G(t; \mu) = \phi(\mu/2 - t/\mu) > 0$, every $\mu \ge 2k$ is also admissible. Necessity proves minimality. ■

Although the profiles agree at $t = 0$ for this smallest GDP parameter, the proof gives $\delta_L(t; r) < \delta_G(t; \mu_L(r))$ for every finite $t > 0$. Thus the result identifies the tightest Gaussian trade-off lower bound for this Laplace family; it does not identify its actual trade-off function with a Gaussian one.

Corollary 15.2 (HDP correspondence and inverse GDP calibration) *For $0 < h < 2$, let $r_h > 0$ be the unique solution of $(1 + r_h)e^{-r_h} = 1 - h/2$. The minimum scalar Laplace scale attaining h-HDP is $b_H(h) = \Delta/(2r_h)$. At that scale, the smallest GDP parameter is*

$$\mu_h^L = 2\Phi^{-1}\left(1 - \frac{1}{2}e^{-r_h}\right). \tag{81}$$

Every scalar Laplace mechanism satisfying h-HDP therefore also satisfies μ_h^L-GDP.

Conversely, for a prescribed $\mu > 0$, put

$$q_\mu = 2\Phi(-\mu/2), \qquad r_\mu = -\log q_\mu. \tag{82}$$

Then M_b is μ-GDP if and only if

$$b \ge b_{\mathrm{GDP}}(\mu) := \frac{\Delta}{2r_\mu}. \tag{83}$$

The noise variance and attained Hellinger level at this minimum scale are, respectively,

$$\mathrm{Var}(Z_{b_{\mathrm{GDP}}}) = \frac{\Delta^2}{2r_\mu^2}, \qquad h_L^{\mathrm{GDP}}(\mu) = 2\{1 - (1 + r_\mu)e^{-r_\mu}\} = 2\{1 - q_\mu(1 - \log q_\mu)\}. \tag{84}$$

These inverse calibrations recover one another: $b_{\mathrm{GDP}}(\mu_h^L) = b_H(h)$ and $h_L^{\mathrm{GDP}}(\mu_h^L) = h$.

Proof The function $r \mapsto 2\{1-(1+r)e^{-r}\}$ increases strictly on $(0,\infty)$ from 0 to 2, and $r \mapsto \mu_L(r)$ is strictly increasing. Thus the condition $h_L(b) \le h$ is equivalent to $r \le r_h$, and the first assertion follows from Proposition 15.1.

For the inverse, (79) is equivalent to $e^{-r} \ge 2\Phi(-\mu/2)$, hence to $r \le r_\mu$ and $b \ge \Delta/(2r_\mu)$. Substitution into the variance $2b^2$ and (77) proves (84). Finally, $2\Phi(-\mu_h^L/2) = e^{-r_h}$, which proves the inverse identities. ■

Remark 15.3 (Scope of the inverse-TV expression) *For an additive Gaussian mechanism with a common dataset-independent covariance $\sigma^2 I$, the smallest GDP parameter is Δ/σ, and its exact TV distance is $2\Phi\{\Delta/(2\sigma)\} - 1$. Together with Proposition 15.1, this shows that in either this Gaussian family or the common-scale scalar Laplace family, the smallest GDP parameter equals $2\Phi^{-1}\{(1+\eta)/2\}$, with η the exact worst-case TV distance.*

Consequently, within either specified family, h-HDP implies GDP with parameter

$$2\Phi^{-1}\left(\frac{1+\sqrt{h-h^2/4}}{2}\right), \qquad 0 < h < 2.$$

For $0 < h < 1$, replacing $\sqrt{h-h^2/4}$ by $\sqrt{h}$ gives a further valid relaxation. These conclusions use the verified family-specific profile comparisons; they are not consequences of TV control alone. For the scalar Laplace family, (81) *retains the exact Hellinger information and can be used instead of this relaxation.*

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